Lower Bounds in Data-Driven Decision Making: Twenty-Five Applications of Le Cam's Two-point Method

Jingkai Huang* Yanjun Han† Zhengyuan Zhou‡

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Abstract

This survey provides a readable account of Le Cam's two-point method and its applications to proving lower bounds for data-driven decision making problems. A potent tool that has long been utilized in statistics for establishing fundamental limits in estimation problems, Le Cam's method provides an elegant framework for establishing the fundamental difficulty of learning by reducing it to binary hypothesis testing, where the learner cannot distinguish between two constructed environments sufficiently well.

In this survey, we bring this toolkit to the attention of these communities and demonstrate its versatility across a range of such data-driven decision making problems, either online where data arrives sequentially or offline where data has been collected into a single batch. We present twenty-five applications – by no means exhaustive but hopefully representative of the richness in the problem landscape and hence a reflection of this method's power – where lower bounds are established on the data efficiency metrics: regret when the problem is online and sample complexity when the problem is offline. Through these examples, we delineate a unified blueprint for constructing indistinguishable problem instances with separated optimal decisions, offering a principled approach to quantifying regret and sample complexity lower bounds using Le Cam, which offers an analytical template that can guide future lower bound constructions in problems of a similar nature. Although nothing is really new in this survey (except for a few typos), we provide streamlined and simplified proofs (which may sometimes differ from its original published version) wherever possible. Through this, we hope to equip the current and future generations of researchers and students in these and related quantitative fields with the most fundamental tool for proving lower bounds.

*New York University; jh9959@nyu.edu
†New York University; yanjunhan@nyu.edu
‡New York University; zz26@nyu.edu

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1 Introduction

In recent years, there has been growing interest in understanding the fundamental statistical limits of data-driven decision-making problems across a range of operations management (OM) domains, e.g., inventory control, revenue management, and treatment planning. These problems share a common informational structure: a decision-maker must learn from data to make informed decisions that minimize cost or maximize reward.

Lower bound techniques play a central role in this landscape. They not only characterize the best achievable performance of any algorithm, but also guide the design of provably optimal methods. Among the available techniques, a particularly powerful and widely applicable one is *Le Cam's two-point method*.

1.1 Le Cam: From Statistical Estimation to Decision-Making

Le Cam's two-point method is a classical approach for deriving minimax lower bounds in statistical decision theory. Its central idea is to reduce a general estimation problem to a binary hypothesis test between two carefully constructed probabilistic environments. If these two scenarios are statistically hard to distinguish, then no estimator can perform well on both, implying an inevitable performance gap.

This technique has long been foundational in the statistical learning literature, where it is used to study the limits of inference under various model classes and noise assumptions; see Tsybakov (2009); Le Cam and Yang (2000); Hastie et al. (2009); Wu (2017) for in-depth tutorials on Le Cam from the perspective of statistics and information theory, and Loh (2017) for a survey on lower bound techniques in statistical learning. The idea of constructing lower bounds based on hypothesis testing was pioneered by LeCam (1973); Ibragimov and Khas' minskii (1977); Yu (1997), and was subsequently developed into a versatile tool that has been applied across a wide range of statistical and information-theoretic problems. Applications span a broad range of problems, including entropy estimation (Valiant, 2008; Wu and Yang, 2016; Acharya et al., 2016); robust statistics (Chen et al., 2016, 2018; Diakonikolas et al., 2018); high-dimensional linear regression (Raskutti et al., 2011; Cai and Guo, 2017); linear functional estimation for sparse Gaussian means (Collier et al., 2017); and nonparametric functional estimation (Bickel and Ritov, 1988; Birgé and Massart, 1995; Robins et al., 2017), among others. These examples collectively highlight the breadth and maturity of lower bound techniques in statistical theory. From classical parametric models to robust high-dimensional inference, these tools offer a principled way to characterize the intrinsic difficulty of estimation tasks. Their strength lies in reducing complex inference questions to fundamental testing problems, revealing how model structure and noise interact to limit achievable accuracy. As such, they form a foundational component of modern statistical methodology.

In that context, the central focus is on estimation accuracy—how close an estimator is to the truth—under a fixed data-generating distribution. However, in operations manage-

ment (OM), learning is not an end in itself, but a means to making effective decisions. For instance, one may not aim to recover the true demand distribution, but instead to determine an order quantity or price that performs well under uncertainty. This fundamental difference shifts the objective from minimizing estimation error to minimizing **decision loss**, such as regret or excess cost.

Crucially, this shift is not merely a change in loss function—it fundamentally alters the learning environment. In OM problems, the decisions taken by the learner can directly influence the distribution of the observed data. That is, the data-generating process is not fixed ex ante but evolves endogenously through the interaction between the learner's policy and the environment. In contrast, traditional statistical estimation problems assume passive observation from a static distribution, and therefore do not account for such feedback dynamics. This decision-dependent nature of observations introduces significant analytical challenges when deriving lower bounds in OM, and is a core source of complexity that sets it apart from classical statistical settings.

While Le Cam's method has appeared in numerous OM papers as a lower bound technique, its applications remain fragmented and domain-specific. To our knowledge, there is no unified treatment of how this method operates across decision-making problems in OM. This survey aims to fill that gap. We provide twenty-four representative applications of Le Cam's method across online and offline learning problems in OM, identify recurring proof strategies, and highlight how the same core idea adapts to diverse modeling choices and feedback structures.

1.2 Structure of the Survey

Throughout the survey, we demonstrate how Le Cam's two-point approach serves as a unifying lens to understand the statistical difficulty of learning in OM. Here we present the organization of the survey and describe the twenty-four applications we cover. We begin in Section 2 with a standalone treatment of the Le Cam's two-point method, including its formal statement, statistical intuition, and supporting information-theoretic tools. We illustrate the method with a warm-up example from stochastic convex optimization, and provide a proof template that will be reused across applications. Then, the applications are categorized into six thematic domains, each corresponding to the remaining sections of this survey:

Inventory Control (Section 3): we discuss the online repeated newsvendor control under various demand distributions including discrete demand (Application 1), continuous demand (Application 2), and local-separated continuous demand (Application 3); and sample complexity lower bounds of offline newsvendor policy learning under full feedback (Application 4) and censored feedback (Application 5).

Revenue Management (Section 4): we discuss the repeated first-price auction (Appli-

cation 6), and its variant of joint bidding and evaluation auction (Application 7); discrete-time dynamic pricing (Application 8), and continuous time dynamic pricing with inventory constraint (Application 9); and the assortment optimization under MNL-choice model without capacity constraint (Application 10), with capacity constraint (Application 11).

Online Bandits Problem (Section 5): we discuss the classic stochastic two-armed bandits (Application 12) with its static batched (Application 13) and dynamic batched (Application 14) variants; the stochastic multi-armed bandits (Application 15); and the linear bandits (Application 16) with its finitely-many arms (Application 17) variants further under the static batched setting (Application 18) and the dynamic batched setting (Application 19).

Offline Policy Learning (Section 6): we discuss the sample complexity lower bound of (contextual) treatment (two-action) policy learning (Application 20); multi-action policy learning (Application 21) and its variant under adaptively collected data (Application 22); distributionally robust policy learning (Application 23).

Advanced Applications in Operation Management (Section 7): we discuss a scheduling queuing system (Application 24) and an adversarial MNL assortment optimization problem (Application 25) in the last section.

Across all sections, we aim to expose the core construction that underlies Le Cam's two-point method: two environments with (i) separated optimal decisions and (ii) nearly indistinguishable distributions, and how this idea manifests under different operational settings. Each application is presented in a self-contained way, with recurring proof strategies and technical tools highlighted across domains.

The organization of the survey follows a deliberate progression—from simpler to more intricate settings—designed to reflect the increasing analytical complexity in applying Le Cam's method. As discussed earlier, a key distinction between classical statistical estimation and decision-making problems in OM lies in the interaction between decisions and the environment. When the learner's decision does not influence the data-generating process, the lower bound analysis closely resembles traditional estimation problems, and the application of Le Cam's method is relatively straightforward. In contrast, when decisions do affect the environment—either directly or through adaptive feedback—the statistical structure becomes inherently dynamic. This introduces fundamental difficulties in lower bound derivations, such as decision-induced distribution shift and feedback coupling. It is precisely these interactions that differentiate OM problems and make Le Cam-based analysis more subtle and context-specific.

Reflecting this distinction, our Applications 1–7 correspond to settings in which the learner's decision does not meaningfully influence the environment (except in Applications 5 and 7, where the interaction exists but remains structurally simple). In these settings, the

observations can be viewed as passively drawn, and the lower bound construction resembles classical hypothesis testing in statistics literatures.

From Applications 8–25 onward, the landscape changes. The decision-maker's actions begin to influence future observations—sometimes indirectly, sometimes explicitly—leading to endogenous data collection and richer feedback structures. The presence of such interaction complicates distinguishing the environments, thereby making the analysis substantially more challenging. This simple-to-complex sequencing not only facilitates exposition but also mirrors the increasing depth of statistical difficulty introduced by decision-dependent feedback—one of the defining features of OM settings when viewed through the lens of Le Cam's method.

Finally, in Section 7, we conclude the survey with a broader discussion of lower bound techniques beyond Le Cam's method. While this survey focuses on Le Cam's two-point approach as a unifying framework, other tools such as Fano's inequality, Assouad's inequality, van Trees inequality also play important roles in establishing statistical hardness across OM domains. We highlight when and why these tools may be preferred, and review recent theoretical advances that reinterpret Le Cam's method through the lens of convex duality and linear programming, shedding light on its "simple but powerful" nature. This final discussion situates our treatment within the broader landscape of lower bound methodology and points toward future theoretical developments in understanding the statistical limits of data-driven decision-making.

2 Foundation of Le Cam's Two-Point Method

This section lays the methodological foundation for the lower bound arguments developed throughout the survey. We begin in Section 2.1 by presenting Le Cam's two-point method—a classical reduction from minimax lower bounds to a binary testing problem—and introduce the separation condition that underlies its use in decision problems. This formalism will serve as a reusable blueprint for the rest of the survey.

To concretize how the method is used in practice, we present in Section 2.2 a warm-up example from stochastic convex optimization. This example walks through the full process of constructing two problem instances that are statistically hard to distinguish, yet yield significantly different optimal decisions. As a result, any algorithm must suffer a non-negligible regret on at least one instance. Through this example, we highlight a recurring technical challenge in applying Le Cam's method: although total variation distance is the key quantity controlling distinguishability, it is rarely tractable to compute exactly in high-dimensional or sequential settings.

Motivated by this challenge, we conclude the section in Section 2.3 by assembling an information-theoretic toolkit that enables indirect control of total variation distance via more analyzable divergences, such as the Kullback–Leibler divergence. These inequalities, such as Pinsker's and Bretagnolle–Huber, will be used repeatedly in the survey to derive clean, tractable lower bounds. Together, the ideas in this section establish a general blueprint that supports the diverse, domain-specific constructions developed in the remainder of the survey. Before proceeding, we first provide a summary of notations that will be applied repeatedly throughout this survey in the following table:

Table 1: Notation Used throughout the Survey

Symbol	**Description**
T	Total length of time horizon (in online problems)
$t \in [T]$	Time index of period t
N	Total number of historical sample size (in offline problems)
$\mathcal{A}$	Action (decision) space
$a_t \in \mathcal{A}$	Action (decision) at period t
$\mathcal{H}_t$	History up to time t
$\pi \in \Pi$	Policy mapping from history to actions
$\theta, v \in \Theta$	Environment parameters
$\mathbb{P}_v$	Data-generating distribution under environment v
F_v, f_v	CDF and PDF of distribution $\mathbb{P}_v$
$\mathcal{R}(T; \pi, v)$	Regret of policy π under environment v over horizon T
$\|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}}$	Total Variation divergence of $\mathbb{P}$ and $\mathbb{Q}$
$\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q})$	Kullback-Leibler divergence of $\mathbb{P}$ and $\mathbb{Q}$

2.1 Le Cam's Two-Point Method

Le Cam's two-point method is a classical technique in statistical decision theory that provides a lower bound on the performance of any estimator or policy, by reducing the problem to a binary hypothesis testing problem between two parameter values. The core idea is simple but powerful: if two instances of the problem are hard to distinguish from the observed data, then no estimator can perform well simultaneously on both.

Suppose we are given a family of possible environments or instances Θ, and we wish to design an estimator or policy $\hat{\theta}$ that minimizes some loss function $\ell(\theta, \hat{\theta})$, which measures how far $\hat{\theta}$ is from the true underlying parameter θ. The *minimax risk* is defined as

$$\mathcal{R}^* = \inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_\theta[\ell(\theta, \hat{\theta})],$$

where the expectation is taken over the randomness of the observed data generated under parameter θ. Directly analyzing $\mathcal{R}^*$ is often challenging. Le Cam's two-point method provides a strategy to lower bound $\mathcal{R}^*$ by considering only two carefully chosen parameters $\theta_1, \theta_2 \in \Theta$, and reducing the problem to a binary hypothesis testing task. To use the method, we first introduce the following separation condition between θ_1 and θ_2:

Definition 2.1 (Separation Condition)**.** We say that the pair (θ_1, θ_2) satisfies a separation condition with separation parameter $\Delta > 0$ if

$$\inf_{\hat{\theta}} \left\{ \ell(\theta_1, \hat{\theta}) + \ell(\theta_2, \hat{\theta}) \right\} \geq \Delta.$$

In words, no estimator $\hat{\theta}$ can simultaneously be close to both θ_1 and θ_2 without incurring a total loss of at least Δ.

Once such a pair (θ_1, θ_2) is found that satisfies a separation condition, the next step is to evaluate how difficult it is to distinguish between the two induced distributions $\mathbb{P}_1$ and $\mathbb{P}_2$, where $\mathbb{P}_i$ denotes the distribution of the observed data under parameter θ_i. Intuitively, if the two distributions are statistically indistinguishable, i.e., no statistical test can reliably tell them apart, then any estimator must incur a non-negligible error on at least one of the two instances. To formalize this notion of indistinguishability, we quantify the "closeness" between $\mathbb{P}_1$ and $\mathbb{P}_2$ using a family of divergence measures known as the f-divergence.

Definition 2.2. Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability distributions over a common measurable space $(\mathcal{X}, \mathcal{F})$, with $\mathbb{P}$ absolutely continuous with respect to $\mathbb{Q}$ (denoted $\mathbb{P} \ll \mathbb{Q}$). Then for any convex function $f : (0, \infty) \to \mathbb{R}$ such that it is strictly convex at 1 and $f(1) = 0$, the f-divergence of Q from P is defined as

$$\mathrm{D}_f(\mathbb{P} \| \mathbb{Q}) \triangleq \mathbb{E}_Q \left[f\left(\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}} \right) \right].$$

And for two examples of the f-divergence:

- For $f(x) \triangleq \frac{1}{2}|x-1|$, the *Total Variation (TV) divergence (distance)* is defined as

$$\|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}} \triangleq \mathrm{D}_f(\mathbb{P}\|\mathbb{Q}) = \frac{1}{2}\int |\mathrm{d}\mathbb{P} - \mathrm{d}\mathbb{Q}|.$$

- For $f(x) \triangleq x \log x$, the *Kullback–Leibler (KL) divergence* is defined as

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q}) \triangleq \mathrm{D}_f(\mathbb{P}\|\mathbb{Q}) = \mathbb{E}_{\mathbb{P}}\left[\log \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}\right].$$

The Le Cam's Two-Point method gives the following lower bound based on the total variation (TV) distance between distributions $\mathbb{P}_1$ and $\mathbb{P}_2$.

Theorem 2.1 (Le Cam's Two-Point Method)**.** *Fix any θ_1, θ_2 in the parametric space (determines $\mathbb{P}_1$ and $\mathbb{P}_2$, respectively). Suppose the Separation Condition holds with separation parameter $\Delta > 0$, then:*

$$\mathcal{R}^\star \geq \inf_{\hat{\theta}} \max_{\theta \in \{\theta_1, \theta_2\}} \mathbb{E}_\theta[\ell(\theta, \hat{\theta})] \geq \frac{\Delta}{2} \cdot (1 - \|\mathbb{P}_1 - \mathbb{P}_2\|_{\mathrm{TV}}).$$

Proof of Theorem 2.1. The result follows from the following inequalities:

$$\begin{aligned}
\mathcal{R}^\star &= \inf_{\hat{\theta}} \max_{\theta \in \Theta} \mathbb{E}_\theta[\ell(\theta, \hat{\theta})] \geq \inf_{\hat{\theta}} \max_{\theta \in \{\theta_1, \theta_2\}} \mathbb{E}_\theta[\ell(\theta, \hat{\theta})] \\
&\geq \inf_{\hat{\theta}} \frac{\mathbb{E}_{\theta_1}[\ell(\theta_1, \hat{\theta})] + \mathbb{E}_{\theta_2}[\ell(\theta_2, \hat{\theta})]}{2} \\
&= \frac{1}{2} \inf_{\hat{\theta}} \left[\int \ell(\theta_1, \hat{\theta}) \mathrm{d}\mathbb{P}_{\theta_1} + \int \ell(\theta_2, \hat{\theta}) \mathrm{d}\mathbb{P}_{\theta_2}\right] \\
&\geq \frac{1}{2} \int \inf_{\hat{\theta}} \left\{\ell(\theta_1, \hat{\theta}) + \ell(\theta_2, \hat{\theta})\right\} \min\{\mathrm{d}\mathbb{P}_{\theta_1}, \mathrm{d}\mathbb{P}_{\theta_2}\} \\
&\geq \frac{\Delta}{2} \cdot (1 - \|\mathbb{P}_1 - \mathbb{P}_2\|_{\mathrm{TV}}).
\end{aligned}$$

where the last inequality follows from the separation condition and the definition of TV-distance such that $\|\mathbb{P}_1 - \mathbb{P}_2\|_{\mathrm{TV}} = 1 - \int \min\{\mathrm{d}\mathbb{P}_1, \mathrm{d}\mathbb{P}_2\}$. □

The intuition behind Le Cam's two-point method is simple yet powerful. If two parameters θ_1 and θ_2 satisfy a separation condition, then any estimator must, in effect, distinguish between them to perform well. However, if the data distributions $\mathbb{P}_1$ and $\mathbb{P}_2$ corresponding to these two parameters are close in total variation distance, then distinguishing between them based on observed data is fundamentally difficult. As a result, any estimator, regardless of its form, must incur a minimum expected loss. The lower bound on performance thus arises from the interplay between how far apart the optimal decisions are (captured by the separation Δ) and how hard it is to tell the two scenarios apart statistically (captured by the total variation distance $\|\mathbb{P}_1 - \mathbb{P}_2\|_{\mathrm{TV}}$).

2.2 Warm-up Application: Stochastic Convex Optimization

To illustrate how Le Cam's method can be applied in deriving lower bounds for OM problems, we start with an illustrative example of stochastic convex optimization.

Application 0: Stochastic Convex Optimization

Consider a fundamental problem in stochastic convex optimization: the learner aims to optimize an unknown function $f : [-1, 1] \to [-1, 1]$ belonging to a function class $\mathcal{F}$, where $\mathcal{F}$ denotes the set of all convex and 1-Lipschitz continuous functions. The objective is to find a point that approximately minimizes f, based on noisy observations gathered in the history.

The interaction between the learner and the environment proceeds sequentially over T rounds. In each round $t = 1, \cdots, T$, the learner randomly samples a query point $x_t \sim \mathbb{P}_x$, and receives two noisy feedback signals $y_t \sim \mathbb{P}_y$ and $z_t \sim \mathbb{P}_z$, where y_t approximates the function value $f(x_t)$ and z_t approximates the gradient $f'(x_t)$. Formally, the feedback satisfies

$$\mathbb{E}[y_t] = f(x_t), \quad \mathbb{E}[z_t] = f'(x_t),$$

where the expectation is over the noise distribution in observations. Assume that x_t, y_t and z_t all take values in $[-1, 1]$. Denote the joint distribution of (x_t, y_t, z_t) as $\mathbb{P} \triangleq \mathbb{P}_x \times \mathbb{P}_y \times \mathbb{P}_z \in \mathcal{P}$, where $\mathcal{P}$ is the distribution class that satifies our previous settings. After collecting T i.i.d. observations $\{(x_t, y_t, z_t)\}_{t=1}^T$, the learner produces a final decision $\hat{x} \in [-1, 1]$. The performance of the learner is evaluated through the optimization error:

$$L(f, \hat{x}) = f(\hat{x}) - \min_{x \in [-1,1]} f(x),$$

which measures how far the output $\hat{x}$ is from minimizing f over the feasible domain. We are interested in its minimax optimization error defined as:

$$\inf_{\hat{x}} \sup_{f \in \mathcal{F}, \mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}^T}[L(f, \hat{x})],$$

where $\mathbb{P}^T$ stands for the joint distribution of $\{(x_t, y_t, z_t)\}_{t=1}^T$; the infimum is taken over all possible learning strategies (i.e., mappings from observed history to output decisions); the supremum is taken over all possible functions f in the function class $\mathcal{F}$ and all possible distributions $\mathbb{P}$ in the distribution class $\mathcal{P}$.

Our goal is to establish a fundamental lower bound for the stochastic convex optimization problem. Specifically, we will prove the following lower bound result.

Theorem 2.2. *For the stochastic convex optimization problem introduced before, for any data-driven decision $\hat{x}$,*

$$\sup_{f \in \mathcal{F}, \mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}^T}[L(f, \hat{x})] = \Omega\left(\frac{1}{\sqrt{T}}\right),$$

i.e., no learner can achieve optimization error smaller than $\Omega(1/\sqrt{T})$ in this setting.

A Template for Lower Bound Proofs via Le Cam

We now illustrate the general paradigm of applying Le Cam's two-point method to derive minimax lower bounds, using the stochastic convex optimization problem introduced above as a guiding example. The core idea is to first (1) construct two problem instances—specifically, two distributions—that are statistically indistinguishable under noisy observations, yet have well-separated minimizers; then (2) verify the separation condition under the constructed instances to make sure that no learner can perform well on both instances simultaneously, thereby inducing a fundamental limit on achievable optimization accuracy; (3) apply the Le Cam's two-point method to derive a lower bound w.r.t. the divergence between two distribution instances; and at last (4) calculate the divergence (lower bound) and finish the proof. We present the details step by step as follows:

Step 1. Instance Construction. We first construct two candidate functions f_1 and f_2 over $[-1, 1]$ as follows:

$$f_1(x) = \frac{x}{\sqrt{T}}, \quad f_2(x) = -\frac{x}{\sqrt{T}}.$$

It can be easily verified that both f_1 and f_2 are convex, 1-Lipschitz, and map to $[-1, 1]$ as required. At each round t, let x_t be uniformly sampled from $[-1, 1]$, and conditional on x_t, the learner further receives observations (y_t, z_t) independently generated according to:

$$y_t = \begin{cases} +1 & \text{w.p. } \frac{1+f(x_t)}{2}, \\ -1 & \text{w.p. } \frac{1-f(x_t)}{2}, \end{cases} \quad z_t = \begin{cases} +1 & \text{w.p. } \frac{1+f'(x_t)}{2}, \\ -1 & \text{w.p. } \frac{1-f'(x_t)}{2}, \end{cases}$$

where $f \in \{f_1, f_2\}$.

Step 2. Separation Verification. Under the previous construction, we can verify that $L(f_1, x_1^*) = L(f_2, x_2^*) = -\frac{1}{\sqrt{T}}$, where x_1^* and x_2^* denote the optimal solutions for f_1 and f_2, respectively. Besides,

$$\inf_{\hat{x}} \{L(f_1, \hat{x}) + L(f_2, \hat{x})\} = \inf_{\hat{x}} \left\{ \frac{\hat{x} + 1}{\sqrt{T}} + \frac{-\hat{x} + 1}{\sqrt{T}} \right\} = 0$$

then we have the separation condition holds:

$$\inf_{\hat{x}} \{L(f_1, \hat{x}) + L(f_2, \hat{x})\} - (L(f_1, x_1^*) + L(f_2, x_2^*)) = \frac{2}{\sqrt{T}},$$

with the separation parameter $\Delta = \frac{2}{\sqrt{T}}$.

Step 3. Applying Le Cam. Then, by applying the Le Cam's two-point method in Theorem 2.1, we have for any $\hat{x}$,

$$\sup_{f \in \mathcal{F}, \mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}^T}[L(f, \hat{x})] \geq \frac{1}{\sqrt{T}} \left(1 - \|\mathbb{P}_1^T - \mathbb{P}_2^T\|_{\mathrm{TV}}\right), \tag{1}$$

where $\mathbb{P}_1^T$ and $\mathbb{P}_2^T$ are the probability of observation $\{(x_t, y_t, z_t)\}_{t=1}^T$ under the underlying functions f_1 and f_2, respectively.

Thus, to provide the lower bound of $\sup_{f\in\mathcal{F},\mathbb{P}\in\mathcal{P}} \mathbb{E}_{\mathbb{P}^T}[L(f,\hat{x})]$, it is sufficient to focus on the value of TV distance between $\mathbb{P}_1^T$ and $\mathbb{P}_2^T$. However, as it is hard to calculate $\|\mathbb{P}_1^T - \mathbb{P}_2^T\|_{\mathrm{TV}}$ directly, and we therefore introduce several useful tools in the following section, which will be applied throughout the survey.

2.3 The Information-Theoretic Toolkit

Recall that in Le Cam's two-point method, the lower bound on the minimax risk is expressed in terms of the TV distance between two probability distributions $\mathbb{P}_1$ and $\mathbb{P}_2$.

However, computing the total variation distance directly can be technically involved, especially in high-dimensional or sequential settings, where the joint distribution over observations becomes complex. To derive more easy-to-calculate lower bounds, we will rely on the KL divergence of two distributions instead, which is often more tractable to compute and analyze.

To facilitate analysis, we therefore seek to bound the TV distance via the KL divergence. This substitution enables us to derive useful and computable lower bounds without having to directly evaluate the TV distance. The rest of this section introduces several classical inequalities that relate KL divergence to total variation, forming the core of our information-theoretic toolkit.

Lemma 2.1. *Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability distributions over the same measurable space $(\Omega, \mathcal{F})$. The TV distance satisfies:*

(a) Pinsker's inequality:

$$\|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}}^2 \le \frac{1}{2}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q}). \tag{2}$$

(b) Bretagnolle-Huber inequality: let $A \in \mathcal{F}$ be an arbitrary event, $A^c = \Omega\backslash A$ is the complement of A, then

$$\mathbb{P}(A) + \mathbb{Q}(A^c) \ge \frac{1}{2}\exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q})), \tag{3}$$

which implies the following upper bounds on the TV distance:

$$\|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}} \le \sqrt{1 - \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q}))} \le 1 - \frac{1}{2}\exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q})).$$

Lemma 2.2 (Data processing inequality)**.** *Consider a channel that produces Y given X based on the law $\mathbb{P}_{Y|X}$ (shown below). If $\mathbb{P}_Y$ is the distribution of Y when X is generated by $\mathbb{P}_X$ and $\mathbb{Q}_Y$ is the distribution of Y when X is generated by $\mathbb{Q}_X$, then for any f-divergence $\mathrm{D}_f(\cdot\|\cdot)$,*

$$\mathrm{D}_f(\mathbb{P}_Y\|\mathbb{Q}_Y) \le \mathrm{D}_f(\mathbb{P}_X\|\mathbb{Q}_X).$$

Lemma 2.3 (Properties of the KL divergence)**.** *Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability distributions over a common measurable space $(\mathcal{X}, \mathcal{F})$,*

- $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q}) \geq 0$*;*
- *If $\mathbb{P}$ and $\mathbb{Q}$ are product measures, $\mathbb{P} = \otimes_{i=1}^{n}\mathbb{P}_i$, $\mathbb{Q} = \otimes_{i=1}^{n}\mathbb{Q}_i$, then $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q}) = \sum_{i=1}^{n}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_i\|\mathbb{Q}_i)$;*
- $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q})$ *is jointly convex in its arguments $\mathbb{P}$ and $\mathbb{Q}$.*

Proofs for these lemmas are left in the Appendix A. Based on the tools introduced before, we introduce the last step in our template for lower bound proofs via Le Cam:

Step 4. Calculating KL between Environments. we are now able to further lower bound $1 - \|\mathbb{P}_1^T - \mathbb{P}_2^T\|_{\mathrm{TV}}$ through:

$$1 - \|\mathbb{P}_1^T - \mathbb{P}_2^T\|_{\mathrm{TV}} \overset{(i)}{\geq} \frac{1}{2\sqrt{T}} \exp\left(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^T\|\mathbb{P}_2^T)\right)$$
$$\overset{(ii)}{=} \frac{1}{2\sqrt{T}} \exp\left(-\sum_{t=1}^{T} \mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1(y_t, z_t|x_t)\|\mathbb{P}_2(y_t, z_t|x_t))\right),$$

with (i) following the Bretagnolle-Huber inequality in Lemma 2.1; (ii) following the property of KL divergence in Lemma 2.3, where we have $\mathbb{P}_k^T = \prod_{t=1}^{T}\mathbb{P}_x(x_t)\cdot\mathbb{P}_k(y_t, z_t|x_t)$ for any $k \in \{1, 2\}$, and that $\mathbb{P}_x(x_t)$ are identical in the two distribution instances.

Note that $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1(y_t, z_t|x_t)\|\mathbb{P}_2(y_t, z_t|x_t)) = \mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1(y_t|x_t)\|\mathbb{P}_2(y_t|x_t)) + \mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1(z_t|x_t)\|\mathbb{P}_2(z_t|x_t))$ by the conditional independence property of y_t and z_t given x_t. Then, for any given x_t and $T > 4$,

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1(y_t|x_t)\|\mathbb{P}_2(y_t|x_t)) = f_1(x_t)\cdot\log\frac{1+f_1(x_t)}{1-f_1(x_t)} \leq 2f_1(x_t)^2 + 2f_1(x_t)^3 \leq \frac{2}{T} + \frac{2}{T^{3/2}},$$

where the inequality comes from the fact that $\log\frac{1+x}{1-x} \leq 2x + 2x^2$ for $x \in (0, 1/2)$. Similarly, we can also derive

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1(z_t|x_t)\|\mathbb{P}_2(z_t|x_t)) \leq \frac{2}{T} + \frac{2}{T^{3/2}}.$$

Combine the KL results and taking them back to (1), we have

$$\begin{aligned}\sup_{f\in\mathcal{F}}\mathbb{E}[L(f, \hat{x})] &\geq \frac{1}{\sqrt{T}}\left(1 - \|\mathbb{P}_1^T - \mathbb{P}_2^T\|_{\mathrm{TV}}\right)\\ &\geq \frac{1}{2\sqrt{T}}\exp\left(-4\cdot(1 + T^{-1/2})\right)\\ &\geq \frac{\exp(-6)}{2\sqrt{T}} = \Omega(1/\sqrt{T}),\end{aligned}$$

which completes our proof for Theorem 2.2.

2.4 Summary and Bibliographic Remarks

This section introduced Le Cam's two-point method as a foundational technique for deriving minimax lower bounds in decision-making problems. The core strategy—constructing two indistinguishable environments that induce different optimal decisions—forms the basis for a wide range of lower bound arguments throughout the survey.

To operationalize the method, we presented a warm-up example in stochastic convex optimization and highlighted the structure of a reusable four-step routine: (i) construct a pair of problem instances; (ii) verify separation condition; (iii) apply Le Cam's two-point method; and (iv) upper bound the KL-divergence via information-theoretic inequalities. The accompanying toolkit, including Pinsker's inequality, the Bretagnolle–Huber bound, and KL decomposition for product measures, will appear frequently in subsequent sections.

Lower bounds for online convex optimization have been extensively studied in the optimization and learning theory literature; see Shalev-Shwartz et al. (2012) and Hazan et al. (2016) for comprehensive treatments. Our warm-up application is simplified from the oracle-based framework proposed in Agarwal et al. (2009), which develops information-theoretic lower bounds under stochastic gradient oracles. In their setting, the learner operates over a decision space $\mathbb{R}^d$ and actively selects query points x_t based on past feedback, aiming to minimize the expected loss using a sequential algorithm. They establish a lower bound of $\Omega(\sqrt{d/T})$, even when the learner has access to unbiased function and gradient values. Besides, several follow-up works have extended the oracle-based framework to richer feedback models. For example, Duchi et al. (2015) study zero-order optimization with noisy function evaluations, and Singer and Vondrák (2015) characterize the information-theoretic limits under the biased oracles. These results further highlight how oracle structure impacts the statistical hardness of convex optimization problems in the oracle-based framework.

In the sections ahead, we will apply this general methodology to increasingly complex operational problems, where the interaction between learning and decision-making creates new challenges beyond classical estimation.

3 Inventory Control

Inventory control is a central problem in operations management, arising in a wide range of applications from retail and supply chain planning to production systems and service operations. At its core, inventory management involves making decisions about how much to order or replenish in the presence of uncertain demand, typically with the goal of minimizing costs associated with inventory holding or backlogging.

In this section, we study the fundamental statistical limits of inventory control through the lens of the Le Cam's two-point method. While many classical results in inventory theory focus on structural properties of optimal policies under known demand distributions, recent research has shifted attention to the data-driven setting—where the underlying demand distribution is unknown and must be learned from past observations. This shift necessitates a deeper understanding of the regret and sample complexity associated with learning-based inventory policies.

We divide our analysis into two parts: in Section 3.1, we analyze online inventory problems where demand is revealed sequentially, and the decision-maker must update her ordering decisions over time based on observed outcomes; in Section 3.2, we investigate the sample complexity of learning near-optimal inventory policies from historical data, which highlights the fundamental information-theoretic limits of inventory optimization in the offline setting. Together, these sections showcase the versatility of Le Cam's method in revealing how hard it is to learn or adapt inventory policies from data crossing both online and offline contexts.

Before proceeding, we first summarize the notations tailored for Section 3, with its detailed definitions introduced later in the corresponding sections.

Table 2: Notation Used throughout Section 3

Symbol	**Description**
$\mathcal{P}$	Demand distribution class
$\mathbb{P} \in \mathcal{P}$	(Bounded) demand distribution
$D_t \sim \mathbb{P}$	Random demand at period t
M	Upper bound of demand
h, b	Holding and lost-sales cost parameters
ρ	Critical quantile level $\frac{b}{b+h}$
q_t	Inventory decision at time t
$C(q_t; \mathbb{P})$	Single-period cost under q_t and distribution $\mathbb{P}$

3.1 Online Inventory Control: Regret Analysis

In this section, we delve into the online inventory control problem, focusing on scenarios where a decision-maker must make sequential inventory decisions under uncertainty about

future demand. We will mainly focus on the repeated newsvendor setting.

Consider a repeated newsvendor problem over a finite time horizon of T periods. In each period $t \in [T]$, the sequence of event is as follows:

(1) **Inventory Decision**: An inventory order decision $q_t \geq 0$ is made and arrives instantaneously (assume zero lead time).

(2) **Demand Realization**: A random demand D_t drawn independently from an unknown distribution $\mathbb{P}$ occurs and is **fully observed** by the decision-maker.

(3) **Cost Evaluation**: If $q_t \geq D_t$, the decision-maker incurs a holding cost h per unit of unsold inventory $(q_t - D_t)$; if $q_t < D_t$, the decision-maker incurs a lost-sales cost b per unit of unmet demand $(D_t - q_t)$. The expected single-period cost at period t is $C(q_t; \mathbb{P}) \triangleq h \cdot \mathbb{E}_{\mathbb{P}}[q_t - D_t]^+ + b \cdot \mathbb{E}_{\mathbb{P}}[D_t - q_t]^+$.

Let $\mathcal{H}_t \triangleq (q_1, D_1, \cdots, q_{t-1}, D_{t-1})$ denote the history observations up to time t and policy $\pi = (q_1, \cdots, q_T)$ is a mapping from history $\mathcal{H}_t$ to inventory decision q_t for $t = 1, \cdots, T$. The total expected inventory cost over T periods under policy π is given by $\mathbb{E}\left[\sum_{t=1}^{T} C\left(q_t; \mathbb{P}\right)\right]$, where the expectation is taken w.r.t. the randomness of history decisions and observations.

We note that when the decision-maker has access to the distribution $\mathbb{P}$, then the optimal order decision $q_t^\star$ at each period can be directly obtained through $\min_q C(q; \mathbb{P})$, and we have $q_t^\star = \{q : \inf_q F(q) \geq \rho\}$ through simple calculus, where $\rho = \frac{b}{b+h}$ is the critical ratio. Since the optimal order quantity is time-independent, we abbreviate it as $q^\star$ and the regret is given by:

$$\mathcal{R}(T; \pi, \mathcal{P}) = \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}\left[\sum_{t=1}^{T} \left(C\left(q_t; \mathbb{P}\right) - C(q^\star; \mathbb{P})\right)\right],$$

where $\mathcal{P}$ is the distribution space of demand D_t with its support bounded by $[0, M]$, and we will construct the lower bounds of $\mathcal{R}(T; \pi, \mathcal{P})$ under different demand distribution classes $\mathcal{P}$. Specifically, we will consider:

(i) Discrete distribution $\mathcal{P}_d$,

(ii) Continuous distribution class $\mathcal{P}_c$,

(iii) Continuous *local-separated* distribution class that $\mathcal{P}_c(\varepsilon, \alpha) \triangleq \{\mathbb{P} \in \mathcal{P}_c : f(q) \geq \varepsilon, \forall q \in [q^\star - \alpha, q^\star + \alpha]\}$ for some $\varepsilon, \alpha > 0$. Clearly, $\mathcal{P}_c(\varepsilon, \alpha) \subseteq \mathcal{P}_c$.

Application 1: Repeated newsvendor under $\mathcal{P}_d$

We begin with repeated newsvendor learning under the discrete demand. The following Theorem restates the lemma 4 of Besbes and Muharremoglu (2013).

Theorem 3.1. *Under the settings introduced previously, for any policy $\pi \in \Pi$,*

$$\mathcal{R}(T;\pi,\mathcal{P}_d) = \Omega((b+h)M\sqrt{T}).$$

Proof of Theorem 3.1.

Step 1. Instance Construction. First, construct Instance $v = 1$ ($\mathbb{P}_1$) and Instance $v = 2$ ($\mathbb{P}_2$) with CDF F_1 and F_2 as follows:

$$F_1(q) = \begin{cases} \rho + \Delta & \text{if } q = 0, \\ 1 & \text{if } q = M, \end{cases} \quad F_2(q) = \begin{cases} \rho - \Delta & \text{if } q = 0, \\ 1 & \text{if } q = M, \end{cases}$$

for some small $\Delta > 0$ that will be determined later. By the construction, in Instance 1, the optimal order quantity is $q^\star_{F_1} = 0$ and in Instance 2, it is $q^\star_{F_2} = M$. Thus the two distributions are hard to distinguish, while their optimal solutions are far apart.

Step 2. Separation Verification. Next, we examine the separation condition under the two constructed instances. We have:

$$C(q;\mathbb{P}_1) - C(q^\star_{F_1};\mathbb{P}_1) \geq \frac{M\Delta(b+h)}{4}, \ \forall q > M/2, \tag{4}$$

$$C(q;\mathbb{P}_2) - C(q^\star_{F_2};\mathbb{P}_2) \geq \frac{M\Delta(b+h)}{4}, \ \forall q \leq M/2, \tag{5}$$

with detailed calculations of (4) and (5) left in Appendix B of this section. In other words, any policy cannot perform both well in the two instances, and the separation condition holds with:

$$\inf_q \{C(q;\mathbb{P}_1) + C(q;\mathbb{P}_2)\} - C(q^\star_{F_1};\mathbb{P}_1) - C(q^\star_{F_2};\mathbb{P}_2) \geq \frac{M\Delta(b+h)}{4}. \tag{6}$$

Step 3. Applying Le Cam. Now, we can rely on the Le Cam's two-point method and derive that for any policy π,

$$\begin{aligned} \mathcal{R}(T;\pi,\mathcal{P}_d) &\overset{(i)}{\geq} \frac{(b+h)M\Delta}{8} \sum_{t=1}^{T} \left(1 - \|\mathbb{P}_1^{t-1} - \mathbb{P}_2^{t-1}\|_{\mathrm{TV}}\right) \\ &\overset{(ii)}{\geq} \frac{(b+h)M\Delta}{16} \sum_{t=1}^{T} \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{t-1} \| \mathbb{P}_2^{t-1})), \end{aligned} \tag{7}$$

where (i) comes from the Le Cam's Two-Point Method in Theorem 2.1; (ii) is according to the Bretagnolle-Huber inequality, and here $\mathbb{P}_1^{t-1}$ and $\mathbb{P}_2^{t-1}$ denote the distribution of history $\mathcal{H}_{t-1}$ under policy π.

Step 4. Calculating KL between Environments. The last step remains in calculating the KL-divergence in (7):

$$\begin{aligned}
\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{t-1}\|\mathbb{P}_2^{t-1}) &= \mathbb{E}_{\mathbb{P}_1}\left[\log\frac{\mathbb{P}_1\left(q_1, D_1, \cdots, q_{t-1}, D_{t-1}\right)}{\mathbb{P}_2\left(q_1, D_1, \cdots, q_{t-1}, D_{t-1}\right)}\right] \\
&\overset{(i)}{=} (t-1)\mathbb{E}_{\mathbb{P}_1}\left[\log\frac{\mathbb{P}_1(D_1)}{\mathbb{P}_2(D_1)}\right] \\
&= (t-1)\left[(\rho+\Delta)\log\left(\frac{\rho+\Delta}{\rho-\Delta}\right) + (1-\rho-\Delta)\log\left(\frac{1-\rho-\Delta}{1-\rho+\Delta}\right)\right] \\
&\overset{(ii)}{\leq} (t-1)\left[2(\rho+\Delta)\left(\rho^{-1}\Delta + \rho^{-2}\Delta^2\right) - 2(1-\rho-\Delta)(1-\rho)^{-1}\Delta\right] \\
&= c_0\cdot(t-1)\Delta^2,
\end{aligned}$$

for $c_0 \triangleq 2\rho^{-1} + (1-\rho)^{-1} + \rho^{-2}$. (i) is due to the i.i.d. nature of demand D_t, and that the inventory decision will not affect how the demand is generated; (ii) use the fact that $2x \leq \log\frac{1+x}{1-x} \leq 2x + 2x^2$ for $x \in (0, 1/2)$ and Δ is small so that $\Delta \leq \min\{\rho, 1-\rho\}/2$.

Taking the KL-divergence upper bound back to (7), we obtain

$$\begin{aligned}
\mathcal{R}(T;\pi,\mathcal{P}_d) &\geq \frac{(b+h)M\Delta}{16}\cdot\sum_{t=1}^{T}\left[\exp(-c_0(t-1)\Delta^2)\right] \\
&\geq \frac{(b+h)M\Delta}{16}\cdot\sum_{t=1}^{\min\left\{\lfloor\Delta^{-2}\rfloor, T\right\}}\exp(-c_0) \\
&\geq \frac{\exp(-c_0)}{16}\cdot(b+h)M\min\left\{\Delta^{-1}, \Delta T\right\}.
\end{aligned}$$

Note that if Δ can be arbitrary close to 0, we can choose $\Delta = T^{-1/2}$ to maximize the regret lower bound, which leads to $\Omega((b+h)M\sqrt{T})$; if Δ is restricted to be lower bounded by some $\varepsilon > 0$ (i.e., $f(q) \geq \varepsilon > 0$ for any q), we choose $\Delta = \varepsilon$ and the regret is now lower bounded by a constant $\Omega((b+h)M/\varepsilon)$. Both regret lower bounds are tight for the repeated newsvendor control with discrete support. □

Application 2: Repeated newsvendor under $\mathcal{P}_c$

Here we consider the repeated newsvendor learning under the continuous demand. The information structure is very similar to that of the discrete case, and here we will mainly emphasis the different construction for a continuous distribution other than the two-point distributions constructed previously. The following Theorem restates the proposition 1 of Zhang et al. (2020).

Theorem 3.2. *Under the settings introduced previously, for any policy $\pi \in \Pi$,*

$$\mathcal{R}(T;\pi,\mathcal{P}_c) = \Omega((b+h)M\sqrt{T}).$$

Proof of Theorem 3.2. The idea in providing the regret lower bound for $\mathcal{P}_c$ is very similar to that in its discrete counterpart $\mathcal{P}_d$. Following the same analysis routine:

Step 1. Instance Construction. We introduce two continuous distribution instances $\mathbb{P}_1$ and $\mathbb{P}_2$ with CDF F_1 and F_2 as follows respectively:

$$F_1(q) = \begin{cases} \left(\frac{4\rho}{M} + \frac{4\Delta}{M}\right) q, & \text{for } 0 \le q < \frac{M}{4}, \\ \rho + \Delta, & \text{for } \frac{M}{4} \le q < \frac{3M}{4}, \\ \left(\frac{4(1-\rho)}{M} - \frac{4\Delta}{M}\right)\left(q - \frac{3M}{4}\right) + \rho + \Delta, & \text{for } \frac{3M}{4} \le q < M, \\ 1, & \text{for } q = M, \end{cases}$$

and

$$F_2(q) = \begin{cases} \left(\frac{4\rho}{M} - \frac{4\Delta}{M}\right) q, & \text{for } 0 \le q < \frac{M}{4}, \\ \rho - \Delta, & \text{for } \frac{M}{4} \le q < \frac{3M}{4}, \\ \left(\frac{4(1-\rho)}{M} + \frac{4\Delta}{M}\right)\left(q - \frac{3M}{4}\right) + \rho - \Delta, & \text{for } \frac{3M}{4} \le q < M, \\ 1, & \text{for } q = M, \end{cases}$$

for some small $\Delta > 0$ that will be determined later. By the construction, the optimal order quantities $q^\star_{F_1}$ and $q^\star_{F_2}$ are within $(0, M/4)$ and $(3M/4, M)$, respectively.

Step 2. Separation Verification. Through some calculations, we still have the expected single-period cost satisfies (4) and (5), thus leading to the separation condition (6).

Step 3. Applying Le Cam. Therefore, (7) still holds for $\mathcal{R}(T; \pi, \mathcal{P}_c)$, i.e.,

$$\mathcal{R}(T; \pi, \mathcal{P}_c) \ge \frac{(b+h)M\Delta}{16} \sum_{t=1}^{T} \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{t-1} \| \mathbb{P}_2^{t-1})).$$

Step 4. Calculating KL between Environments. The KL divergence $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{t-1} \| \mathbb{P}_2^{t-1})$ can be upper bounded through a similar routine. We omit the details and derive $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{t-1} \| \mathbb{P}_2^{t-1}) \le c_0 \cdot (t-1)\Delta^2$ for the same c_0 defined before. Thus, we conclude that nothing changes (except the construction) compared with the $\mathcal{P}_d$ case and we get the same tight regret lower bound $\mathcal{R}(\mathcal{P}_c, T) = \Omega((b+h)M\sqrt{T})$. □

Application 3: Regret Lower Bound under $\mathcal{P}_c(\varepsilon, \alpha)$

Here we consider the repeated newsvendor learning problem under a restricted class of local-separated continuous demand distributions, where the distribution is required to have a density lower bounded in a neighborhood of the optimal order quantity. This local separation condition ensures that the demand distribution is "well-behaved" near the optimal solution, which facilitates sharper regret bounds that depend on (ε, α).

Compared to the general continuous class $\mathcal{P}_c$, this restriction rules out distributions that are overly flat or sparse near the decision threshold, and therefore allows us to derive instance-dependent bounds that interpolate between the worst-case and favorable cases. The proof follows the same four-step Le Cam template as in the previous settings, but employs a more localized construction to ensure that the KL divergence remains controlled while maintaining sufficient separation in cost. The following Theorem restates the theorem 1 of Lin et al. (2022).

Theorem 3.3. *Under the settings introduced previously, for any policy $\pi \in \Pi$, suppose that $\alpha\varepsilon \leq \min\left\{\frac{1}{3}\rho, \frac{1}{2}(1-\rho)\right\}$ and $4\alpha + 1 \leq M$ hold, then*

$$\mathcal{R}(T; \pi, \mathcal{P}_c(\varepsilon, \alpha)) = \Omega\left((b+h)\min\left\{\frac{1}{\varepsilon\alpha}, \sqrt{T}\right\}\right).$$

Proof of Theorem 3.3. Now, we aim to show how the distribution parameters ε and α will influence the lower bound. The intuition why these parameters constitute the difficulty in learning is that: when ε or α is small, the potential demand distribution can be flat around the optimal quantity. The construction of hard instances follow the idea.

Step 1. Instance Construction. We construct the following two distribution instances with CDF F_1 and F_2 as follows:

$$F_1(q) = \begin{cases} \dfrac{\rho}{\alpha} \cdot q, & \text{for } 0 \leq q \leq \alpha, \\ \rho + \varepsilon \cdot (q - \alpha), & \text{for } \alpha < q \leq 2\alpha, \\ \rho + \alpha\varepsilon, & \text{for } 2\alpha < q \leq 2\alpha + 1, \\ \rho + \varepsilon \cdot (q - \alpha - 1), & \text{for } 2\alpha + 1 < q \leq 3\alpha + 1, \\ \rho + 2\alpha\varepsilon + \dfrac{1 - \rho - 2\alpha\varepsilon}{\alpha} \cdot (q - 3\alpha - 1), & \text{for } 3\alpha + 1 < q \leq 4\alpha + 1. \end{cases}$$

and

$$F_2(q) = \begin{cases} \left(\dfrac{\rho}{\alpha} - 2\varepsilon\right) \cdot q, & \text{for } 0 \leq q \leq \alpha, \\ \rho - 2\alpha\varepsilon + \varepsilon \cdot (q - \alpha), & \text{for } \alpha < q \leq 2\alpha, \\ \rho - \alpha\varepsilon, & \text{for } 2\alpha < q \leq 2\alpha + 1, \\ \rho + \varepsilon \cdot (q - 3\alpha - 1), & \text{for } 2\alpha + 1 < q \leq 3\alpha + 1, \\ \rho + \dfrac{1 - \rho}{\alpha} \cdot (q - 3\alpha - 1), & \text{for } 3\alpha + 1 < q \leq 4\alpha + 1. \end{cases}$$

It is easy to verify that both $\mathbb{P}_1, \mathbb{P}_2 \in \mathcal{P}_c(\varepsilon, \alpha)$. We also have $q^\star_{F_1} = \alpha$ and $q^\star_{F_2} = 3\alpha + 1$.

Step 2. Separation Verification. We have:

$$C(q; \mathbb{P}_1) - C(q^\star_{F_1}; \mathbb{P}_1) \geq \frac{(b+h)\alpha\varepsilon}{2}, \ \forall q > 2\alpha + 1/2, \tag{8}$$

$$C(q; \mathbb{P}_2) - C(q^\star_{F_2}; \mathbb{P}_2) \geq \frac{(b+h)\alpha\varepsilon}{2}, \ \forall q \leq 2\alpha + 1/2, \tag{9}$$

under our construction, where we leave the detailed calculations in Appendix B as well. Thus, the separation condition holds:

$$\inf_q \{C(q; \mathbb{P}_1) + C(q; \mathbb{P}_2)\} - C(q^\star_{F_1}; \mathbb{P}_1) - C(q^\star_{F_2}; \mathbb{P}_2) \geq \frac{(b+h)\alpha\varepsilon}{2}. \tag{10}$$

Step 3. Applying Le Cam. Following the routine in deriving (7), we have for any policy $\pi \in \Pi$,

$$\mathcal{R}(T; \pi, \mathcal{P}_c(\varepsilon, \alpha)) \geq \frac{(b+h)\alpha\varepsilon}{4} \cdot \sum_{t=1}^{T} \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{t-1} \| \mathbb{P}_2^{t-1})). \tag{11}$$

Step 4. Calculating KL between Environments. At last, we calculate the KL-divergence in (11):

$$\begin{aligned}
&\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{t-1} \| \mathbb{P}_2^{t-1}) = (t-1) \int_0^{4\alpha+1} f_1(q) \log \frac{f_1(q)}{f_2(q)} \mathrm{d}q \\
&= (t-1) \left[\rho \log \left(1 + \frac{2\alpha\varepsilon}{\rho - 2\alpha\varepsilon} \right) + (1 - \rho - 2\alpha\varepsilon) \log \left(1 - \frac{2\alpha\varepsilon}{1-\rho} \right) \right] \\
&\overset{(i)}{\leq} 4(t-1) \cdot (\alpha\varepsilon)^2 \left(\frac{1}{\rho - 2\alpha\varepsilon} + \frac{1}{1-\rho} \right) \\
&\overset{(ii)}{\leq} 4(t-1) \cdot (\alpha\varepsilon)^2 \left(\frac{3}{\rho} + \frac{1}{1-\rho} \right)
\end{aligned}$$

with (i) follows from $\log(1+x) \leq x$ for any $x > -1$; (ii) follows from the assumption that $\alpha\varepsilon \leq \rho/3$. Let $c_1 \triangleq 4 \cdot \left(\frac{3}{\rho} + \frac{1}{1-\rho} \right)$ and taking the KL result back to (11):

$$\begin{aligned}
\mathcal{R}(T; \pi, \mathcal{P}_c(\varepsilon, \alpha)) &\geq \frac{(b+h)\alpha\varepsilon}{4} \cdot \sum_{t=1}^{\min\{\lfloor \alpha\varepsilon \rfloor^{-2}, T\}} \exp(-c_1) \\
&\geq \frac{\exp(-c_1)}{4} \cdot (b+h) \min\{(\alpha\varepsilon)^{-1}, (\alpha\varepsilon) \cdot T\}.
\end{aligned}$$

For $T > (\alpha\varepsilon)^{-1}$, we have the lower bound $\Omega((b+h)\sqrt{T})$; otherwise, we derive $\Omega((b+h) \cdot (\alpha\varepsilon)^{-1})$. □

Note that for $\mathbb{P}$ in distribution class $\mathcal{P}_c(\varepsilon, \alpha)$, the regret lower bound does not explicitly depends on the demand upper bound M, and it is the local flatness parameters ε, α that truly matters.

We also emphasize that the $\Omega\left(\frac{1}{\alpha\varepsilon}\right)$ lower bound derived here is not tight in general. In fact, a recent work by Lyu et al. (2024) shows that the minimax regret under such locally flat models is lower bounded by $\Omega\left(\frac{\log T}{\varepsilon} + \frac{1}{\alpha\varepsilon}\right)$. This suggests that while our construction accurately captures the dependence on the flatness parameters and is tight on $\alpha\varepsilon$, it omits the $\Omega\left(\log T/\varepsilon\right)$ term that accounts for the cumulative complexity of learning over a horizon of T rounds. We will return to this point in the Bibliographic Remarks Section 3.3 at the end of this section, where we further discuss the tight bound and related developments.

3.2 Offline Inventory Policy Learning: Sample Complexity

In the offline inventory control setting, decision-makers rely on historical data to inform their inventory decisions, without the opportunity to adapt based on real-time demand observations. This scenario is particularly relevant in contexts such as seasonal retail planning, product launches, or any situation where decisions can be made based on past data. The central challenge lies in understanding how much historical data is necessary to learn effective inventory policies—a concept formalized through sample complexity analysis.

This section delves into the sample complexity of offline inventory policy learning, focusing on the newsvendor problem and its extensions. We explore how the amount and quality of historical data impact the ability to learn near-optimal policies, considering both fully observed and censored demand scenarios. Additionally, we examine complex inventory systems, such as serial systems, to understand the broader implications of sample complexity in offline settings.

Application 4: Newsvendor under Fully Observable Data

Consider the offline newsvendor problem: the decision-maker have access to historical demand dataset $S_N \triangleq \{d_i\}_{i=1}^N$ which is i.i.d. sampled from a demand distribution $\mathbb{P}$. The objective is to learn a policy $\pi : S_N \to \mathbb{R}^+$ that maps from the history data to the inventory order decision q. The expected inventory cost is still denoted as $C(q; \mathbb{P}) = h \cdot \mathbb{E}_{\mathbb{P}}[q - D]^+ + b \cdot \mathbb{E}_{\mathbb{P}}[D - q]^+$, and the optimal order quantity is again $q^\star = \{q : \inf_q F(q) \geq \rho\}$. Assume that with probability at least $1 - \delta$, the data-driven policy π returns a $(1 + \epsilon)$-optimal solution for any $\delta, \epsilon > 0$, i.e.,

$$\mathbb{P}_{S_N}\left(C(\pi; \mathbb{P}) \leq (1 + \epsilon) C(q^*; \mathbb{P})\right) \geq 1 - \delta$$

holds, for any latent demand distribution $\mathbb{P}$. The following Theorem provide least number of samples N it requires for the inequality to hold, which is originated from the theorem 3 of Cheung and Simchi-Levi (2019).

Theorem 3.4. *Given any historical dataset S_N, let π be the $(1+\epsilon)$-optimal solution w.p. at least $1-\delta$ under any latent demand distribution, where $\epsilon \in (0, \frac{1}{20})$ and $\delta \in (0, \frac{1}{4})$. Then π requires at least $N = \Omega\left(\frac{(h+b)\log\frac{1}{4\delta}}{\min\{h,b\}\epsilon^2}\right)$ samples.*

Proof of Theorem 3.4. We start with the case where $h \le b$.

Step 1. Instance Construction. We construct the following two distributions instances with CDF F_1 and F_2 for $\epsilon \in (0, \frac{1}{20})$:

$$F_1(q) = \begin{cases} \frac{b-h}{b+h} & \text{for } q = 0, \\ \frac{b+h\epsilon}{b+h} & \text{for } q = \frac{M}{2}, \\ 1 & \text{for } q = M, \end{cases} \quad F_2(q) = \begin{cases} \frac{b-h}{b+h} & \text{for } q = 0, \\ \frac{b-h\epsilon}{b+h} & \text{for } q = \frac{M}{2}, \\ 1 & \text{for } q = M. \end{cases} \tag{12}$$

By its construction, we have $q^\star_{F_1} = \frac{M}{2}$ and $q^\star_{F_2} = M$.

Step 2. Separation Verification. Differs from the previous examples in which the performance metric for evaluating a policy is the absolute difference in cost, here the performance is measured in terms of relative (multiplicative) difference. As a result, the separation condition used in previous examples no longer directly applies. In particular, a constant absolute gap in cost does not imply a constant relative suboptimality.

Nonetheless, inspired by the same core idea from Le Cam's method, we can verify the separation condition (for relative difference), which we formulate as the following Claim 1:

Claim 1. The $(1+\frac{\epsilon}{20})$-optimal solution sets under $\mathbb{P}_1$ and $\mathbb{P}_2$ are disjoint.

Notice that Claim 1 is in spirit the parallel of (4) and (5), and leave the calculation details in the Appendix B.

Step 3. Applying Information-Theoretic Tool. Thus, for any $(1+\epsilon)$-optimal policy π under any demand distribution w.p. $1-\delta$, define events

$$\mathcal{A}_1 = \left\{\pi \text{ outputs a } \left(1+\frac{\epsilon}{20}\right)\text{-optimal policy for } \mathbb{P}_1\right\},$$
$$\mathcal{A}_2 = \left\{\pi \text{ outputs a } \left(1+\frac{\epsilon}{20}\right)\text{-optimal policy for } \mathbb{P}_2\right\}.$$

According to the definition of π, $\mathbb{P}_1^N(\mathcal{A}_1) \ge 1-\delta$ and $\mathbb{P}_2^N(\mathcal{A}_2) \ge 1-\delta$, where $\mathbb{P}_i^N$ denotes the distribution of observation S_N under distribution instance $\mathbb{P}_i$ for $i \in \{1,2\}$. Further, by the separation condition in Claim 1, $\mathcal{A}_1 \subset \mathcal{A}_2^c$. These together imply

$$\mathbb{P}_1^N(\mathcal{A}_1^c) + \mathbb{P}_2^N(\mathcal{A}_1) \le \mathbb{P}_1^N(\mathcal{A}_1^c) + \mathbb{P}_2^N(\mathcal{A}_2^c) \le 2\delta.$$

By applying the Bretagnolle-Huber inequality:

$$2\delta \ge \mathbb{P}_1^N(\mathcal{A}_1^c) + \mathbb{P}_2^N(\mathcal{A}_1) \ge \frac{1}{2}\exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^N \| \mathbb{P}_2^N)). \tag{13}$$

Step 4. Calculating KL between Environments. At last, we calculate the KL-divergence in (14). The following Claim 2 directly follows the definition of KL and we leave the details in the appendix.

Claim 2. $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^N \| \mathbb{P}_2^N) \leq \dfrac{8Nh\epsilon^2}{b+h}$.

Combining the results together, we get $N \geq \frac{b+h}{8h\epsilon^2} \log \frac{1}{4\delta}$. Similarly, for $b \leq h$, we adopt the same analysis and derive $N \geq \frac{b+h}{8b\epsilon^2} \log \frac{1}{4\delta}$. Thus it requires $N = \Omega\left(\frac{(b+h)\log \frac{1}{4\delta}}{\min\{h,b\}\epsilon^2}\right)$ for any $(1+\epsilon)$-optimal policy π. □

Application 5: Newsvendor under Censored Data

Consider the offline newsvendor problem with censored demand: the decision-maker has access to a historical dataset $\tilde{S}_N \triangleq \{(q_i, \tilde{d}_i)\}_{i=1}^N$, where $q_i \in \mathbb{R}^+$ denotes the order quantity in the i-th observation, and $\tilde{d}_i = \min\{q_i, d_i\}$ is the observed (censored) sales, with $d_i \sim \mathbb{P}$ drawn i.i.d. from an unknown demand distribution. Specifically, we assume that q_i is also drawn i.i.d. from an unknown distribution $\mathbb{P}_v^\kappa$ that satisfies $\mathbb{P}_v^\kappa(q > q^*) \geq \kappa$ for some constant $\kappa > 0$.

The objective is again to learn a policy π that maps the historical censored dataset to an order decision, aiming to minimize the newsvendor cost with respect to the unknown true demand distribution. We want to further answer the question: how many censored samples are required to guarantee that the learned decision is near-optimal? The following Theorem restates theorem 1 of Fan et al. (2022).

Theorem 3.5. *Given any censored historical dataset $\tilde{S}_N$ sampled accordingly following demand distribution $\mathbb{P}$ and decision policy $\mathbb{P}_v^\kappa$, let π be the $(1+\epsilon)$-optimal policy w.p. at least $1-\delta$ under any latent demand distribution, where $\epsilon \in (0, \frac{1}{20})$ and $\delta \in (0, \frac{1}{4})$. Then π requires at least $N = \Omega\left(\frac{(b+h)\log \frac{1}{4\delta}}{\kappa \min\{h,b\}\epsilon^2}\right)$ samples.*

Proof of Theorem 3.5. The only difference compared with the observable data setting is that now we need to construct a decision policy distribution $\mathbb{P}_v^\kappa$ in addition, to make the distribution instances even harder to distinguish.

Step 1. Instance Construction. We apply the same construction of demand distributions $\mathbb{P}_1$ and $\mathbb{P}_2$ as that in (12). We further construct the decision policy distribution $\mathbb{P}_v^\kappa$ as:

$$\mathbb{P}_v^\kappa(q=0) = 1-\kappa,\ \mathbb{P}_v^\kappa(q=M+1) = \kappa.$$

Denote $\tilde{D} \triangleq \min\{D, q\} \sim \tilde{\mathbb{P}}$ as the observed sales quantity and $\tilde{F}_1$, $\tilde{F}_2$ as the corresponding

CDFs under two instances. By the construction, we have

$$F_1(q) = \begin{cases} 1-\kappa+\kappa\frac{b-h}{b+h} & \text{for } q=0, \\ 1-\kappa+\kappa\frac{b+h\epsilon}{b+h} & \text{for } q=\frac{M}{2}, \\ 1 & \text{for } q=M, \end{cases} \quad F_2(q) = \begin{cases} 1-\kappa+\kappa\frac{b-h}{b+h} & \text{for } q=0, \\ 1-\kappa+\kappa\frac{b-h\epsilon}{b+h} & \text{for } q=\frac{M}{2}, \\ 1 & \text{for } q=M. \end{cases}$$

Step 2. Separation Verification. As the instances $\mathbb{P}_1$ and $\mathbb{P}_2$ are exactly the same as before, we still have the Claim 1 holds (which does not involve observation issues):
Claim 1. The $(1+\frac{\epsilon}{20})$-optimal solution sets under $\mathbb{P}_1$ and $\mathbb{P}_2$ are disjoint.

Step 3. Applying Information-Theoretic Tool. The definition of $\mathcal{A}_1$ and $\mathcal{A}_2$ are the same, except that now it will be equipped with the measure $\tilde{\mathbb{P}}_1^N$ and $\tilde{\mathbb{P}}_2^N$ so that $\tilde{\mathbb{P}}_1^N(\mathcal{A}_1) \geq 1-\delta$ and $\tilde{\mathbb{P}}_2^N(\mathcal{A}_2) \geq 1-\delta$ as the policy π are learned based on the censored historical data $\tilde{S}_N$. Now we conclude:

$$2\delta \geq \tilde{\mathbb{P}}_1^N(\mathcal{A}_1^c) + \tilde{\mathbb{P}}_2^N(\mathcal{A}_1) \geq \frac{1}{2}\exp(-\mathrm{D}_{\mathrm{KL}}(\tilde{\mathbb{P}}_1^N \| \tilde{\mathbb{P}}_2^N)). \tag{14}$$

Step 4. Calculating KL between Environments. The upper bound of $\mathrm{D}_{\mathrm{KL}}(\tilde{\mathbb{P}}_1^N \| \tilde{\mathbb{P}}_2^N))$ follows a similar routine as we have done in Claim 2, and we leave the details in the appendix. We claim that:

Claim 2′. $\mathrm{D}_{\mathrm{KL}}(\tilde{\mathbb{P}}_1^N \| \tilde{\mathbb{P}}_2^N) \leq \dfrac{8Nh\kappa\epsilon^2}{b+h}$.

Thus, we conclude $N = \Omega\left(\frac{(b+h)\log\frac{1}{4\delta}}{\kappa\min\{h,b\}\epsilon^2}\right)$ for any $(1+\epsilon)$-optimal policy π. □

According to the lower bound, we conclude that the fewer samples that contain uncensored information beyond $q^\star$, the more total samples are needed to learn a good decision. In the extreme case $\kappa \to 0$, the lower bound diverges, indicating that no finite sample size suffices for near-optimal learning.

3.3 Summary and Bibliographic Remarks

We conclude this section by summarizing the regret lower bounds derived in various learning settings for the newsvendor problem. The following table collects the main results:

In addition to the canonical settings in Table 3, several important extensions and generalizations have been explored in the literature.

- For the online inventory control: another widely-considered demand distribution class in inventory literature is the global-separated distribution class $\mathcal{P}_c(\varepsilon)$, where $f(q) \geq \varepsilon > 0$ holds for all $q \in [0, M]$. Note that $\mathcal{P}_c(\varepsilon) \subseteq \mathcal{P}_c(\varepsilon, \alpha)$ and now the newsvendor cost $C(q; \mathbb{P})$ is strongly convex. Based on the property, Besbes and Muharremoglu (2013) propose to use the *van Trees inequality* to establish regret

Table 3: Regret Lower Bounds in Inventory Control

	Setting & Distribution Class	**Lower Bound**
App.1	Online (Full Obs.); Discrete	$\Omega((b+h)M\sqrt{T})$
App.2	Online (Full Obs.); Continuous	$\Omega((b+h)M\sqrt{T})$
App.3	Online (Full Obs.); Local Separation	$\Omega\left((b+h)\min\left\{\frac{1}{\varepsilon\alpha}, \sqrt{T}\right\}\right)$
App.4	Offline (Fully Obs.); Any	$\Omega\left(\frac{(h+b)\log\frac{1}{4\delta}}{\min\{h,b\}\epsilon^2}\right)$
App.5	Offline (Censored Data); Any	$\Omega\left(\frac{(h+b)\log\frac{1}{4\delta}}{\kappa\min\{h,b\}\epsilon^2}\right)$

lower bounds. Unlike Le Cam's method, which reduces the problem to a binary hypothesis test between two close distributions, the van Trees approach (Gill and Levit, 1995) treats learning as a parametric estimation problem and lower bounds the regret via a Bayesian Cramér–Rao bound. Lyu et al. (2024) later point out a subtle issue in the original construction of Besbes and Muharremoglu (2013), and provide a corrected family of distributions that satisfy the required density condition. Under this corrected setup, the regret is shown to be $\Omega\left(\frac{1}{\varepsilon}\log T\right)$. Since our survey focuses on Le Cam's method and its related tools, we omit the technical details of this approach.

Another related result is the regret lower bound under censored observations in the online setting. Besbes and Muharremoglu (2013) show that under the globally separated class $\mathcal{P}_c(\varepsilon)$, the regret is (still) $\Omega(\log T)$. We do not include this example in our survey because the proof, while still using a two-point construction, is not standard and does not follow the classic Le Cam template on which we focus.

Finally, although the repeated online newsvendor setting is simple, its lower bound can serve as a fundamental benchmark for many more complex inventory systems. In fact, it often turns out to be tight in these richer settings, e.g., for the inventory systems with lost-sales, general lead time, perishable items (Zhang et al., 2018; Agrawal and Jia, 2019; Zhang et al., 2020); and a comprehensive tutorial on inventory management in Chen et al. (2022b) for detailed discussions.

- For the offline inventory control: Zhang et al. (2025) study the sample complexity lower bound for a two-stage serial inventory system. They follow the routine discussed in this section, with a more involved analysis in deriving the separation condition (due to the complexity nature of cost function) and also derive a lower bound of order $\Omega(\frac{1}{\epsilon^2})$. Besides the relative difference metric discussed here, literature also studies the absolute difference counterpart following the standard Le Cam's routine in this section (Qin et al., 2023).

  However, there still exists some open questions for the offline sample complexity lower bound: the current result holds for any distribution classes, which may be

conservative in some restricted distribution classes. For example, Levi et al. (2015) and Zhang et al. (2025) observe that the Sample Average Approximation (SAA) method can achieve sample complexity upper bound of order $O(\frac{1}{\epsilon})$, while a matched distribution-dependent lower bound remains to be constructed.

4 Revenue Management: Auction, Pricing, and Assortment

Revenue management problems lie at the heart of operations research, economics, and algorithmic decision-making. Among the most fundamental of these problems, we discuss the **auction**, **pricing**, and **assortment** optimization, each of which plays a central role in real-world markets. Though these problems differ in formulation and application, they share a common structure: a decision-maker must make revenue-maximizing choices under uncertainty about preferences or demand, typically in a data-driven or online setting.

From a methodological perspective, these problems are unified by the tools used to analyze and design learning algorithms. In particular, Le Cam's two-point method has emerged as a powerful tool to characterize regret and sample complexity lower bounds across these revenue management domains. The essential idea is to construct pairs of difficult-to-distinguish environments that lead to optimal pricing, auction formats, or product assortments, and use information-theoretic inequalities to derive fundamental limits.

This section surveys lower bounds for auction, pricing and assortment learning problems, highlighting how Le Cam's method applies across these settings. While the specifics differ—from continuous versus discrete choices, or single-agent versus market settings—the core ideas around indistinguishability, estimation versus decision-making, and the role of information constraints are shared. We begin in Section 4.1 with the canonical problem of repeated first-price auction, then in Section 4.2, we discuss the dynamic pricing problem. We turn to the assortment optimization in Section 4.3. Before proceeding, we summarize the notations tailored for Section 4 below, with its detailed definitions introduced later.

Table 4: Notation Used throughout Section 4

Symbol	**Description**
b_t	Learner's bid at time t
m_t	Highest other bid (HOB) at time t
$\mathcal{P}, \mathcal{Z}$	Price space and parameter space
$p_t \in \mathcal{P}$	Price at time t
$\mathbf{y}_t \in \{0,1\}^t$	History of customer purchasing decisions
$\mathbf{z} \in \mathcal{Z}$	Demand parameter
$d(p;\mathbf{z})$	Demand function under price p and parameter $\mathbf{z}$
$r(p;\mathbf{z})$	Expected revenue under price p and parameter $\mathbf{z}$
$p^\star(\mathbf{z})$	Optimal price under demand parameter $\mathbf{z}$
$\mathcal{L} = (\mathcal{P}, \mathcal{Z}, d)$	Demand function class
$S_t \in [N]$	Assortment set at time t
$i_t \in S_t \cup \{0\}$	Customer's choice at time t
v_i	Preference parameter for item i
$S^\star$	Optimal assortment in hindsight

4.1 Online Auction

In this section, we consider the problem of learning to bid in repeated online auction, a fundamental model of strategic interaction under uncertainty. Unlike pricing or assortment problems where the seller controls the mechanism and observes customer responses, here the learner acts as a participant in a competitive environment, submitting bids against unknown opponents. This setting is particularly relevant in digital advertising, real-time markets, and procurement, where auction outcomes depend not only on one's own decision but also on stochastic competitor behavior.

We study a setting where the highest other bid (HOB) is drawn i.i.d. from an unknown distribution and is fully observed after each auction, regardless of win or loss. This full-feedback assumption allows the learner to update their beliefs about the environment after each round and distinguishes our formulation from more restricted feedback models. Nevertheless, the core difficulty remains: the optimal bid depends on the unknown distribution of competition, and any deviation from the optimal bidding policy incurs regret.

We begin our analysis with the repeated first-price auction under unknown HOB distributions, and show how Le Cam's two-point method can be applied to establish regret lower bounds in this setting. The construction illustrates how distributional indistinguishability and bid-dependent rewards jointly limit the learner's ability to achieve optimal performance.

Application 6: Repeated First-Price Auction

First-price auctions represent a modern and strategically rich setting within revenue management. Like dynamic pricing, they offer a mechanism to extract value under uncertain demand conditions, but with a key structural shift: instead of setting a posted price and observing buyer responses, the learner now takes the role of a bidder, strategically submitting prices (bids) in response to unknown competition.

At each round $t \in [T]$, the learner observes a private valuation $v_t \in [0, 1]$, and selects a bid $b_t \in [0, 1]$. The auction outcome depends on the HOB $m_t \sim \mathbb{P}$, drawn i.i.d. from an unknown distribution $\mathbb{P} \in \mathcal{P}$ for distribution class $\mathcal{P}$ with CDF F. The learner receives the good and pays b_t if $b_t \geq m_t$, and nothing otherwise. Crucially, the learner only observes whether they won, and if they lose, they may observe the value of m_t; otherwise, the competing bid is censored. For simplicity, in this survey we will only discuss the case where m_t can be fully observed. The expected single-period reward for bidding b_t given value v_t at period t is

$$r(v_t, b_t) = (v_t - b_t) \cdot F(b_t),$$

with benchmark to be the clairvoyant policy $b^\star(v_t) \triangleq \arg\max_b r(v_t, b_t)$ with full knowledge of $\mathbb{P}$. The learner aims to minimize the cumulative regret:

$$\mathcal{R}(T; \pi, v, \mathcal{P}) \triangleq \mathbb{E}\left[\sum_{t=1}^{T} \left(r\left(v_t, b^\star\left(v_t\right)\right) - r\left(v_t, b_t\right)\right)\right],$$

where $\pi = (b_1, \cdots, b_T)$ is the learner's bidding policy, and the expectation is taken w.r.t. the randomness over private values, policy, and the induced bidding outcomes. The following theorem restates that of theorem 9 in Han et al. (2025).

Theorem 4.1. *Consider a special case where $v_t \equiv 1$ and the HOBs m_t are always revealed at the end of each round, then for any bidding price policy π,*

$$\sup_{\mathbb{P}\in\mathcal{P}} \mathcal{R}(T;\pi,v,\mathcal{P}) = \Omega(\sqrt{T}),$$

where the supremum is taken over all possible $\mathbb{P}(\cdot)$ of the HOB m_t.

Proof of Theorem 4.1. We have the distribution construction of m_t very similar in spirits to that of Application 1:

Step 1. Instance Construction. We first construct two instances $\mathbb{P}_1$ and $\mathbb{P}_2$ with CDF F_1 and F_2 supported on $[0,1]$ respectively:

$$F_1(y) = \begin{cases} 0 & \text{if } 0 \le y < \frac{1}{3} \\ \frac{1}{2}+\Delta & \text{if } \frac{1}{3} \le y < \frac{2}{3} \\ 1 & \text{if } y \ge \frac{2}{3} \end{cases}, \qquad F_2(y) = \begin{cases} 0 & \text{if } 0 \le y < \frac{1}{3} \\ \frac{1}{2}-\Delta & \text{if } \frac{1}{3} \le y < \frac{2}{3} \\ 1 & \text{if } y \ge \frac{2}{3} \end{cases},$$

for some small $\Delta > 0$ that will be determined later.

Step 2. Separation Verification. Let $r_1(v_t, b_t)$ and $r_2(v_t, b_t)$ be the expected reward averaged under CDF F_1 and F_2, respectively. It is straightforward to check that

$$\begin{aligned} \max_{b\in[0,1]} r_1(v_t,b) &= \max_{b\in[0,1]} (1-b)F_1(b) = \frac{1+2\Delta}{3}, \\ \max_{b\in[0,1]} r_2(v_t,b) &= \max_{b\in[0,1]} (1-b)F_2(b) = \frac{1}{3}, \\ \max_{b\in[0,1]} (r_1(v_t,b)+r_2(v_t,b)) &= \max_{b\in[0,1]} (1-b)(F_1(b)+F_2(b)) = \frac{2}{3}. \end{aligned}$$

Hence, for any $b_t \in [0,1]$, we have the separation condition holds:

$$\max_{b\in[0,1]} r_1(v_t,b) + \max_{b\in[0,1]} r_2(v_t,b) - r_1(v_t,b_t) - r_2(v_t,b_t) \ge \frac{2\Delta}{3}.$$

Step 3. Applying Le Cam. Then, for any policy π,

$$\begin{aligned} \sup_{\mathbb{P}\in\mathcal{P}} \mathcal{R}(T;\pi,\mathcal{P}) &\ge \frac{\Delta}{3} \sum_{t=1}^{T} \left(1 - \|\mathbb{P}_1^{t-1} - \mathbb{P}_2^{t-1}\|_{\mathrm{TV}}\right) \\ &\ge \frac{\Delta}{6} \sum_{t=1}^{T} \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{t-1} \| \mathbb{P}_2^{t-1})) \\ &\ge \frac{T\Delta}{6} \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{T} \| \mathbb{P}_2^{T})). \end{aligned}$$

Step 4. Calculating KL between Environments. Due to the similar construction of CDFs F_1 and F_2 to that in Application 1, we conclude its KL-divergence $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^T \| \mathbb{P}_2^T) \leq 32T\Delta^2$. Then, by choosing $\Delta = 1/\sqrt{T}$, we derive the lower bound. □

Application 7: Joint Bidding and Estimation under First-Price Auction

Now, we consider a more involved repeated first-price auction setting in which the learner's observed private value v also depends on the bidding outcome. At each round $t \in [T]$, the learner first selects a bid $b_t \in [0, 1]$, and competes against the HOB $m_t \sim \mathbb{P}$, drawn i.i.d. from the unknown distribution $\mathbb{P}$. The learner then observe one of the two random variables of private valuation:

- $v_{t,1} \in [0, 1]$: representing the private value she receives if she wins the auction,
- $v_{t,0} \in [0, 1]$: representing the private value she receives if she loses the auction,

with $v_{t,1} > v_{t,0}$ holds w.p.1. and that $v_{t,1}$, $v_{t,0}$ and m_t are independently distributed. The expected single-period reward for bidding b_t given value $\boldsymbol{v}_t = (v_{t,0}, v_{t,1})$ at period t is now given by:

$$r_t(\boldsymbol{v}_t, b_t) = (v_{t,1} - v_{t,0} - b_t) \cdot F(b_t) + v_{t,0},$$

and we denote the clairvoyant policy that bids $b^\star \triangleq \arg\max_b \mathbb{E}[r(\boldsymbol{v}_t, b_t)]$ with full knowledge of $\mathbb{P}$ and the distribution of $\boldsymbol{v}_t$ as the benchmark (now we have expectation taken w.r.t. the joint distribution of $\boldsymbol{v}_t$). The learner aims to minimize the cumulative regret:

$$\mathcal{R}(T; \pi, v, \mathcal{P}) \triangleq \mathbb{E}\left[\sum_{t=1}^{T} \left(r\left(\boldsymbol{v}_t, b^\star\left(\boldsymbol{v}_t\right)\right) - r\left(\boldsymbol{v}_t, b_t\right)\right)\right].$$

We note that, in contrast to the previous classic repeated first-price auction setting on the private value, where $v_{t,0} \equiv 0$ and $v_{t,1}$ can be observed before bidding (thus we treat it as a time-invariant constant in the lower bound construction), now the learner has to jointly learn $(v_{t,0}, v_{t,1})$ and the distribution of m_t in order to determine a good policy. Thus, we also need to incorporate the randomness of $\boldsymbol{v}_t$ in our lower bound construction. The following Theorem restates that of Theorem B.1 in Wen et al. (2025).

Theorem 4.2. *Consider a special instance with i.i.d.* $v_{t,1} \sim \mathrm{Bern}(\mu)$ *for an unknown* $\mu \in (0, 1)$, $v_{t,0} \equiv 0$, *and the HOBs* m_t *are always revealed at the end of each round, then for any bidding price policy* π,

$$\sup_{\mathbb{P} \in \mathcal{P}, \mu \in (0,1)} \mathcal{R}(T; \pi, v, \mathcal{P}) = \Omega(\sqrt{T}),$$

where the supremum is taken over all possible $\mathbb{P}(\cdot)$ *of the HOB* m_t *and all Bernoulli distribution parameter* $\mu \in (0, 1)$.

Proof of Theorem 4.2.

Step 1. Instance Construction. We let the distribution of HOB m_t follows:

$$m_t \sim \frac{1}{2}\,\mathrm{Unif}\left(\left[0, \frac{1}{100}\right]\right) + \frac{1}{2}\,\mathrm{Unif}\left(\left[\frac{1}{8} - \frac{1}{200}, \frac{1}{8} + \frac{1}{200}\right]\right),$$

with its CDF denoted as F. We further construct two instances on the distribution of $v_{t,1}$: in Instance 1, we set $\mu_1 = \frac{1}{4} - \Delta$; and in Instance 2, we set $\mu_2 = \frac{1}{4} + \Delta$, for some small $\Delta > 0$ that will be determined later.

Step 2. Separation Verification. We use $\mathbb{E}_1[r(v_t, b_t)]$ and $\mathbb{E}_2[r(v_t, b_t)]$ to denote the expected single-period reward taken w.r.t. $v_{t,1} \sim \mathrm{Bern}(\mu_1)$ and $v_{t,1} \sim \mathrm{Bern}(\mu_2)$, respectively. It is straightforward to check that $\mathbb{E}_i[r(\boldsymbol{v}_t, b_t)] = (\mu_i - b)F(b)$ for $i \in \{1, 2\}$, and we have for any $t \in [T]$:

$$\begin{aligned}
&\max_{b\in[0,1]} \mathbb{E}_1\left[r(\boldsymbol{v}_t, b)\right] = \max_{b\in[0,1]} (\mu_1 - b)F(b) = \frac{1}{8} - \frac{1}{200} - \frac{\Delta}{2},\\
&\max_{b\in[0,1]} \mathbb{E}_2\left[r(\boldsymbol{v}_t, b)\right] = \max_{b\in[0,1]} (\mu_2 - b)F(b) = \frac{1}{8} - \frac{1}{200} + \Delta,\\
&\max_{b\in[0,1]} \mathbb{E}_1\left[r(\boldsymbol{v}_t, b)\right] + \mathbb{E}_2\left[r(\boldsymbol{v}_t, b)\right] = \max_{b\in[0,1]} 2 \cdot \left(\frac{1}{4} - b\right) F(b) = \frac{1}{4} - \frac{1}{100}.
\end{aligned}$$

Hence, for any $b_t \in [0, 1]$, we have the separation condition holds:

$$\max_{b\in[0,1]} \mathbb{E}_1\left[r(\boldsymbol{v}_t, b)\right] + \max_{b\in[0,1]} \mathbb{E}_2\left[r(\boldsymbol{v}_t, b)\right] - \mathbb{E}_1\left[r(\boldsymbol{v}_t, b_t)\right] - \mathbb{E}_2\left[r(\boldsymbol{v}_t, b_t)\right] \geq \frac{\Delta}{2}.$$

Step 3. Applying Le Cam. We can thus directly apply the Le Cam's Two-Point method and derive that for any policy π:

$$\begin{aligned}
\sup_{\mathbb{P}\in\mathcal{P}, \mu\in(0,1)} \mathcal{R}(T; \pi, v, \mathcal{P}) &\geq \frac{\Delta}{4} \sum_{t=1}^{T} \left(1 - \|\mathbb{P}_1^{t-1} - \mathbb{P}_2^{t-1}\|_{\mathrm{TV}}\right)\\
&\geq \frac{\Delta}{8} \sum_{t=1}^{T} \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{t-1} \| \mathbb{P}_2^{t-1}))\\
&\geq \frac{T\Delta}{8} \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^{T} \| \mathbb{P}_2^{T})),
\end{aligned}$$

where now we use $\mathbb{P}_1^T$ and $\mathbb{P}_2^T$ to denote the joint distribution of $\{\boldsymbol{v}_t, m_t\}_{t=1}^T$ under the two instances.

Step 4. Calculating KL between Environments. Note that as the distributions of m_t are identical among the two instances, and that $v_{t,0}$ is deterministic, we thus have:

$$\mathrm{D_{KL}}(\mathbb{P}_1^T \| \mathbb{P}_2^T) \leq T \mathrm{D_{KL}}(\mathrm{Bern}(\mu_1) \| \mathrm{Bern}(\mu_2)) \leq \frac{64}{3} T \Delta^2.$$

Then, by choosing $\Delta = \frac{1}{\sqrt{T}}$, we derive the corresponding lower bound.

□

4.2 Dynamic Pricing

In this section, we delve into the dynamic pricing problem, where a seller sequentially posts prices to maximize revenue under uncertain demand. Dynamic pricing is a cornerstone of modern revenue management, used extensively in industries such as retail, airlines, and online platforms. The defining feature of this setting is that the seller actively controls the price path over time while learning from purchase outcomes, which are shaped by latent demand characteristics.

What distinguishes dynamic pricing from the previous auction applications is the direct link between the decision (price) and the consumer's binary response (purchase or not) which serves as the primary feedback for learning, and this is the fundamental difficulty compared with standard statistical estimation problems, which we have mentioned before in Section 1, that is, the decision will also affect the environments.

From a theoretical standpoint, this interplay poses deep challenges in both algorithm design and regret analysis. In particular, the regret lower bounds often depend on how the optimal price shifts with underlying demand parameters and how sensitive the revenue function is to pricing deviations. Le Cam's method proves effective in isolating this difficulty by constructing pairs of demand models that induce different optimal prices but generate nearly identical purchase data, thus limiting the learner's ability to adapt.

We begin our formal treatment with the following discrete-time pricing model under binary demand feedback and unknown demand functions, and examine its information-theoretic regret limits via Le Cam's method.

Application 8: Discrete-time Dynamic Pricing

We assume that customers arrive in discrete time steps. At each time $t = 1, \cdots, T$, the seller offers a price $p_t \in \mathcal{P} = [p_{\min}, p_{\max}]$, and observes a binary feedback $y_t \in \{0, 1\}$, indicating whether the customer purchases the product.

The buyer's purchase probability is governed by a demand function $d(p; \mathbf{z}) = \mathbb{P}_{\mathbf{z}}(v_t \geq p)$, where $\mathbf{z} \in \mathcal{Z} \subset \mathbb{R}^d$ is an unknown demand parameter and v_t represents the customer's utility. The expected revenue at price p is $r(p; \mathbf{z}) = p \cdot d(p; \mathbf{z})$, and the optimal price is defined as $p^\star(\mathbf{z}) \in \arg\max_{p \in \mathcal{P}} r(p; \mathbf{z})$.

For any $t \geq 1$, we denote by $\mathbf{y}_t = (y_1, \ldots, y_t) \in \{0, 1\}^t$ a history of customer purchasing decisions. A pricing policy $\pi = (p_1, \cdots, p_T)$ maps the past purchase history to a price

offered in each round, i.e., $p_t : \{0,1\}^{t-1} \to \mathcal{P}$. For any policy π and $\mathbf{z} \in \mathcal{Z}$, let $\mathbb{P}_{\mathbf{z}}^t : \{0,1\}^t \to [0,1]$ denote the probability distribution of the customer responses $\mathbf{y}_t$ under demand parameter $\mathbf{z}$ (and policy π); that is, for all $\mathbf{y}_t = (y_1, \ldots, y_t) \in \{0,1\}^t$,

$$\mathbb{P}_{\mathbf{z}}^t(\mathbf{y}_t) = \prod_{s=1}^{t} d(p_s; \mathbf{z})^{y_s} (1 - d(p_s; \mathbf{z}))^{1-y_s}.$$

For a given problem class $\mathcal{L} = (\mathcal{P}, \mathcal{Z}, d)$, and a demand parameter $\mathbf{z} \in \mathcal{Z}$, the cumulative regret of a pricing policy π is defined as:

$$\mathcal{R}(T; \pi, \mathbf{z}, \mathcal{L}) = \sum_{t=1}^{T} \mathbb{E}_{\mathbf{z}} \left[r\left(p^\star(\mathbf{z}); \mathbf{z}\right) - r\left(p_t; \mathbf{z}\right) \right],$$

which measures the cumulative loss in expected revenue relative to the optimal static price $p^\star(\mathbf{z})$. Our goal is to study the fundamental limits of learning in this model, by deriving minimax regret lower bounds over $\mathcal{L} = (\mathcal{P}, \mathcal{Z}, d)$. We impose the following regularity conditions on the model: (i) bounded demand: there exists some positive constants such that $0 < d_{\min} \leq d(p; \mathbf{z}) \leq d_{\max} < 1 \quad$ for all $p \in \mathcal{P}, \mathbf{z} \in \mathcal{Z}$; (ii) unique optimal price: the revenue function $r(p; \mathbf{z})$ has a unique maximizer $p^\star(\mathbf{z}) \in \mathcal{P}$ for each $\mathbf{z} \in \mathcal{Z}$; (iii) Lipschitz continuity of optimal price: $|p^\star(\mathbf{z}) - p^\star(\bar{\mathbf{z}})| \leqslant L\|\mathbf{z} - \bar{\mathbf{z}}\|$ for all $\mathbf{z}, \bar{\mathbf{z}} \in \mathcal{Z}$; and (iv) smoothness of the revenue function: the function $p \mapsto r(p; \mathbf{z})$ is twice differentiable. The following Theorem restates the theorem 3.1 of Broder and Rusmevichientong (2012).

Theorem 4.3. *Under the previous regularity assumptions, we can define a problem class $\mathcal{L}_{GenLB} = (\mathcal{P}, \mathcal{Z}, d)$ by letting $\mathcal{P} = [3/4, 5/4], \mathcal{Z} = [1/3, 1]$, and $d(p; z) = 1/2 + z - zp$ that satisfies these assumptions. Then for any policy π setting prices in $\mathcal{P}$, and any $T \geq 2$,*

$$\sup_{z \in \mathcal{Z}} \mathcal{R}(T; \pi, z, \mathcal{L}_{GenLB}) = \Omega(\sqrt{T}).$$

Proof of Theorem 4.3.
Step 1. Instance Construction. Here we construct two instances (demand functions) with demand parameters $z_1 = 1/2$ and $z_2 = z_1 + \Delta$ for some small $\Delta > 0$ that will be determined later. Note that $z_1 = 1/2$ is chosen so that the corresponding optimal price $p^*(z_1) = 1$ is uninformative: all demand curves in the class intersect at this price and yield identical demand, making it impossible for any policy to distinguish between z_1 and z_2 without pricing away from $p^*(z_1)$, and here we choose z_2 to be slightly deviate from z_1, so that the two instances are hard to distinguish.

Step 2. Separation Verification. We define two intervals $C_{z_1}, C_{z_2} \subset \mathcal{P}$ as follows:

$$C_{z_1} = \left\{ p : |p^\star(z_1) - p| \leq \frac{\Delta}{12} \right\}, \; C_{z_2} = \left\{ p : |p^\star(z_2) - p| \leq \frac{\Delta}{12} \right\}. \tag{15}$$

By the Mean Value Theorem, and the fact that $\frac{d}{dz}p^\star(z) = -\frac{1}{4z^2} \le -\frac{1}{4}$, we have

$$|p^\star(z_1) - p^\star(z_2)| \ge \frac{1}{4}|z_1 - z_2| = \frac{\Delta}{4}.$$

This implies that C_{z_1} and C_{z_2} are disjoint. For each $z \in \{z_1, z_2\}$, if $p \in \mathcal{P}\backslash C_z$, then by the fact that $r'(p^\star(z); z) = 0$ and $r''(p; z) = -2z \le -\frac{2}{3}$, the instantaneous regret is at least

$$r\left(p^\star(z); z\right) - r(p; z) \ge \frac{1}{3}\left(p - p^\star(z)\right)^2 \ge \frac{\Delta^2}{3(12)^2}. \tag{16}$$

Thus, we have verified the separation condition via (16), i.e., no pricing policy can perform well in both the two instances.

Step 3. Applying Le Cam. By directly applying the Le Cam, we can conclude that

$$\begin{aligned}\sup_{z\in\mathcal{Z}} \mathcal{R}(T; \pi, z, \mathcal{L}_{\text{GenLB}}) &\ge \frac{\Delta^2}{6(12)^2}\sum_{t=1}^{T}\left(1 - \|\mathbb{P}_{z_1}^{t-1} - \mathbb{P}_{z_2}^{t-1}\|_{\text{TV}}\right) \ge \frac{\Delta^2}{(12)^3}\sum_{t=1}^{T}\exp(-\mathrm{D}_{\text{KL}}(\mathbb{P}_{z_1}^{t-1}\|\mathbb{P}_{z_2}^{t-1})) \\ &\ge \frac{T\Delta^2}{(12)^3}\exp(-\mathrm{D}_{\text{KL}}(\mathbb{P}_{z_1}^{T}\|\mathbb{P}_{z_2}^{T})). \end{aligned} \tag{17}$$

Step 4. Calculating KL between Environments. Now we focus on the calculation of the KL-divergence $\mathrm{D}_{\text{KL}}(\mathbb{P}_{z_1}^{T}\|\mathbb{P}_{z_2}^{T})$. Note that it can be further decomposed using the Chain Rule of KL-divergence as $\mathrm{D}_{\text{KL}}\left(\mathbb{P}_{z_1}^{T}\|\mathbb{P}_{z_2}^{T}\right) = \sum_{s=1}^{T}\mathrm{D}_{\text{KL}}\left(\mathbb{P}_{z_1}^{s}\|\mathbb{P}_{z_2}^{s} \mid \mathbf{y}_{s-1}\right)$. For each component, we can conclude:[1]

$$\begin{aligned}\mathrm{D}_{\text{KL}}(\mathbb{P}_{z_1}^{s}\|\mathbb{P}_{z_2}^{s} \mid \mathbf{y}_{s-1}) &= \sum_{\mathbf{y}_s\in\{0,1\}^s}\mathbb{P}_{z_1}^{s}(\mathbf{y}_s)\log\left(\frac{\mathbb{P}_{z_1}^{s}(y_s \mid \mathbf{y}_{s-1})}{\mathbb{P}_{z_2}^{s}(y_s \mid \mathbf{y}_{s-1})}\right) \\ &= \sum_{\mathbf{y}_{s-1}\in\{0,1\}^{s-1}}\mathbb{P}_{z_1}^{s-1}(\mathbf{y}_{s-1})\sum_{y_s\in\{0,1\}}\mathbb{P}_{z_1}^{s}(y_s \mid \mathbf{y}_{s-1})\log\left(\frac{\mathbb{P}_{z_1}^{s}(y_s \mid \mathbf{y}_{s-1})}{\mathbb{P}_{z_2}^{s}(y_s \mid \mathbf{y}_{s-1})}\right) \\ &\overset{(i)}{\le} \frac{1}{d(p_s; z_2)(1 - d(p_s; z_2))}\sum_{\mathbf{y}_s\in\{0,1\}^{s-1}}\mathbb{P}_{z_1}^{s-1}(\mathbf{y}_{s-1})\left(d(p_s; z_1) - d(p_s; z_2)\right)^2 \\ &\overset{(ii)}{\le} \frac{16}{3}\sum_{\mathbf{y}_s\in\{0,1\}^{s-1}}\mathbb{P}_{z_1}^{s-1}(\mathbf{y}_{s-1})\left(d(p_s; z_1) - d(p_s; z_2)\right)^2 \\ &\overset{(iii)}{\le} \frac{16}{3}(z_1 - z_2)^2 \cdot \sum_{\mathbf{y}_s\in\{0,1\}^{s-1}}\mathbb{P}_{z_1}^{s-1}(\mathbf{y}_{s-1})\cdot\left(p^\star(z_1) - p_s\right)^2 \\ &= \frac{16}{3}(z_1 - z_2)^2 \cdot \mathbb{E}_{z_1}\left[\left(p^\star(z_1) - p_s\right)^2\right], \end{aligned}$$

[1]Note that here we only require $z_1 = 1/2$ as in the construction, while z_2 can be different from our that in our construction and can take arbitrary $z \in \mathcal{Z}$.

with (i) comes from the KL-divergence between two Bernoulli distributions (see Appendix C for details); (ii) comes from the assumption of $\mathcal{P}$ and $\mathcal{Z}$ such that $d(p;z) \in [1/4, 3/4]$ for all $p \in \mathcal{P}$ and $z \in \mathcal{Z}$; (iii) is based on the construction of z_1, which leads to $\left(d\left(p_s; z_1\right) - d\left(p_s; z\right)\right)^2 \leq \left(z_1 - z\right)^2 \left(p^*\left(z_1\right) - p_s\right)^2$. Then, summing over all s, we obtain:

$$\begin{aligned}
\mathrm{D_{KL}}\left(\mathbb{P}_{z_1}^T \| \mathbb{P}_{z_2}^T\right) &\leq \frac{16}{3}\left(z_1 - z_2\right)^2 \sum_{s=1}^{T} \mathbb{E}_{z_1}\left[\left(p^\star(z_1) - p_s\right)^2\right] \\
&\leq 16\left(z_1 - z_2\right)^2 \sum_{s=1}^{T} \mathbb{E}_{z_1}\left[r\left(p^\star(z_1); z_1\right) - r\left(p_s; z_1\right)\right] \\
&= 16\left(z_1 - z_2\right)^2 \mathcal{R}(T; \pi, z_1, \mathcal{L}_{\mathrm{GenLB}}),
\end{aligned}$$

where the second inequality follows from the inequality (16). In addition, since $\mathcal{R}(T; \pi, z_2, \mathcal{L}_{\mathrm{GenLB}})$ is non-negative and $z_2 - z_1 = \Delta$ by definition,

$$\mathcal{R}(T; \pi, z_1, \mathcal{L}_{\mathrm{GenLB}}) + \mathcal{R}(T; \pi, z_2, \mathcal{L}_{\mathrm{GenLB}}) \geq \frac{1}{16\Delta^2} \cdot \mathrm{D_{KL}}\left(\mathbb{P}_{z_1}^T \| \mathbb{P}_{z_2}^T\right). \tag{18}$$

Finally, by taking $\Delta = T^{-1/4}$, combining results (17) and (18), and using the inequality $x + e^{-x} \geq 1$, we obtain there exists a constant $c_0 > 0$ such that:

$$\begin{aligned}
\sup_{z \in \mathcal{Z}} \mathcal{R}(T; \pi, z, \mathcal{L}_{\mathrm{GenLB}}) &\geq \frac{1}{2}[\mathcal{R}(T; \pi, z_1, \mathcal{L}_{\mathrm{GenLB}}) + \mathcal{R}(T; \pi, z_2, \mathcal{L}_{\mathrm{GenLB}})] \\
&\geq c_0 \sqrt{T}\left[\exp(-\mathrm{D_{KL}}(\mathbb{P}_{z_1}^T \| \mathbb{P}_{z_2}^T)) + \mathrm{D_{KL}}(\mathbb{P}_{z_1}^T \| \mathbb{P}_{z_2}^T)\right] = \Omega(\sqrt{T}).
\end{aligned}$$

□

Compared to previous applications of Le Cam's method in this survey, the current construction for dynamic pricing introduces a qualitatively different flavor. In earlier examples, the two instances were constructed symmetrically, each playing an equivalent analytical role in the proof. That symmetry allowed for a straightforward verification of separation and KL bounds between the two environments.

In contrast, the lower bound argument in this pricing example is inherently asymmetric. The first instance, corresponding to $z_1 = 1/2$, plays a special role: at this parameter value, the optimal price $p^\star(z_1) = 1$ is deliberately chosen to be uninformative, in the sense that it yields identical demand probabilities across a family of demand functions, making it statistically impossible to distinguish nearby demand curves without incurring regret. The second instance, $z_2 = z_1 + \Delta$, is then chosen to be sufficiently close to preserve indistinguishability but different enough to induce regret if misestimated.

This asymmetry also leads to a distinct analysis pathway. Here we do not derive a closed-form upper bound for the KL divergence between z_1 and z_2. Instead, leveraging the special structure of the revenue function, we relate the KL divergence to the expected

regret incurred under z_1. This coupling is crucial: it allows us to close the loop by showing that either the learner incurs large regret, or it gains enough information to distinguish the instances, formalized using the inequality $x + e^{-x} \geq 1$, which ensures that at least one of the two terms (regret or divergence) must be substantial.

Overall, this example highlights how problem-specific structural properties such as uninformative actions and parameter sensitivity can be used to construct asymmetric and more nuanced lower bound arguments within the Le Cam framework. We will again apply this idea in the next continuous-time dynamic pricing application.

Application 9: Continuous-time Pricing with Inventory Constraint

Now, we consider the dynamic pricing problem faced by a monopolist over a finite selling horizon $[0, T]$. The seller begins the season with a fixed initial inventory of $x \in \mathbb{N}$ units and cannot replenish the inventory throughout the season. The objective is to dynamically adjust the price over time to maximize expected revenue while depleting the inventory by time T.

Customers arrive according to a Poisson process with an instantaneous demand rate λ_t, which is solely a function of the price $p(t)$ offered at time t. That is, we write $\lambda_t = \lambda(p(t))$, where $\lambda(\cdot)$ is the demand rate function. There is no salvage value for any leftover inventory at the end of the horizon.

The set of feasible prices is again an interval $[\underline{p}, \bar{p}] \subset \mathbb{R}_+$, and there exists an extended price $p_\infty > \bar{p}$ such that $\lambda(p_\infty) = 0$, meaning that at very high prices, demand vanishes. We make the following standard structural assumptions on the demand rate function parallel to the discrete-time setting, with the difference that we are now focusing on a (more narrow) Poisson demand process: (i) Bounded demand parameter: $|\lambda(p)| \leq M$ for all $p \in [\underline{p}, \bar{p}]$; (ii) Lipschitz continuity: $\lambda(p)$ and $r(\lambda(p); z)$ are all Lipschitz continuous; (iii) Strict concavity of the revenue function: the revenue function $r(\lambda(p); z)$ is twice differentiable with bounded $r''(\lambda(p); z)$ for all p and z.

In the dynamic pricing problem with limited inventory, our goal is to evaluate how well a pricing policy π performs compared to the best possible policy under the true (but unknown) demand function. At each time $t \in [0, T]$, the decision-maker selects a price $p(t) \in [\underline{p}, \bar{p}] \cup \{\infty\}$, subject to the inventory constraint:

$$\int_0^T dN^\pi(s) \leq x, \quad \text{w.p.1},$$

where $N^\pi(t) = N\left(\int_0^t \lambda(p(s))ds\right)$ denotes the cumulative demand up to time t using policy π. The expected revenue of a pricing policy π is thus defined as:

$$R(T; \pi, x, \lambda) \triangleq \mathbb{E}\left[\int_0^T p(s)dN^\pi(s)\right],$$

with the expectation taken w.r.t. the distribution of historical sales pasts. However, finding the optimal expected revenue directly is complex due to the stochastic nature of demand even when the demand function λ is known. To simplify, we use a deterministic relaxation that replaces the Poisson demand process with its expected value as follows.

$$R^D(T; x, \lambda) \triangleq \sup_{p(s)} \left\{ \int_0^T p(s)\lambda(p(s))ds : \int_0^T \lambda(p(s))ds \leq x \right\}.$$

According to the literatures (e.g., Gallego and Van Ryzin 1994), it gives an upper bound on the expected revenue for any admissible pricing policy π, i.e., $R(T; \pi, x, \lambda) \leq R^D(T; x, \lambda)$. We then define a normalized regret of policy π under demand λ as:

$$\mathcal{R}(T; \pi, x, \lambda) = 1 - \frac{R(T; \pi, x, \lambda)}{R^D(T; x, \lambda)}$$

which measures how far the policy π is from the deterministic upper bound. Our objective is to minimize the worst-case regret over a family of demand functions $\lambda \in \Gamma$. That is,

$$\inf_{\pi \in \mathcal{P}} \sup_{\lambda \in \Gamma} \mathcal{R}(T; \pi, x, \lambda), \tag{19}$$

where $\mathcal{P}$ is the class of all admissible pricing policies (i.e., pricing space). Due to the analysis difficulty for (19), we will adopt the asymptotic performance analysis: consider a regime in which both the size of the initial inventory, as well as the potential demand, grow proportionally large. In a problem with size N, the initial inventory and the demand function are given by:

$$x_N = Nx, \text{ and } \lambda_N(\cdot) = N\lambda(\cdot),$$

and correspondingly, we have $R^D(T, N; x, \lambda) \triangleq R^D(T; Nx, N\lambda) = N \cdot R^D(T, x, \lambda)$ to be the deterministic optimal solution for the problem with size N and $R(T, N; \pi, x, \lambda) \triangleq R(T; \pi, Nx, N\lambda) = N \cdot R(T; \pi, x, \lambda)$ to be the expected value of a pricing policy π when it is applied to a problem with size N. This models large-scale systems where both supply and potential demand grow proportionally—capturing a "fluid limit" perspective. Under this scaling, the cumulative demand (and thus potential regret) grows with N, making it possible to study asymptotic efficiency via normalized regret for the size-N problem defined by:

$$\mathcal{R}(T, N; \pi, x, \lambda) = 1 - \frac{R(T, N; \pi, x, \lambda)}{R^D(T, N; x, \lambda)}.$$

We note that in contrast to the previous discrete-time pricing application where the regret analysis focuses on T, the number of rounds, the N-scaling model reflects a complementary perspectives: the former emphasizes finite-time performance in sequential learning, while the latter emphasizes the large-scale asymptotics in stochastic control problems. For simplicity, we can normalize $T = 1$ and study how the regret grows with the size of the problem

N, and we further simplify the regret and expected rewards as $\mathcal{R}(N;\pi,x,\lambda)$, $R(N;\pi,x,\lambda)$, and $R^D(N;x,\lambda)$, respectively. The following Theorem restates the theorem 2 of Wang et al. (2014).

Theorem 4.4. *Define a problem class by letting $\mathcal{P} = [1/2, 3/2]$, $\mathcal{Z} = [1/3, 2/3]$ and let $\lambda(p;z) = 1/2 + z - zp$ for $z \in \mathcal{Z}$. Assume that $x = 2$, then for any admissible pricing policy $\pi \in \mathcal{P}$ and any $N \geq 1$, the regret can be lower bounded by:*

$$\sup_{z \in Z} \mathcal{R}(N;\pi,x,\lambda) = \Omega(1/\sqrt{N}).$$

Proof of Theorem 4.4. It is easy to verify that the problem class satisfies our regularity conditions. Next, we will follow a similar idea as that in the proof of Theorem 4.3 in constructing the two demand instances.

Step 1. Instance Construction. Again, we construct demand parameters $z_1 = 1/2$ and $z_2 = z_1 + \Delta$ for some small $\Delta > 0$ that will be determined later. Now the optimal price $p^*(z_1) = 1$ under z_1 is still the "uninformative" price.

Step 2. Separation Verification. All the analysis remains the same to that before, in that we just replace the demand model $d(p;z)$ in Application 8 with a Poisson demand process with parameter $\lambda(p;z)$, and we can derive the same separation result in (16):

$$r\left(p^\star(z);z\right) - r(p;z) \geq \frac{1}{3}\left(p - p^\star(z)\right)^2 \geq \frac{\Delta^2}{3(12)^2}. \tag{20}$$

Step 3. Applying Information-Theoretical Tools. As the regret definition has already changed under the new problem setting, we need to start from the beginning. We have:

$$\begin{aligned}
\mathcal{R}(N;\pi,x,\lambda(\cdot;z_1)) &\overset{(i)}{\geq} \frac{\mathbb{E}_{z_1}\int_0^1 \left\{r\left(p^\star\left(z_1\right);z_1\right) - r\left(p(s);z_1\right)\right\}\mathrm{d}s}{\mathbb{E}_{z_1}\int_0^1 r\left(p^\star\left(z_1\right);z_1\right)\mathrm{d}s} \\
&\overset{(ii)}{\geq} \frac{\Delta^2}{6(12)^2}\cdot\mathbb{E}_{z_1}\int_0^1 \mathbf{1}\left(p(s)\in C_{z_2}\right)\mathrm{d}s \\
&= \frac{\Delta^2}{6(12)^2}\cdot\int_0^1 \mathbb{P}^s_{z_1}\left(p(s)\in C_{z_2}\right)\mathrm{d}s,
\end{aligned} \tag{21}$$

with (i) holds because we have relaxed the inventory constraint when using π here, and (ii) holds according to the separation condition in (20), together with C_{z_1} defined in (15) before, and the fact that $r\left(p^\star\left(z_1\right);z_1\right) = 1/2$. Here $\mathbb{P}^s_{z_1}$ is the probability measure of demand under policy π up to time s with demand function parameter z_1. We could conclude a similar lower bound to instance under z_2 and derive:

$$\mathcal{R}(N;\pi,x,\lambda(\cdot;z_2)) \geq \frac{\Delta^2}{3(12)^2}\cdot\int_0^1 \mathbb{P}^s_{z_2}\left(p(s)\notin C_{z_2}\right)\mathrm{d}s, \tag{22}$$

where now we upper bounded the denominator $r\left(p^{\star}\left(z_2\right); z_2\right)$ by 1. Then, by summing the two regrets (21), (22) and applying the Bretagnolle-Huber inequality:

$$\begin{aligned}\sup_{z\in Z}\mathcal{R}(N;\pi,x,\lambda) &\geq \frac{1}{2}(\mathcal{R}(N;\pi,x,\lambda(\cdot;z_1))+\mathcal{R}(N;\pi,x,\lambda(\cdot;z_2)))\\ &\geq \frac{\Delta^2}{2\cdot 12^3}\cdot\int_0^1 \exp\left(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}^s_{z_1}\|\mathbb{P}^s_{z_2})\right)\mathrm{d}s. \end{aligned} \tag{23}$$

Step 4. Calculating KL between Environments. Next, we examine the KL-divergence between $\mathbb{P}^s_{z_1}$ and $\mathbb{P}^s_{z_2}$, which is the major difference compared with the discrete setting. By its definition, we have:

$$\begin{aligned}&\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}^1_{z_1}\|\mathbb{P}^1_{z_2}\right)=\mathbb{E}_{z_1}\left[\int_0^1 N\cdot\lambda(p(s);z)\left(\frac{\lambda\left(p(s);z_1\right)}{\lambda(p(s);z_2)}\log\frac{\lambda\left(p(s);z_1\right)}{\lambda(p(s);z_2)}+1-\frac{\lambda\left(p(s);z_1\right)}{\lambda(p(s);z_2)}\right)\mathrm{d}s\right]\\ &=\mathbb{E}_{z_1}\left[\int_0^1 N\cdot\left(\frac{1}{2}+z_1-z_1p(s)\right)\left(-\log\frac{\frac{1}{2}+z_2-z_2p(s)}{\frac{1}{2}+z_1-z_1p(s)}-1\right)+N\cdot\left(\frac{1}{2}+z_2-z_2p(s)\right)\mathrm{d}s\right].\end{aligned} \tag{24}$$

First, we notice that $\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}^1_{z_1}\|\mathbb{P}^1_{z_2}\right)\geq\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}^s_{z_1}\|\mathbb{P}^s_{z_2}\right)$ for any $s\in[0,1]$ through some straightforward calculations following (24) and our construction of the distribution and decision class. Then, the regret in (23) can be further lower bounded by:

$$\sup_{z\in Z}\mathcal{R}(N;\pi,x,\lambda)\geq\frac{\Delta^2}{2\cdot 12^3}\cdot\exp\left(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}^1_{z_1}\|\mathbb{P}^1_{z_2})\right). \tag{25}$$

Next, based on the facts that $-\log x-1\leq -x+\frac{(x-1)^2}{2(2-x)}$ for any $x\in(0,2)$, and that $\frac{2}{3}\leq\frac{\frac{1}{2}+z-zp(s)}{\frac{1}{2}+z_1-z_1p(s)}\leq 2$ for any $z\in[1/3,2/3]$ and $p(s)\in[1/2,3/2]$, we can simplify (24) through:

$$\begin{aligned}\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}^1_{z_1}\|\mathbb{P}^1_{z_2}\right)&\leq N\mathbb{E}_{z_1}\int_0^1\frac{1}{2\left(2-\frac{1/2+z_1-z_1p(s)}{1/2+z_2-z_2p(s)}\right)}\cdot\frac{\left(z_1-z_2\right)^2\left(1-p(s)\right)^2}{\left(1/2+z_1-z_1p(s)\right)^2}\mathrm{d}s\\ &=N\mathbb{E}_{z_1}\int_0^1\frac{\left(z_1-z_2\right)^2\left(1-p(s)\right)^2}{\left(1/2+z_1-z_1p(s)\right)^2}\mathrm{d}s\\ &\leq 16N(z_1-z_2)^2\cdot\mathbb{E}_{z_1}\int_0^1(1-p(s))^2\mathrm{d}s.\end{aligned}$$

Now, for the instance under z_1, we can conclude that:

$$\begin{aligned}\mathcal{R}(N;\pi,x,\lambda(\cdot;z_1)) &\geq \frac{\mathbb{E}_{z_1}\int_0^1 \{r\left(p^\star\left(z_1\right);z_1\right)-r\left(p(s);z_1\right)\}\,\mathrm{d}s}{\mathbb{E}_{z_1}\int_0^1 r\left(p^\star\left(z_1\right);z_1\right)\mathrm{d}s}\\ &\overset{(i)}{\geq} \frac{2}{3}\mathbb{E}_{z_1}\int_0^1 (1-p(s))^2\mathrm{d}s\\ &\geq \frac{1}{24N\Delta^2}\cdot \mathrm{D_{KL}}\left(\mathbb{P}^1_{z_1}\|\mathbb{P}^1_{z_2}\right) \qquad (26)\end{aligned}$$

where (i) is by $-4/3 \leq r''(p;z) \leq -2/3$ for all p and z. Combining (25) and (26), and by choosing $\Delta = N^{-1/4}$, we derive the lower bound. □

In contrast to the discrete-time setting where regret is measured over a fixed number of periods, the continuous-time pricing problem frames regret in terms of the problem size N, which reflects the scale of both initial inventory and potential demand. This shift captures a key modeling difference: here, learning is constrained not by time but by inventory. Once the inventory is exhausted, the learner can no longer influence or observe outcomes, regardless of how much time remains. As a result, increasing the time horizon does not mitigate uncertainty unless inventory also increases.

Besides, since the regret is defined relative to a deterministic benchmark that also scales with N, the lower bound appears as $\Omega(1/\sqrt{N})$, rather than the $\Omega(\sqrt{N})$ rate seen in unnormalized formulations. This normalization reflects the diminishing marginal impact of uncertainty in large-scale systems and underscores how problem structure-especially resource constraints-can dictate both the form and scaling of regret in learning problems.

4.3 Assortment Optimization

We now turn to the third pillar of revenue management: assortment optimization, where the goal is to select subsets of products to offer in each period to maximize expected revenue. Unlike pricing or auctions, assortment problems involve combinatorial decisions over sets of items, often under capacity constraints, and are common in retail, recommendation, and online platforms.

The learning challenge arises from unknown customer preferences that drive stochastic choice behavior. Since only offered items generate feedback, and decisions affect what is observed, inferring optimal assortments becomes a complex interaction of exploration and exploitation. Capacity constraints further complicate the structure of the optimal policy and its regret analysis.

In this section, we examine regret lower bounds under the multinomial logit (MNL) model, with and without capacity constraints. Using Le Cam's two-point method, we construct preference vectors that induce different optimal assortments while remaining statistically indistinguishable, thereby revealing fundamental limits of learning in this setting.

Application 10: Uncapacitated MNL Assortment

We begin with the dynamic assortment selection problem under the MNL-choice model with uncapacitated assortment decision set. This model assumes that, at each time period $t = 1, \cdots, T$, a retailer selects a subset $S_t \subseteq [M]$ of M items to display to a customer. The customer then makes a probabilistic choice $i_t \in S_t \cup \{0\}$, where $i_t = 0$ denotes no purchase. If an item $i_t \in S_t$ is purchased, the retailer collects revenue $r_{i_t} \in [0, 1]$. The no-purchase option is normalized to yield zero revenue.

The purchasing choice follows the MNL model: each item $i \in [M]$ is associated with an unknown preference parameter $v_i \geq 0$, and the probability that a customer selects item $j \in S_t \cup \{0\}$ is given by:

$$\mathbb{P}\left(i_t = j \mid S_t\right) = \frac{v_j}{1 + \sum_{k \in S_t} v_k}, \tag{27}$$

where $v_0 = 1$ is the normalized utility of the no-purchase option. The expected revenue for assortment S_t is thus:

$$r\left(S_t\right) = \sum_{j \in S_t} \frac{r_j v_j}{1 + \sum_{k \in S_t} v_k}.$$

The retailer's objective is to learn from observed customer choices and select assortment policy $\pi = (S_1, \cdots, S_T)$ to minimize the cumulative regret:

$$\mathcal{R}(T; \pi, v) = \mathbb{E} \sum_{t=1}^{T} \left[r\left(S^\star\right) - r\left(S_t\right)\right],$$

where $S^\star \in \arg\max_{S \subseteq [M]} r(S)$ is the optimal assortment in hindsight, and the expectation is taken w.r.t. the randomness of the policy and the induced observations of purchasing decisions. The following Theorem restates the theorem 4 of Wang et al. (2018).

Theorem 4.5. *For the uncapacitated MNL assortment optimization introduced before, for any assortment policy π,*

$$\sup_{v} \mathcal{R}(T; \pi, v) = \Omega(\sqrt{T}).$$

Proof of Theorem 4.5.

Step 1. Instance Construction. Note that now we need to construct both the revenue and utility parameters. We construct revenue parameters $\{r_i\}_{i=1}^M$ as $r_1 = 1, r_2 = 1/2$ and $r_3 = \cdots = r_M = 0$ for all the environments. The utility parameters are construct accordingly via:

$$\begin{aligned} \boldsymbol{v}_a: &\quad v_1^a = 1 - \Delta, v_2^a = 1, v_3^a = \cdots = v_M^a = 0; \\ \boldsymbol{v}_b: &\quad v_1^b = 1 + \Delta, v_2^b = 1, v_3^b = \cdots = v_M^b = 0, \end{aligned}$$

for some sufficiently small $\Delta > 0$ that will be determined later. Thus the two instances are distinguished by the utility $\boldsymbol{v}_a$ and $\boldsymbol{v}_b$.

Step 2. Separation Verification. We first verify the optimal assortment decision under full information for the two instances through some straightforward calculations. Note that including any items $i \in \{3, \cdots, M\}$ in the assortment decision S will not affect the expected revenue at all (as their revenue and utility are all zeros), we thus focus on $S \subseteq \{1, 2\}$.

- For $\boldsymbol{v}_a$: now $S_a^\star = \{1, 2\}$ and the optimal expected revenue is $r_{\boldsymbol{v}_a}(S_a^\star) = \dfrac{1.5 - \Delta}{3 - \Delta}$;
- For $\boldsymbol{v}_b$: now $S_b^\star = \{1\}$ and the optimal expected revenue is $r_{\boldsymbol{v}_b}(S_b^\star) = \dfrac{1 + \Delta}{2 + \Delta}$.

For any assortment decision $S \subseteq \{1, 2\}$ in instance a, we have:

$$\begin{aligned} r_{\boldsymbol{v}_a}(S_a^\star) - r_{\boldsymbol{v}_a}(S) &= \frac{1.5 - \Delta}{3 - \Delta} - \frac{v_1^a \cdot \mathbf{1}\{1 \in S\} + \frac{1}{2} \cdot \mathbf{1}\{2 \in S\}}{1 + (v_1^a \cdot \mathbf{1}\{1 \in S\} + \mathbf{1}\{2 \in S\})} \\ &\geq \left(\frac{1.5 - \Delta}{3 - \Delta} - \frac{1 - \Delta}{2 - \Delta}\right) \mathbf{1}\{2 \notin S\} \geq \frac{\Delta}{12} \mathbf{1}\{2 \notin S\}. \end{aligned} \tag{28}$$

For any assortment decision $S \subseteq \{1, 2\}$ in instance b, we have:

$$\begin{aligned} r_{\boldsymbol{v}_b}(S_b^\star) - r_{\boldsymbol{v}_b}(S) &= \frac{1 + \Delta}{2 + \Delta} - \frac{v_1^b \cdot \mathbf{1}\{1 \in S\} + \frac{1}{2} \cdot \mathbf{1}\{2 \in S\}}{1 + \left(v_1^b \cdot \mathbf{1}\{1 \in S\} + \mathbf{1}\{2 \in S\}\right)} \\ &\geq \frac{1}{2} \left(\frac{1 + \Delta}{2 + \Delta} - \frac{1.5 + \Delta}{3 + \Delta}\right) \mathbf{1}\{2 \in S\} \geq \frac{\Delta}{48} \mathbf{1}\{2 \in S\}. \end{aligned} \tag{29}$$

Combining (28) and (29), we have the separation condition holds for any assortment decision S:

$$r_{\boldsymbol{v}_a}(S_a^\star) + r_{\boldsymbol{v}_b}(S_b^\star) - r_{\boldsymbol{v}_a}(S) - r_{\boldsymbol{v}_a}(S) \geq \frac{\Delta}{48}.$$

Step 3. Applying Le Cam. We have:

$$\begin{aligned} \sup_v \mathcal{R}(T; \pi, v) &\geq \frac{\Delta}{96} \cdot \sum_{t=1}^{T} \left(1 - \|\mathbb{P}_{\boldsymbol{v}_a}^{t-1} - \mathbb{P}_{\boldsymbol{v}_b}^{t-1}\|_{\mathrm{TV}}\right) \\ &\geq \frac{T\Delta}{192} \cdot \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_a^T \| \mathbb{P}_b^T)), \end{aligned}$$

where $\mathbb{P}_a^T$ and $\mathbb{P}_b^T$ denote the joint distribution of historical assortment decisions under policy π and purchasing actions under $\boldsymbol{v}_a$ (instance a) and $\boldsymbol{v}_b$ (instance b) by period T, respectively.

Step 4. Calculating KL between Environments. We first note that given assortment decision S, the KL-divergence between $\mathbb{P}_a(S)$ and $\mathbb{P}_b(S)$ (i.e., the joint probability of purchasing items in assortment S), are the same for item $i = 2, 3, \cdots, M$, as

$\mathbb{P}_a(S = \{i\}) = \mathbb{P}_b(S = \{i\}) = 1/2$ for $i = 2$ and $\mathbb{P}_a(S = \{i\}) = \mathbb{P}_b(S = \{i\}) = 0$ for $i = 3, \cdots, N$, and the difference lies in the purchase probability of item i. Thus, if $1 \notin S$, $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_a(S) \| \mathbb{P}_b(S)) = 0$. Therefore, we will focus on the S that always includes item 1. The difference of purchase probabilities of item 1 under the two instances can be bounded by:

$$|\mathbb{P}_a(i = 1|S) - \mathbb{P}_b(i = 1|S)| = \frac{2\Delta \cdot (1 + \mathbf{1}\{2 \in S\})}{(2 + \mathbf{1}\{2 \in S\})^2 - \Delta^2} \leq \frac{2\Delta}{2 + \mathbf{1}\{2 \in S\}}.$$

If item 2 is also in S, then its difference of purchase probabilities under the two instances can also be bounded by:

$$\begin{aligned} |\mathbb{P}_a(i = 2|S) - \mathbb{P}_b(i = 2|S)| &= \frac{2\Delta}{(2 + \mathbf{1}\{2 \in S\})^2 - \Delta^2} \\ &\leq \frac{2\Delta}{(1 + \mathbf{1}\{2 \in S\})(2 + \mathbf{1}\{2 \in S\})}. \end{aligned}$$

For the zero purchase probabilities under the two instances, we have:

$$|\mathbb{P}_a(i = 0|S) - \mathbb{P}_b(i = 0|S)| \leq \frac{2\Delta}{4 - \Delta^2}.$$

Then, by the definition of KL-divergence $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_a(S) \| \mathbb{P}_b(S))$ and the upper bound inequality for two a categorical distributions we introduce in Lemma C.3 in the Appendix, we have for any S:

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_a(S) \| \mathbb{P}_b(S)) \leq \frac{12\Delta^2}{2 + \mathbf{1}\{2 \in S\}} \leq 6\Delta^2. \tag{30}$$

At last, by using the chain rule of KL-divergence, we can conclude:

$$\begin{aligned} \mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_a^T \| \mathbb{P}_b^T\right) &= \mathbb{E}_a\left[\log \frac{\mathrm{d}\mathbb{P}_a\left(S_1, i_1, \cdots, S_T, i_T\right)}{\mathrm{d}\mathbb{P}_b\left(S_1, i_1, \cdots, S_T, i_T\right)}\right] \\ &= \mathbb{E}_a\left[\sum_{t=1}^{T} \log \frac{\mathrm{d}\mathbb{P}_a\left(i_t|S_t\right)}{\mathrm{d}\mathbb{P}_b\left(i_t|S_t\right)}\right] \\ &\overset{(i)}{\leq} 6T\Delta^2, \end{aligned}$$

where (i) is from (30). By choosing $\Delta = 1/\sqrt{T}$ and combining the results together, we derive the regret lower bound. □

Application 11: Capacitated MNL Assortment

In contrast to the uncapacitated setting where the retailer may offer any subset of products, the capacitated MNL assortment problem imposes a cardinality constraint on the offered

assortment. Specifically, at each round $t = 1, \cdots, T$, the retailer selects a subset $S_t \subseteq [M]$ such that $|S_t| \leq K$ for some known capacity constraint $K \in \mathbb{N}$. This models practical limitations in shelf space, display slots, or recommendation panels.

As in the uncapacitated case, each product $i \in [M]$ has an unknown preference parameter $v_i \geq 0$ and a known reward $r_i \in [0, 1]$. The customer makes a choice $i_t \in S_t \cup \{0\}$ according to the standard MNL model (27), with expected revenue under an offered assortment S_t still given by

$$r\left(S_t\right) = \sum_{j \in S_t} \frac{r_j v_j}{1 + \sum_{k \in S_t} v_k},$$

but the action space $S \subseteq [N] : |S| \leq K$ is now combinatorially constrained. Let $S^\star \in \arg\max_{S:|S|\leq K} r(S)$ be the optimal assortment in hindsight. Then the regret of a learning policy $\pi = (S_1, \ldots, S_T)$ is still defined as:

$$\mathcal{R}(T; \pi, v) = \mathbb{E} \sum_{t=1}^{T} \left[r\left(S^\star\right) - r\left(S_t\right)\right].$$

The following Theorem restates the theorem 1 of Chen and Wang (2018).

Theorem 4.6. *For the capacitated MNL assortment optimization introduced before, suppose $K \leq M/4$ and $M \leq 100T$, then for any assortment policy π,*

$$\sup_v \mathcal{R}(T; \pi, v) = \Omega(\sqrt{MT}).$$

Proof of Theorem 4.6.
Step 1. Instance Construction. We let $r_1 = \cdots = r_M = 1$ be fixed. For any given K-sized assortment S, we construct the preference vector $\boldsymbol{v}_S$ through:

$$\boldsymbol{v}_S \triangleq \begin{cases} \dfrac{1+\Delta}{K} & \text{if } i \in S, \\ \dfrac{1}{K} & \text{otherwise,} \end{cases}$$

for some small $\Delta > 0$ that will be determined later. Denote $\mathcal{S}_K$ as all subsets of $[M]$ of size K. Note that $|\mathcal{S}_K| = \binom{M}{K}$ and hence we have constructed at most $\binom{M}{K}$ different $\boldsymbol{v}_S$, Under the construction, we can easily verify that $S^\star(\boldsymbol{v}_S) = S$, so we can simply restrict our attention to the decision space of $\mathcal{S}_K$.

Step 2. Separation Verification. For some fixed $S_0 \in \mathcal{S}_K$ and utility parameter $\boldsymbol{v}_{S_0}$, next, we will show that for any $\tilde{S}_t \in \mathcal{S}_K$ that differs significantly from S_0, the single-period expected revenue loss is large under $\boldsymbol{v}_{S_0}$. Recall that by its construction, we have

$$r(S_0) = \frac{1+\Delta}{2+\Delta},$$

and that

$$r(\tilde{S}_t) = \frac{1 + (1-\delta)\Delta}{2 + (1-\delta)\Delta},$$

where we define $1 - \delta \triangleq \left(|\tilde{S}_t \cap S_0|/K\right)$ as the coverage rate of S_0 using $\widetilde{S}_t$. Subsequently,

$$r(S_0) - r(\tilde{S}_t) = \frac{\delta\Delta}{(2+\Delta)(2+(1-\delta)\Delta)} \geq \frac{\delta\Delta}{9}, \tag{31}$$

for any small $0 < \Delta \leq 1/2$.

Step 3. Applying Information-Theoretical Tool. By the previous constructions, we have already shown that the optimal policy falls within $\tilde{S}_t \in \tilde{\pi}$, and we will thus focus on the regret under the policy class $\tilde{\pi}$ instead of π. Define $\tilde{M}_i \triangleq \sum_{t=1}^{T} \mathbf{1}\{i \in \tilde{S}_t\}$ as the number of times item i is included in the assortment under policy $(\tilde{S}_1, \cdots, \tilde{S}_T)$. The regret $\sup_v \mathcal{R}(T; \tilde{\pi}, v)$ can be decomposed and lower bounded through:

$$\begin{aligned}
\sup_v \mathcal{R}(T;\pi,v) &\geq \frac{1}{|\mathcal{S}_K|} \sum_{S\in\mathcal{S}_K} \mathbb{E}_{\boldsymbol{v}_S}\left[\sum_{t=1}^{T}\left(r(S) - r(\tilde{S}_t)\right)\right] \\
&\overset{(i)}{\geq} \frac{1}{|\mathcal{S}_K|} \sum_{S\in\mathcal{S}_K} \sum_{i\notin S} \mathbb{E}_{\boldsymbol{v}_S}[\tilde{M}_i] \cdot \frac{\Delta}{9K} \\
&\overset{(ii)}{=} \frac{\Delta}{9}\left(T - \underbrace{\frac{1}{|\mathcal{S}_K|}\sum_{S\in\mathcal{S}_K}\frac{1}{K}\sum_{i\in S}\mathbb{E}_{\boldsymbol{v}_S}[\tilde{M}_i]}_{\text{(a)}}\right),
\end{aligned} \tag{32}$$

where (i) comes from the separation in (31), and (ii) holds due to the fact that $\sum_{i=1}^{M} \mathbb{E}_{\boldsymbol{v}_S}[\tilde{M}_i] = TK$ for any S.

Next, let $\mathcal{S}_{K-1}^{(i)} = \mathcal{S}_{K-1} \cap \{S \subseteq [M] : i \notin S\}$ be all subsets of size $K-1$ that do not include i. Re-arranging summation order we have

$$\begin{aligned}
\frac{1}{|\mathcal{S}_K|}\sum_{S\in\mathcal{S}_K}\frac{1}{K}\sum_{i\in S}\mathbb{E}_{\boldsymbol{v}_S}[\tilde{M}_i] &= \frac{1}{K}\sum_{i=1}^{M}\frac{1}{|\mathcal{S}_K|}\sum_{S\in\mathcal{S}_K, i\in S}\mathbb{E}_{\boldsymbol{v}_S}[\tilde{M}_i] \\
&= \frac{1}{K}\sum_{i=1}^{M}\frac{1}{|\mathcal{S}_K|}\sum_{S'\in\mathcal{S}_{K-1}^{(i)}}\mathbb{E}_{\boldsymbol{v}_{S'\cup\{i\}}}[\tilde{M}_i],
\end{aligned} \tag{33}$$

and we want to bound the discrepancy of the expected $\tilde{M}_i$ under different instances. Notice that for the S' above (which does not include item i):

$$\begin{aligned}
|\mathbb{E}_{\boldsymbol{v}_{S'}}[\tilde{M}_i] - \mathbb{E}_{\boldsymbol{v}_{S'\cup\{i\}}}[\tilde{M}_i]| &\le \sum_{j=0}^{T} j \cdot |\mathbb{P}_{\boldsymbol{v}_{S'}}(\tilde{N}_i = j) - \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}}(\tilde{N}_i = j)| \\
&\le T \cdot \sum_{j=0}^{T} |\mathbb{P}_{\boldsymbol{v}_{S'}}(\tilde{N}_i = j) - \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}}(\tilde{N}_i = j)| \\
&\le T \cdot \|\mathbb{P}_{\boldsymbol{v}_{S'}} - \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}}\|_{\mathrm{TV}} \\
&\overset{(i)}{\le} T \cdot \sqrt{\frac{1}{2}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\boldsymbol{v}_{S'}} \| \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}})}, \qquad (34)
\end{aligned}$$

where in (i) we use the Pinsker's inequality. Taking (33) and (34) back to (32), we have

$$\text{(a)} \le \frac{1}{K}\sum_{i=1}^{M} \frac{1}{|\mathcal{S}_K|} \sum_{S'\in\mathcal{S}_{K-1}^{(i)}} \left(\mathbb{E}_{\boldsymbol{v}_{S'}}[\tilde{M}_i] + T \cdot \sqrt{\frac{1}{2}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\boldsymbol{v}_{S'}} \| \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}})} \right). \qquad (35)$$

The first term in RHS of (35) can be upper bounded through:

$$\begin{aligned}
&\frac{1}{K}\sum_{i=1}^{M} \frac{1}{|\mathcal{S}_K|} \sum_{S'\in\mathcal{S}_{K-1}^{(i)}} \mathbb{E}_{\boldsymbol{v}_{S'}}[\tilde{M}_i] = \frac{1}{|\mathcal{S}_K|} \sum_{S'\in\mathcal{S}_{K-1}} \frac{1}{K} \sum_{i\notin S'} \mathbb{E}_{\boldsymbol{v}_{S'}}[\tilde{M}_i] \\
&\le \frac{1}{|\mathcal{S}_K|} \sum_{S'\in\mathcal{S}_{K-1}} \frac{1}{K} \sum_{i=1}^{M} \mathbb{E}_{\boldsymbol{v}_{S'}}[\tilde{M}_i] \\
&= \frac{|\mathcal{S}_{K-1}|}{K\,|\mathcal{S}_K|} \cdot TK = \frac{\binom{M}{K-1}}{K\binom{M}{K}} \cdot TK = \frac{TK}{M-K+1} \le \frac{T}{3},
\end{aligned}$$

with the last inequality holds due to the assumption that $K \le M/4$. Taking these results back to (32):

$$\sup_{v} \mathcal{R}(T;\pi,v) \ge \frac{\Delta}{9}\left(\frac{2T}{3} - \frac{T}{|\mathcal{S}_K|} \sum_{S'\in\mathcal{S}_{K-1}} \frac{1}{K} \sum_{i\notin S'} \sqrt{\frac{1}{2}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\boldsymbol{v}_{S'}} \| \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}})} \right). \qquad (36)$$

Step 4. Calculating KL between Environments. Now we will upper bound the KL-divergence $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\boldsymbol{v}_{S'}} \| \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}})$. Define $M_i \triangleq \sum_{t=1}^{T} \mathbf{1}\{i \in S_t\}$. Note that because $S_t \subseteq \tilde{S}_t$, we have $M_i \le \tilde{M}_i$ almost surely and hence $\sum_{i=1}^{M} \mathbb{E}_{\boldsymbol{v}_S}[M_i] \le \sum_{i=1}^{M} \mathbb{E}_{\boldsymbol{v}_S}[\tilde{M}_i] = TK$ for all $S \subseteq [M]$. It is clear that for any $S_t \subseteq [M]$, $|S_t| \le K$ such that $i \notin S_t$, we have $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\boldsymbol{v}_{S'}}(\cdot \mid S_t) \| \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}}(\cdot \mid S_t)) = 0$. Therefore, we shall focus only on those $S_t \subseteq [M]$

with $i \in S_t$, which happens for $\mathbb{E}_{\boldsymbol{v}_{S'}}[M_i]$ epochs in expectation. For any such S_t, define $K' \triangleq |S_t| \le K$, $J \triangleq |S_t \cap S'| \le K-1$, and $a \triangleq 1 + \frac{K'}{K} \in (1,2]$. We then have for the no-purchase probability:

$$|\mathbb{P}_{\boldsymbol{v}_{S'}}(0|S_t) - \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}}(0|S_t)| = \left|\frac{1}{a+J\Delta/K} - \frac{1}{a+(J+1)\Delta/K}\right| \le \frac{\Delta}{K};$$

for the purchase probability of item $1 \le j \le M$ that $j \ne i$:

$$|\mathbb{P}_{\boldsymbol{v}_{S'}}(j|S_t) - \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}}(j|S_t)| \le \frac{1+\Delta}{K}\left|\frac{1}{a+J\Delta/K} - \frac{1}{a+(J+1)\Delta/K}\right| \le \frac{2\Delta}{K^2};$$

and for the purchase probability of item i:

$$\begin{aligned}
&|\mathbb{P}_{\boldsymbol{v}_{S'}}(i|S_t) - \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}}(i|S_t)| \le \left|\frac{1}{K}\frac{1}{a+J\Delta/K} - \frac{1+\Delta}{K}\frac{1}{a+(J+1)\Delta/K}\right| \\
&\le \frac{\Delta}{K}\frac{1}{a+(J+1)\Delta/K} + \frac{1}{K}\left|\frac{1}{a+J\Delta/K} - \frac{1}{a+(J+1)\Delta/K}\right| \\
&\le \frac{\Delta}{K} + \frac{1}{K}\cdot\frac{\Delta}{K} \le \frac{\Delta}{K^2} + \frac{\Delta}{K} \le \frac{4\Delta}{K}.
\end{aligned}$$

Besides, we also have $\mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}}(0|S_t) \ge \frac{1}{3}$ and $\mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}}(j|S_t) \ge \frac{1}{3K}$ for all $j \ge 1$, which can be easily verified according to our previous settings. By using the upper bound result on KL-divergence between categorical distributions in Lemma C.3, we conclude:

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\boldsymbol{v}_{S'}} \| \mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}}) \le \frac{3\Delta^2}{K^2} + 3K\cdot\frac{4J\Delta^2}{K^4} + 3K\cdot\frac{16\Delta^2}{K^2} \le \frac{63\Delta^2}{K^2}. \tag{37}$$

Then, we have the KL term in (36):

$$\begin{aligned}
\frac{1}{|\mathcal{S}_K|}\sum_{S'\in\mathcal{S}_{K-1}}\frac{1}{K}\sum_{i\notin S'}\sqrt{\frac{1}{2}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\boldsymbol{v}_{S'}}\|\mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}})} &\overset{(i)}{\le} \frac{|\mathcal{S}_{K-1}|}{K\,|\mathcal{S}_K|}\cdot\max_{S'\in\mathcal{S}_{K-1}}\sum_{i\notin S'}\sqrt{\frac{1}{2}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\boldsymbol{v}_{S'}}\|\mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}})} \\
&= \max_{S'\in\mathcal{S}_{K-1}}\frac{1}{M-K+1}\sum_{i\notin S'}\sqrt{\frac{1}{2}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\boldsymbol{v}_{S'}}\|\mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}})} \\
&\overset{(ii)}{\le} \max_{S'\in\mathcal{S}_{K-1}}\sqrt{\frac{1}{2(M-K+1)}\sum_{i\notin S'}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\boldsymbol{v}_{S'}}\|\mathbb{P}_{\boldsymbol{v}_{S'\cup\{i\}}})} \\
&\overset{(iii)}{\le} \max_{S'\in\mathcal{S}_{K-1}}\sqrt{\frac{1}{2(M-K+1)}\sum_{i\notin S'}\mathbb{E}_{\boldsymbol{v}_{S'}}[M_i]\cdot\frac{63\Delta^2}{K}} \\
&\le \max_{S'\in\mathcal{S}_{K-1}}\sqrt{\frac{63\Delta^2}{2K(M-K+1)}\sum_{i=1}^{M}\mathbb{E}_{\boldsymbol{v}_{S'}}[M_i]} \le \sqrt{\frac{63T\Delta^2}{M}},
\end{aligned}$$

where (i) is by Hölder's inequality; (ii) is by Jensen's inequality and the concavity of the square root; (iii) is by our (37). By choosing $\Delta = 0.05\sqrt{M/T}$ (now based on the assumption that $M \leq 100T$, we have $\Delta \leq 0.5$ holds), we can show that the RHS of (36) can be lower bounded by $\Omega(\sqrt{MT})$. □

4.4 Summary and Bibliographic Remarks

We conclude this section by summarizing the regret lower bounds derived for various revenue management settings. These problems, while diverse in structure, all exhibit a common challenge: learning under demand or preference uncertainty. The following table summarizes the settings and the corresponding lower bounds:

Table 5: Regret Lower Bounds in Revenue Management

	Problem Setting	**Lower Bound**
App.8	Discrete-time Pricing	$\Omega(\sqrt{T})$
App.9	Continuous-time Pricing with Constraint	$\Omega(1/\sqrt{N})$
App.6	First-Price Auction (FPA)	$\Omega(\sqrt{T})$
App.7	Joint Bidding and Evaluation FPA	$\Omega(\sqrt{T})$
App.10	Uncapacitated MNL Assortment	$\Omega(\sqrt{T})$
App.11	Capacitated MNL Assortment	$\Omega(\sqrt{MT})$
App.25	Adversarial Capacitated Assortment	$\Omega(\sqrt{M^K T})$

These canonical examples illustrate how Le Cam's method yields minimax regret lower bounds of order $\Omega(\sqrt{T})$ across a variety of revenue management problems. Despite the different feedback models, the common proof strategy of Le Cam still applies: construct two environments that are hard to distinguish and whose optimal decisions diverge. The regret then follows from the tension between learning and earning under uncertainty.

Several natural extensions are not included in our survey but are worth mentioning.

- For the auction: while our discussion focuses on the repeated first-price auction with full feedback, there is a broader literature on regret lower bounds in auction learning, particularly under more restricted information settings. Learning in Repeated Auctions have been extensively studied, see Nedelec et al. (2022) for a survey.

  Specifically, we note that other than the first-price auction, Weed et al. (2016) consider a repeated second-price auction (the highest bidder wins the item but pays the second-highest bid). Based on the Le Cam's method, they also provide a regret lower bound of $\Omega(\sqrt{T})$ under the adversarial private value setting.

- For the dynamic pricing: it is worth noting that the dynamic pricing have been thoroughly discussed in the literature spanning a wide range of models, assumptions,

and application domains see Van Ryzin and Talluri (2005), Talluri and Van Ryzin (2006), Chen et al. (2022b) for the tutorial.

Among the literatures, we note that Kleinberg and Leighton (2003) also provide a $\Omega(\sqrt{T})$ regret lower bounds for the non-parametric discrete-time dynamic pricing under minimal regularity assumptions. Their construction, however, does not rely on Le Cam's method and instead uses combinatorial arguments. Moreover, their framework excludes "uninformative prices", a key feature in our Le Cam-based constructions.

Another important line of work studies parametric dynamic pricing, where the demand is governed by some parametric model (e.g., linear, generalized linear). In this setting, regret lower bounds have been derived using tools beyond Le Cam's method, most notably the van Trees inequality, as we have mentioned before in the discussion of Section 3.3. It connects the Bayes risk (under squared loss) to the Fisher information in both the prior and the likelihood, allowing one to quantify how hard it is to estimate a parameter even with optimal learning. For example, by translating the pricing problem into the Bayesian parametric estimating and utilizing the van Trees inequality, Keskin and Zeevi (2014) also provide a $\Omega(\sqrt{T})$ lower bound for discrete-time dynamic pricing problem; Broder and Rusmevichientong (2012) and Keskin and Zeevi (2014) provide both provide a $\Omega(\log T)$ lower bound for the "well-separated" demand function class (that precludes the existence of "uninformative" price). Other examples using van Trees inequality including but not limitted to: Ban and Keskin (2021) for dynamic pricing with high-dimensional data with regret lower bound $\Omega(\sqrt{T})$; Chen et al. (2020) for joint pricing and inventory control with limited price changes with regret lower bound $\Omega(\log T)$; Chen et al. (2022a) for pricing with infrequent inventory replenishment, etc.

- For the assortment optimization: we refer to Chen et al. (2022b) and Heger and Klein (2024) for a comprehensive survey on assortment optimization models. Note that research on MNL-based assortment optimization has become especially active in the past decade, driven by its tractable structure and practical relevance. However, despite this surge of interest, formal regret lower bounds in the learning setting remain relatively limited. For the capacitated MNL assortment problem, Agrawal et al. (2016) first established a regret lower bound of $\Omega(\sqrt{MT/K})$. Their construction, while insightful, is not tight however. In our survey, we provide a sharper analysis that gives a tight regret lower bound for this setting.

5 Online Decision Making with Bandits Feedback

The online multi-armed bandit (MAB) problem is a fundamental model for sequential decision-making under uncertainty. At each round $t \in [T]$, a learner selects an action $a_t \in \mathcal{A}$, where $\mathcal{A}$ denotes the action set, and receives a random reward $r_t \in [0, 1]$ generated from an unknown distribution associated with the chosen arm. The learner's goal is to make sequential decisions so as to maximize cumulative reward or, equivalently, minimize regret relative to the best fixed action in hindsight. We define the regret of a policy π after T rounds under bandit environment v as:

$$\mathcal{R}(T; \pi, v) \triangleq \max_{a \in \mathcal{A}} \mathbb{E}\left[\sum_{t=1}^{T} r_t(a)\right] - \mathbb{E}\left[\sum_{t=1}^{T} r_t\left(a_t\right)\right], \tag{38}$$

where $r_t(a)$ denotes the reward of action a at round t generated under environment v, and the expectation is taken over the *randomness in both the rewards and the learner's policy*.

The section is organized as follows: in Section 5.1, we begin with the simplest case where $|\mathcal{A}| = 2$. We review classical instance-independent and instance-dependent lower bounds, and also study the batched variant where the learner only updates decisions at fixed intervals; in Section 5.2, we extend the analysis to the general multi-armed case with $|\mathcal{A}| = K > 2$. We discuss regret lower bounds in both the standard and batched settings and illustrate how the complexity scales with the number of arms; in Section 5.3, we further consider a structured action space $\mathcal{A} \subset \mathbb{R}^d$, where the expected reward takes a linear form $\mu(a) = \langle \theta^\star, a \rangle$ for an unknown parameter $\theta^\star$. We discuss minimax regret bounds under both full-feedback and batched feedback assumptions as well. Throughout, we emphasize how Le Cam's method can be tailored to derive tight lower bounds in each of these settings. These results provide insight into the inherent statistical difficulty of online decision-making and set performance benchmarks for algorithm design. We first summarize the notations used throughout this section below:

Table 6: Notation Used throughout Section 5

Symbol	**Description**
K	Number of arms in bandits problem
μ_a	Mean reward of arm a
$\theta \in \mathbb{R}^d$	Linear parameter vector for linear bandits
d	Dimension of linear parameter vector
$x_{t,a}$	Contextual vector of arm a at period t
η_t	Random noise of reward at period t
M	Number of batches in batched bandits problem
$\mathcal{T}$	Batched grid decision $(t_1, \cdots, t_M)$

5.1 Two-Armed Problems

We begin our discussion of online bandit problems with the simplest possible setting: the two-armed bandit. Despite its simplicity, this case already captures the fundamental difficulty of balancing exploration and exploitation in sequential decision-making. We begin our discussion of online bandit problems with the simplest possible setting: the two-armed bandit. Despite its simplicity, this case already captures the fundamental difficulty of balancing exploration and exploitation in sequential decision-making. Moreover, two-armed bandits frequently arise in operations management applications such as dynamic assortment between two products, A/B testing of service policies, or price experimentation between two levels. These simplified formulations provide not only tractable models for analysis, but also foundational building blocks for understanding regret in more complex learning systems.

Application 12: Stochastic Two-Armed Bandits

In the two-armed bandit setting, the learner interacts with an action set $\mathcal{A} = \{1, 2\}$. In each round $t \in [T]$, she selects an arm $a_t \in \mathcal{A}$, and receives a reward $r_{t,a_t} \sim \mathcal{N}(\mu_{a_t}, 1)$, where $\mu_1, \mu_2 \in [0, 1]$ are fixed but unknown means. All rewards are drawn independently across time and arms. The learner's policy $\pi = (a_1, \cdots, a_T)$ depends on the full history of previous actions and observed rewards. The regret of policy π under bandit environment $v \triangleq (\mu_1, \mu_2)$ in (38) is given by:

$$\mathcal{R}(T; \pi, v) \triangleq T \cdot \max\{\mu_1, \mu_2\} - \sum_{t=1}^{T} \mathbb{E}[\mu_{a_t}],$$

which quantifies the expected loss relative to always pulling the best arm.

Theorem 5.1. *For the two-armed bandits problem introduced before, the regret of any policy π can be lower bounded by:*

$$\sup_{v} \mathcal{R}(T; \pi, v) = \Omega(\sqrt{T}).$$

Proof of Theorem 5.1.
Step 1. Instance Construction. We construct two bandit environments $v_1 = (\Delta, 0)$ and $v_2 = (0, \Delta)$ for some small $\Delta > 0$ that will be determined later.

Step 2. Separation Verification. It is straightforward to see that in the bandit environment, for any decision a_t at period t, it cannot perform both well in the two environments v_1 and v_2, and there will be a reward loss of Δ in total.

Step 3. Applying Le Cam. We have for any policy π,

$$\begin{aligned}\sup_v \mathcal{R}(T;\pi,v) &\geq \frac{\Delta}{2}\cdot\sum_{t=1}^{T}\left(1-\|\mathbb{P}_{v_1}^{t-1}-\mathbb{P}_{v_2}^{t-1}\|_{\mathrm{TV}}\right)\\ &\geq \frac{\Delta}{4}\cdot\sum_{t=1}^{T}\exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{v_1}^{t-1}\|\mathbb{P}_{v_2}^{t-1})).\end{aligned}$$

We note that here $\mathbb{P}_{v_i}^{t-1}$ denotes the joint distribution over the random trajectory $\{(a_s,r_s)\}_{s=1}^{t-1}$, induced by sequentially applying the policy π in environment v_i, with each reward $r_s\sim\mathcal{N}(\mu_{a_s},1)$ conditional on the selected arm a_s.

Step 4. Calculating KL between Environments. By the conditional independence (on the history) property of the bandit environment, we have the following decomposition of KL-divergence:

$$\begin{aligned}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{v_1}^{t-1}\|\mathbb{P}_{v_2}^{t-1}) &= \mathbb{E}_{v_1}\left[\sum_{s=1}^{t-1}\log\frac{\mathrm{d}\mathbb{P}_{v_1}(r_s|a_s)}{\mathrm{d}\mathbb{P}_{v_2}(r_s|a_s)}\right]\\ &= (t-1)\cdot[\mathrm{D}_{\mathrm{KL}}(\mathcal{N}(\Delta,1)\|\mathcal{N}(0,1))+\mathrm{D}_{\mathrm{KL}}(\mathcal{N}(0,1)\|\mathcal{N}(\Delta,1))]\\ &\overset{(i)}{=} (t-1)\Delta^2 < T\Delta^2,\end{aligned}$$

where (i) uses the fact that $\mathrm{D}_{\mathrm{KL}}(\mathcal{N}(\Delta,1)\|\mathcal{N}(0,1))=\Delta^2/2$. Thus, by choosing $\Delta=\frac{1}{\sqrt{T}}$, we obtain the regret lower bound. □

Application 13: Batched Stochastic Two-Armed Bandits

In many practical settings, such as clinical trials, large-scale A/B testing, and adaptive experimentation, decisions cannot be updated after every individual observation. Instead, data is collected in batches, and policies are only updated at the end of each batch. This gives rise to the batched bandit setting (also known as the delayed reward setting), where learning is constrained by a pre-specified number of policy update points.

Formally, fix an integer $M\geq 2$ denoting the number of batches. An M-batch policy is a couple $(\mathcal{T},\pi)$ where $\mathcal{T}=\{t_1,t_2,\ldots,t_M\}$ is an increasing sequence of time grid such that $1<t_1<\cdots<t_M=T$, and $\pi=(a_1,\cdots,a_T)$ again denotes the learner's policy. The difference lies in that now each a_t depends only on observations from batches strictly prior to the current one.

Remark 5.1. In the batched setting, we again use $\mathbb{P}_v^{t-1}$ to denote the joint distribution over the action-reward sequence $\{(a_s,r_s)\}_{s=1}^{t-1}$ under environment v and policy π. However, due to the batching constraint, the action a_t is only allowed to depend on the reward collected up to the end of the previous batch-specifically, up to time t_{m-1} if $t\in[t_{m-1},t_m)$.

More precisely, this distribution is supported on the space $\left\{(a_s)_{s=1}^{t-1}, (r_s)_{s=1}^{t_{m-1}}\right\}$. Although actions $\{a_s\}_{s=1}^{t-1}$ are generated up to time $t-1$, the key constraint is that all actions in the current batch t_m must be based solely on feedback available by t_{m-1}. Thus, the effective randomness in the policy is fully determined by the joint distribution over $\{(a_s, r_s)\}_{s=1}^{t_{m-1}}$, and we may equivalently treat $\mathbb{P}_v^{t-1}$ as a distribution supported on this truncated history.

To extend the previous argument to the batched setting, we revisit the same pair of hard-to-distinguish instances, but now exploit the structural constraints of batch-based policies. The key observation is that decisions within each batch must be made simultaneously based on observations from previous batches only. This restricts the learner's ability to adaptively balance exploration and exploitation, and leads to a regret that grows faster than in the fully sequential case. The following Theorem restates the theorem 4 of Perchet et al. (2016):

Theorem 5.2. *Fix $T \geq 2$ and $M \in [T]$, the regret of any M-batch policy $(\mathcal{T}, \pi)$ can be lower bounded by:*

$$\sup_v \mathcal{R}(\mathcal{T}; \pi, v) = \Omega\left(T^{1/\left(2-2^{1-M}\right)}\right).$$

Proof of Theorem 5.2. We note that compared with the standard two-armed setting, the difference lies in calculating the KL between two environments, as we will show later in details.

Step 1. Instance Construction. The construction is the same as that in two-armed bandits setting: we let $v_1 = (\Delta, 0)$ and $v_2 = (0, \Delta)$ for some small $\Delta > 0$.

Step 2. Separation Verification. Note that the separation condition for two-armed bandits under v_1 and v_2 still holds with separation parameter Δ, as it is the single-period property which holds for any a_t.

Step 3. Applying Le Cam. Let $t_0 \triangleq 0$, we have for any M-batch policy $(\mathcal{T}, \pi)$:

$$\begin{aligned}
\sup_v \mathcal{R}(\mathcal{T}; \pi, v) &\geq \frac{\Delta}{2} \cdot \sum_{j=1}^{M} \sum_{t=t_{j-1}+1}^{t_j} \left(1 - \|\mathbb{P}_{v_1}^{t-1} - \mathbb{P}_{v_2}^{t-1}\|_{\mathrm{TV}}\right) \\
&\geq \frac{\Delta}{4} \cdot \sum_{j=1}^{M} \sum_{t=t_{j-1}+1}^{t_j} \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{v_1}^{t-1} \| \mathbb{P}_{v_2}^{t-1})).
\end{aligned}$$

Step 4. Calculating KL between Environments. The major analysis difference between two-armed bandits and its batched variant lies on calculating the KL-divergence between $\mathbb{P}_{v_1}^{t-1}$ and $\mathbb{P}_{v_2}^{t-1}$. Recall that in our Remark 5.1, we have emphasized that $\mathbb{P}_{v_1}^{t-1}$ and $\mathbb{P}_{v_2}^{t-1}$ is distributed on the observations up to the last batch (denoted as t_{j-1}). Thus, for any $t_{j-1} < t \leq t_j$, we have

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{v_1}^{t-1} \| \mathbb{P}_{v_2}^{t-1}) \leq t_{j-1} \cdot \Delta^2 \tag{39}$$

following the previous analysis. We choose a set of separation parameter $(\Delta_1, \cdots, \Delta_M)$ corresponds to each batch and let $\Delta_m = \frac{1}{\sqrt{t_{m-1}}}$ for $m \in [M]$. Recall that the total regret is lower bounded by the regret occurs in a single batch and by choosing the worst batch, i.e.,

$$\begin{aligned} \sup_v \mathcal{R}(\mathcal{T}; \pi, v) &\geq \max_m \sum_{j=1}^{M} \frac{\Delta_m (t_j - t_{j-1})}{4} \exp(-t_{j-1}\Delta_m^2/2) \\ &\geq \max_m \left\{ \frac{t_m}{4\sqrt{e t_{m-1}}} \right\}. \end{aligned} \tag{40}$$

Recall that $t_M = T$ and let $\frac{t_m}{\sqrt{t_{m-1}}} = c$ for constant $c > 0$, we therefore derive the lower bound (with detailed calculations left in Appendix D). □

Application 14: Dynamic Batched Two-Armed Bandits

In the previous subsection, we study the lower bound for batched two-armed bandits under a static grid, that is, the whole grid $\mathcal{T}$ is determined at t_0. Here we consider a more adaptive case where at the end of each batch $m \in [M]$, the decision-maker can also update its decision on the length of the next batch $t_{m+1} - t_m$. We refer to the setting as *dynamic batch learning*.

This setup naturally arises in various applications where decisions are grouped in rounds, and new batches are launched only after aggregating feedback from earlier ones. For instance, in clinical trials, patients are enrolled and treated in cohorts, and interim results are analyzed before adjusting the next phase of the trial. In crowdsourcing platforms, queries are issued in rounds and results arrive asynchronously, limiting the number of interaction rounds. Similarly, in marketing campaigns or simulation-based optimization, decisions are deployed in stages and revised dynamically based on the partial feedback.

Although dynamic grids offer more flexibility than static ones, the following result shows that the improvement is only marginal in the worst case. That is, the problem remains fundamentally constrained by the same trade-off between exploration and exploitation as in the static batch case.

Theorem 5.3. *Fix $T \geq 2$ and $M \in [T]$, the regret of any dynamic M-batch policy $(\mathcal{T}, \pi)$ can be lower bounded by:*

$$\sup_v \mathcal{R}(\mathcal{T}; \pi, v) = \Omega\left(M^{-2} T^{1/\left(2-2^{1-M}\right)}\right).$$

Proof of Theorem 5.3. Note that now we cannot conclude results similar to (40), as t_m is not fixed and based on the realization of observed rewards by t_{m-1}. Thus, we come up with a different idea: we first construct an "ideal" design of grid $\{T_1, \cdots, T_M\}$ and the

parameter distribution space depending on Δ, which are chosen that (i) the difference between expected rewards of two arms is Δ; and (ii) the number of observations up to T_{m-1} is too few for the decision maker to distinguish the two arms. Then, note that the true grid design $\{t_1, \cdots, t_M\}$ of policy π could be different from $\{T_1, \cdots, T_M\}$, and we define for each $m \in [M]$ the "bad" event $B_m = \{t_{m-1} \leq T_{m-1} < T_m \leq t_m\}$. We will show that if at least one B_m occurs with a sufficiently large probability, then we obtain the desired final regret lower bound, and the "if" is further verified by the simple pigeonhole principle.

Step 1. Instance Construction. We first construct the benchmark grid for each $m \in [M]$ through:

$$T_m = \left\lfloor T^{\frac{1-2^{-m}}{1-2^{-M}}} \right\rfloor,$$

and still set $v_1 = (\Delta, 0)$ and $v_2 = (0, \Delta)$ for some small $\Delta > 0$ that will be determined later. This specific construction of the benchmark grid $T_m{}_{m=1}^{M}$ follows the optimal batch allocation in the static batched two-armed bandit setting, as analyzed in the previous analysis. In particular, when the number of batches M is fixed in advance, the regret-minimizing choice of batch endpoints is non-uniform: later batches are allocated more samples to allow refined decisions after earlier exploration. Therefore, we adopt this construction as a benchmark when analyzing dynamic batch learning: any adaptively chosen grid should be compared against this optimal static grid to ensure the lower bound captures the fundamental hardness of the problem.

Step 2. Separation Verification. The single-period expected loss is the same as that of the static batched two-armed bandits with separation parameter $\Delta > 0$.

Step 3. Applying Le Cam. For any dynamic M-batched policy $(\mathcal{T}, \pi)$, by the separation condition, we have for any $m \in [M]$:

$$\begin{aligned}
\sup_{v} \mathcal{R}(\mathcal{T}; \pi, v) &\geq \frac{\Delta}{2} \sum_{t=T_{m-1}+1}^{T_m} \left(1 - \|\mathbb{P}_{v_1}^{t-1} - \mathbb{P}_{v_2}^{t-1}\|_{\mathrm{TV}}\right) \\
&\overset{(i)}{\geq} \frac{\Delta}{2} \sum_{t=T_{m-1}+1}^{T_m} \left(1 - \|\mathbb{P}_{v_1}^{T_m-1} - \mathbb{P}_{v_2}^{T_m-1}\|_{\mathrm{TV}}\right), \qquad (41)
\end{aligned}$$

with (i) comes from the data processing inequality.

Step 4. Calculating KL between Environments. For its total variation distance in (41), now we do not know how many batches are there by period $T_m - 1$, and (41) thus needs to be further lower bounded carefully. Recall the definition of "bad" event

$B_m = \{t_{m-1} \le T_{m-1} < T_m \le t_m\}$: we have

$$\begin{aligned}
1 - \|\mathbb{P}_{v_1}^{T_m-1} - \mathbb{P}_{v_2}^{T_m-1}\|_{\mathrm{TV}} &= \int \min\left(\mathrm{d}\mathbb{P}_{v_1}^{T_m-1}, \mathrm{d}\mathbb{P}_{v_2}^{T_m-1}\right) \\
&\ge \int_{B_m} \min\left(\mathrm{d}\mathbb{P}_{v_1}^{T_m-1}, \mathrm{d}\mathbb{P}_{v_2}^{T_m-1}\right) \\
&= \frac{1}{2}\int_{B_m} \left(\mathrm{d}\mathbb{P}_{v_1}^{T_m-1} + \mathrm{d}\mathbb{P}_{v_2}^{T_m-1} - |\mathrm{d}\mathbb{P}_{v_1}^{T_m-1} - \mathrm{d}\mathbb{P}_{v_2}^{T_m-1}|\right) \\
&\overset{(i)}{=} \frac{1}{2}\int_{B_m} \left(\mathrm{d}\mathbb{P}_{v_1}^{T_{m-1}} + \mathrm{d}\mathbb{P}_{v_2}^{T_{m-1}} - |\mathrm{d}\mathbb{P}_{v_1}^{T_{m-1}} - \mathrm{d}\mathbb{P}_{v_2}^{T_{m-1}}|\right) \\
&\overset{(ii)}{\ge} \mathbb{P}_{v_1}^{T_{m-1}}(B_m) - \frac{3}{2}\|\mathbb{P}_{v_1}^{T_{m-1}} - \mathbb{P}_{v_2}^{T_{m-1}}\|_{\mathrm{TV}} \\
&\overset{(iii)}{\ge} \mathbb{P}_{v_1}^{T_{m-1}}(B_m) - \frac{3}{2}\sqrt{\frac{1}{2}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{v_1}^{T_{m-1}}\|\mathbb{P}_{v_2}^{T_{m-1}})} \\
&\overset{(iv)}{\ge} \mathbb{P}_{v_1}^{T_{m-1}}(B_m) - \frac{3}{2}\sqrt{\frac{1}{2}T_{m-1}\Delta^2},
\end{aligned} \tag{42}$$

where (i) uses the fact that condition on B_m, $\mathrm{d}\mathbb{P}_{v_1}^{T_{m-1}} = \mathrm{d}\mathbb{P}_{v_2}^{T_m-1}$ (as we do not have any observations during the periods); (ii) uses the property that $\|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}} = \frac{1}{2}\int|\mathrm{d}\mathbb{P} - \mathrm{d}\mathbb{Q}|$ and $|\mathbb{P}(A) - \mathbb{Q}(A)| \le \|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}}$ for any random event A; (iii) uses the Pinsker's inequality; (iv) follows from (39).

At last, by the fact that $\sum_{m=1}^{M}\mathbb{P}(B_m) \ge \mathbb{P}\left(\cup_{m=1}^{M}B_m\right) = 1$ for any probability measure $\mathbb{P}$. Hence $\mathbb{P}(B_m) \ge 1/M$ for at least one m. We choose this specific m and let:

$$\Delta = \frac{\sqrt{2}}{3M}\cdot T^{-\frac{1-2^{1-m}}{2(1-2^{-M})}}.$$

Taking the result back to (42):

$$1 - \|\mathbb{P}_{v_1}^{T_m-1} - \mathbb{P}_{v_2}^{T_m-1}\|_{\mathrm{TV}} \ge \mathbb{P}_{v_1}^{T_{m-1}}(B_m) - \frac{3}{2}\sqrt{\frac{1}{2}T_{m-1}\Delta^2} \ge \frac{1}{2M}.$$

Combining the results together, we conclude the lower bound:

$$\sup_v \mathcal{R}(\mathcal{T};\pi,v) \ge \frac{\Delta_m(T_m - T_{m-1})}{4M} \ge \frac{\sqrt{2}}{12M^2}T^{1/\left(2-2^{1-M}\right)}.$$

□

5.2 Multi-Armed Problems

We now turn to the general multi-armed bandit (MAB) setting, a natural extension of the two-armed case that captures the core tension between exploration and exploitation

across a larger decision space. In the MAB model, the learner repeatedly selects from K different arms, each associated with an unknown reward distribution, and seeks to minimize cumulative regret. This setting not only generalizes the two-arm case but also reflects a wide range of operational applications where multiple competing options must be evaluated under uncertainty. The applications include adaptive pricing over a menu of price points, dynamic ad allocation, etc.

The multi-armed setting brings new analytical challenges compared to the two-armed case: as the number of arms K increases, the action space grows linearly, along with the difficulty of identifying the best arm due to the limited feedback and the need for careful exploration. Nevertheless, Le Cam's method remains a useful tool for deriving regret lower bounds in this setting. The applications we consider illustrate how pairwise indistinguishability arguments can still be leveraged to characterize the fundamental hardness of learning in multi-action environments.

Application 15: Stochastic Multi-Armed Bandits

In MAB, the learner interacts with an action set $\mathcal{A} = \{1, 2, \cdots, K\}$ consisting of $K > 2$ arms. At each round $t \in [T]$, she selects an arm $a_t \in \mathcal{A}$, and receives a reward $r_{t,a_t} \sim \mathcal{N}(\mu_{a_t}, 1)$, where the mean vector $\mu = (\mu_1, \cdots, \mu_K) \in [0,1]^K$ is fixed but unknown. All rewards are drawn independently across time and arms.

The learner's policy $\pi = (a_1, \cdots, a_T)$ is adapted to the history of previously selected arms and observed rewards. Again, we denote the bandit environment by the vector $v \triangleq (\mu_1, \ldots, \mu_K)$. The regret of policy π under environment v in (38) is now given by:

$$\mathcal{R}(T; \pi, v) \triangleq T \cdot \max_{a \in \mathcal{A}} \mu_a - \sum_{t=1}^{T} \mathbb{E}\left[\mu_{a_t}\right].$$

The following Theorem just restates the theorem 15.2 of Lattimore and Szepesvári (2020).

Theorem 5.4. *For the multi-armed bandits problem introduced before, the regret of any policy π can be lower bounded by:*

$$\sup_{v} \mathcal{R}(T; \pi, v) = \Omega(\sqrt{KT}).$$

Proof of Theorem 5.4.

Step 1. Instance Construction. We consider a policy-dependent construction scheme: for any fixed policy π, we first construct $v_1 = (\Delta, 0, \cdots, 0)$. Denote $T_k(t; \pi) = \sum_{s=1}^{t} \mathbf{1}(a_s = k)$ as the number of times arm $k \in \mathcal{A}$ was chosen by the learner after the end of round t under policy π. Let $i \in \mathcal{A}/\{1\}$ be the least pulled arm under policy π besides arm 1, since $\sum_{k=1}^{K} \mathbb{E}_v[T_k(T; \pi)] = T$ holds for any bandit environment v, we have $\mathbb{E}_v[T_i(T; \pi)] \leq T/(K-1)$ holds. Then, based on the policy π, we construct $v_2 = (\Delta, 0, \cdots, 2\Delta, 0, \cdots, 0)$, in which the arm i has reward distribution $\mathcal{N}(2\Delta, 1)$. Therefore, the optimal arm in v_1 is

arm 1 and the optimal arm in v_2 is arm $i \neq 1$.

Step 2. Separation Verification. It is easy to verify that the separation condition for multi-armed bandits under v_1 and v_2 still holds with separation parameter Δ.

Step 3. Applying Le Cam. For any policy π, we have

$$\begin{aligned}\sup_v \mathcal{R}(T;\pi,v) &\geq \frac{\Delta}{2}\cdot\sum_{t=1}^{T}\left(1-\|\mathbb{P}_{v_1}^{t-1}-\mathbb{P}_{v_2}^{t-1}\|_{\mathrm{TV}}\right)\\ &\geq \frac{T\Delta}{4}\cdot\exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{v_1}^{T}\|\mathbb{P}_{v_2}^{T})).\end{aligned} \tag{43}$$

Step 4. Calculating KL between Environments. Note that if we again apply the calculation process adopted in the previous two-armed bandits setting, we cannot derive the lower bound with dependence on K. We therefore introduce the following lemma on divergence decomposition, with its proof in Appendix D.

Lemma 5.1 (Divergence Decomposition of Bandit Model)**.** *Let $v = (\mathbb{P}_1, \cdots, \mathbb{P}_K)$ and $v' = (\mathbb{P}'_1, \cdots, \mathbb{P}'_K)$ be the reward distributions associated with two K-armed bandit problems. For any policy π:*

$$\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_v^T\|\mathbb{P}_{v'}^T\right)=\sum_{k=1}^{K}\mathbb{E}_v\left[T_k(T;\pi)\right]\cdot\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_k\|\mathbb{P}'_k\right).$$

According to Lemma 5.1, the total KL divergence of $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{v_1}^T\|\mathbb{P}_{v_2}^T)$ can be decomposed as the sum of KL between the reward distribution of each arm in the two environment times the expected number of times the arm is pulled under policy π. By the construction of v_1 and v_2:

$$\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_{v_1}^T\|\mathbb{P}_{v_2}^T\right)=\mathbb{E}_{v_1}[T_i(T;\pi)]\cdot\mathrm{D}_{\mathrm{KL}}(\mathcal{N}(0,1)\|\mathcal{N}(2\Delta,1))\leq\frac{2T\Delta^2}{K-1}.$$

Finally, by choosing $\Delta=\sqrt{\frac{K}{T}}$ and taking the results back to (43), we derive the lower bound (as $\frac{K}{K-1}<2$ for any $K>2$). □

5.3 Linear Bandits

The linear bandit problem extends the classical multi-armed bandit model by allowing actions to lie in a (contextual-dependent) continuous space, where rewards depend linearly on unknown parameters. This structure enables more efficient learning across actions but also introduces additional complexity in balancing exploration and exploitation. Following the structure of Section 5.1, we will begin with a classic linear bandit setting and a linear contextual bandit with finite arms. Then, we examine its batched variants under both static grid and dynamic grid.

Application 16: Stochastic Linear Bandits

In the linear bandit setting, the learner interacts with a continuous action set $\mathcal{A} = [-1, 1]^d$. At each round $t \in [T]$, she selects an action $a_t \in \mathcal{A}$, and observes a stochastic reward:

$$r_t = a_t^\top \theta + \eta_t,$$

where $\theta \in \mathbb{R}^d$ is an unknown parameter vector, and $\eta_t \sim \mathcal{N}(0, 1)$ is independent standard Gaussian noise. The reward model assumes that the expected reward is linear in the action, and the noise is homoscedastic and independent across time.

The learner's policy $\pi = (a_1, \cdots, a_T)$ depends on the full history of past actions and observed rewards. The environment is fully specified by the unknown vector θ,[2] and the regret of policy π under parameter (environment) θ in (38) is now defined as:

$$\mathcal{R}(T; \pi, \theta) \triangleq T \cdot \max_{a \in \mathcal{A}} a^\top \theta - \sum_{t=1}^{T} \mathbb{E}\left[a_t^\top \theta\right].$$

We restrict the unknown linear parameter θ to have bounded 2-norm. The following Theorem restates theorem 24.1 of Lattimore and Szepesvári (2020).

Theorem 5.5. *For the stochastic linear bandit problem defined above, the regret of any policy π can be lower bounded by:*

$$\sup_{\theta \in \mathbb{R}^d : \|\theta\|_2 \leq 1} \mathcal{R}(T; \pi, \theta) = \Omega(d\sqrt{T}).$$

Proof of Theorem 5.5. Unlike previous examples where the underlying parameter space was one-dimensional, the linear bandit setting introduces a d-dimensional parameter vector, making it insufficient to construct only two problem instances. To address this, we build on the insight developed in earlier proofs by decomposing the d-dimensional problem into d independent one-dimensional subproblems. For each coordinate, we construct two carefully perturbed instances that are statistically close yet lead to different optimal actions along that dimension. The full construction is then obtained by taking the product of these d binary choices, yielding a family of 2^d instances. Below, we detail this construction and use it to establish the desired regret lower bound.

Step 1. Instance Construction. We restrict the parameter space $\Theta = \{-\Delta, \Delta\}^d$ to be a hypercube for some small $\Delta > 0$ that will be determined later. Note that $|\Theta| = 2^d$, that is, now we have 2^d environments in total. For any fixed $i \in [d]$, we further choose two environments $\theta, \theta' \in \Theta$ such that the two vectors differ only in the i-th coordinate, i.e.,

[2] In the remaining part of this section, we will use θ instead of v to denote the bandit environment, as in the linear bandits, θ fully characterizes the environment.

$\theta_j = \theta'_j$ for $j \neq i$ and $\theta_i = -\theta'_i$.

Step 2. Separation Verification. Recall that at any period t, the optimal decision under any environment parameter θ is $(\text{sign}(\theta_1), \cdots, \text{sign}(\theta_d))$, where $\text{sign}(\theta_i) = 1$ for $\theta_i > 0$ and $\text{sign}(\theta_i) = -1$ for $\theta_i < 0$, for all $i \in [d]$.

Then, under the construction of (θ, θ'), any decision $a_t = (a_{t1}, \cdots, a_{td})$ will occur at least Δ reward loss (in a_{ti}) in one of the two environments. Thus, the separation condition holds with separation parameter Δ.

Step 3. Applying Information-Theoretical Tool. Define $p_{\theta i} \triangleq \mathbb{P}_\theta \left(\sum_{t=1}^T \mathbf{1} \left\{ \text{sign}\left(a_{ti}\right) \neq \text{sign}\left(\theta_i\right) \right\} \geq T/2 \right)$. Based on the separation property of the linear bandits problem under the construction, we have

$$\begin{aligned}
&\sup_{\theta \in \mathbb{R}^d : \|\theta\|_2 \leq 1} \mathcal{R}(T; \pi, \theta) \geq \frac{1}{2} \left\{ \mathbb{E}_\theta \left[\sum_{t=1}^T \sum_{i=1}^d \left(\text{sign}\left(\theta_i\right) - a_{ti} \right) \cdot \theta_i \right] + \mathbb{E}_{\theta'} \left[\sum_{t=1}^T \sum_{i=1}^d \left(\text{sign}\left(\theta'_i\right) - a_{ti} \right) \cdot \theta'_i \right] \right\} \\
&\geq \frac{\Delta}{2} \sum_{i=1}^d \left\{ \mathbb{E}_\theta \left[\sum_{t=1}^T \mathbf{1} \left\{ \text{sign}\left(a_{ti}\right) \neq \text{sign}\left(\theta_i\right) \right\} \right] + \mathbb{E}_{\theta'} \left[\sum_{t=1}^T \mathbf{1} \left\{ \text{sign}\left(a_{ti}\right) \neq \text{sign}\left(\theta'_i\right) \right\} \right] \right\} \\
&\overset{(i)}{\geq} \frac{T\Delta}{2} \sum_{i=1}^d \left(p_{\theta i} + p_{\theta' i} \right) \overset{(ii)}{\geq} \frac{T\Delta}{4} \sum_{i=1}^d \exp\left(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_\theta^T \| \mathbb{P}_{\theta'}^T) \right),
\end{aligned} \tag{44}$$

where (i) relies on the Markov inequality; (ii) bases on the fact that $\mathbf{1}\left\{\text{sign}\left(a_{ti}\right) \neq \text{sign}\left(\theta'_i\right)\right\} = \mathbf{1}\left\{\text{sign}\left(a_{ti}\right) = \text{sign}\left(\theta_i\right)\right\}$, and then we applies Bretagnolle–Huber inequality. $\mathbb{P}_\theta^T$ and $\mathbb{P}_{\theta'}^T$ denote the probability of joint observations up to period T under linear parameter θ and θ', respectively.

Step 4. Calculating KL between Environments. According to the Gaussian random reward assumption, we can conclude that

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_\theta^T \| \mathbb{P}_{\theta'}^T) = \frac{1}{2} \sum_{t=1}^T \mathbb{E}_\theta \left[a_t^\top (\theta - \theta') \right]^2 \leq T\Delta^2, \tag{45}$$

and the inequality holds since $a_t \subset [-1, 1]^d$. By letting $\Delta = \frac{1}{\sqrt{T}}$ and taking the results back to (44), we derive the regret lower bound. □

Note that under the same construction, through a slightly more involved analysis, the method can also be extended to prove the regret lower bound for linear bandits with action set $\mathcal{A} = \{\|a\|_2 \leq 1\}$ being a unit ball. Now the actions of the learner in one dimension constrain their choices in other dimensions, but we can still derive a lower bound of $\Omega(d\sqrt{T})$ following similar idea.

Application 17: Linear Contextual Bandits with Finitely Many Arms

We now consider the linear contextual bandit problem with finitely many actions, a widely studied model in sequential decision-making with side information. In this setting, each action is associated with a feature vector, and the expected reward of an action is linear in these features with respect to an unknown parameter vector. The learner's goal is to identify high-reward actions by leveraging contextual information, while balancing exploration and exploitation over time. The setting is formalize below.

Let T be the number of rounds and K the number of possible actions. Let $r_{t,a} \in [0,1]$ be the reward of action a on round t. On each round t for each action a the learner observes all the K feature vectors $x_{t,a} \in \mathbb{R}^d$, with (normalized) $\|x_{t,a}\|_2 \leq 1$. After observing the feature vectors the learner then selects an action a_t and receives reward r_{t,a_t}.

Assume that there exists an unknown weight vector $\theta \in \mathbb{R}^d$ with $\|\theta\|_2 \leq 1$ so that $\mathbb{E}\left[r_{t,a}|x_{t,a}\right] = x_{t,a}^\top \theta$. Let $x_{t,a}$ be the feature vector of action a at step t, and define the regret of policy $\pi = (a_1, \cdots, a_T)$ to be

$$\mathcal{R}(T;\pi,\theta) \triangleq \sum_{t=1}^{T} \left(\max_{a\in[K]} x_{t,a}^\top \theta - \mathbb{E}[x_{t,a_t}^\top \theta] \right),$$

where the expectation is taken w.r.t. all the past rewards which determine the random action a_t. Following theorem restates that in the theorem 2 of Chu et al. (2011).

Theorem 5.6. *For the linear contextual bandits with finitely many arms introduced before, assume $T \geq d^2$, the regret of any policy π under stochastic context can be lower bounded by:*

$$\sup_{\theta\in\mathbb{R}^d:\|\theta\|_2\leq 1} \mathbb{E}_x[\mathcal{R}(T;\pi,\theta)] = \Omega(\sqrt{dT}).$$

Proof of Theorem 5.6.

Step 1. Instance Construction. We can let $K = 2$ for simplicity (note that the lower bound indeed does not depend on K), and construct the space of linear coefficient vectors $\Theta = \{-\Delta, \Delta\}^d$ for some small $\Delta > 0$ that will be determined later (same construction as for linear bandits). For any fixed $i \in [d]$, the two environments $\theta, \theta' \in \Theta$ only varies in the i-th coordinate.

Now we also need to construct the distribution of feature as well: let $x_{t,1}$ be sampled uniformly from $\{\boldsymbol{e}_1, \cdots, \boldsymbol{e}_d\}$, where $\boldsymbol{e}_i$ denotes the vector that all except the i-th coordinate are 0, and the i-th coordinate is 1, for $i \in [d]$; and $x_{t,2} = \mathbf{0}$ (i.e., the zero vector).

Step 2. Separation Verification. Under our construction, the optimal arm when we observe $x_{t,1} = \boldsymbol{e}_i$ is arm 1 when $\theta_i = \Delta$ and arm 2 when $\theta_i = -\Delta$, which leads to a suboptimality gap of at least Δ.

Step 3. Applying Le Cam. First, we notice that the regret can be lower bounded by:

$$\sup_{\theta\in\mathbb{R}^d:\|\theta\|_2\le 1}\mathbb{E}_x[\mathcal{R}(T;\pi,\theta)]\ge\sum_{i=1}^{d}\frac{1}{d}\sum_{\theta\in\Theta}\frac{1}{|\Theta|}\max_{\theta^\star\in\{\theta,\theta'\}}\mathbb{E}_x[\mathcal{R}(T;\pi,\theta^\star)].$$

Following the pairwise decomposition, we now term our attention to $\max_{\theta^\star\in\{\theta,\theta'\}}\mathbb{E}_x[\mathcal{R}(T;\pi,\theta^\star)]$, where θ,θ' only differs in the i-th coordinate. Then, by applying Le Cam:

$$\begin{aligned}\max_{\theta^\star\in\{\theta,\theta'\}}\mathbb{E}_x[\mathcal{R}(T;\pi,\theta^\star)]&\ge\frac{\Delta}{2}\cdot\mathbb{E}_x\sum_{t=1}^{T}\left(1-\|\mathbb{P}_{\theta,x}^{t-1}-\mathbb{P}_{\theta',x}^{t-1}\|_{\mathrm{TV}}\right)\\&\ge\frac{T\Delta}{4}\cdot\mathbb{E}_x\exp\left(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\theta,x}^{T}\|\mathbb{P}_{\theta',x}^{T})\right)\\&\overset{(i)}{\ge}\frac{T\Delta}{4}\cdot\exp\left(-\mathbb{E}_x\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\theta,x}^{T}\|\mathbb{P}_{\theta',x}^{T})\right),\end{aligned}$$

where now $\mathbb{P}_{\theta,x}^{t-1}$ denotes the joint distribution over actions and rewards up to time $t-1$, conditional on the context x, bandit environment θ and policy π; (i) follows the Jensen's inequality.

Step 4. Calculating KL between Environments. Following (45), we have

$$\mathbb{E}_x[\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\theta,x}^{T}\|\mathbb{P}_{\theta',x}^{T})]=\frac{1}{d}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{\theta,x=\boldsymbol{e}_i}^{T}\|\mathbb{P}_{\theta',x=\boldsymbol{e}_i}^{T})\le\frac{T\Delta^2}{d},$$

with the first equality holds according to the construction that θ and θ' only differs in the i-th coordinate, and only when the observation of $x=\boldsymbol{e}_i$ will contribute to the KL-divergence (otherwise the KL is 0). By choosing $\Delta=\sqrt{d/T}$ (here we use the assumption $T\ge d^2$ to ensure $\|\theta\|_2\le 1$), we derive the lower bound:

$$\sup_{\theta\in\mathbb{R}^d:\|\theta\|_2\le 1}\mathbb{E}_x[\mathcal{R}(T;\pi,\theta)]\ge\frac{\sqrt{dT}}{8}\cdot\exp(-1).$$

□

Application 18: Batched Linear Contextual Bandits

Consider the batched variant of the linear bandit problem and we restrict the number of actions $K=2$ (for simplicity) and let $T\ge d^2$.

Now, the decision maker is only allowed to partition the T units into (at most) M batches, and the reward corresponding to each unit in a batch can only be observed at the end of the batch. More specifically, given a maximum batch size M, the decision maker needs to choose a sequential batch learning algorithm that has the following two

components: a grid $\mathcal{T} = \{t_1, \cdots, t_M\}$ and a sequential batch policy $\pi = (a_1, \cdots, a_T)$ as that in the batched two-armed bandits. Let $(\mathcal{T}, \pi)$ be a sequential batch learning algorithm. The definition of regret is the same as that introduced in the previous subsection:

$$\mathcal{R}(\mathcal{T}; \pi, \theta) \triangleq \sum_{t=1}^{T} \left(\max_{i \in \{1,2\}} x_{t,i}^\top \theta - \mathbb{E}[x_{t,a_t}^\top \theta] \right).$$

Remark 5.2. In the batched linear contextual bandit setting, we let $\mathbb{P}_{\theta,x}^{t-1}$ denote the joint distribution over actions and rewards up to time $t-1$, conditional on a fixed sequence of contexts $x = \{(x_{s,1}, x_{s,2})\}_{s=1}^T$, under bandit environment θ and policy π. This makes explicit the fact that the contextual sequence is stochastic and sampled in advance from the environment, while the rewards are generated based on θ and the learner's actions. Again, the effective support of $\mathbb{P}_{\theta,x}^{t-1}$ lies in the space $\left\{ (a_s)_{s=1}^{t-1}, (x_{s,1}, x_{s,2})_{s=1}^{t-1}, (r_s)_{s=1}^{t_{m-1}} \right\}$, and this highlights a difference from the batched two-armed bandits setting: here, the action selection at time t depends not only on observed rewards but also on full context vectors.

Theorem 5.7 (Theorem 2 of Han et al. (2020))**.** *Fix $K = 2$, for any $T \geq d^2$ and $M \in [T]$, the regret of any M-batch policy $(\mathcal{T}, \pi)$ under stochastic context can be lower bounded by:*

$$\sup_{\theta: \|\theta\|_2 \leq 1} \mathbb{E}_x \left[\mathcal{R}(\mathcal{T}; \pi, \theta) \right] = \Omega \left(\sqrt{dT} \left(\frac{T}{d^2} \right)^{\frac{1}{2(2^M - 1)}} \right).$$

Proof of Theorem 5.7. Recall that in the proof for the batched two-armed bandits, a key step is to decompose the total regret into M batches as in (40). Similarly, for the batched linear setting, we first establish the following key lemma.

Lemma 5.2. *For any fixed grid $0 = t_0 < t_1 < \cdots < t_M = T$ and any $\Delta \in [0, 1]$, the following minimax lower bound holds for any policy π under this grid:*

$$\sup_{\theta: \|\theta\|_2 \leq 1} \mathbb{E}_x \left[\mathcal{R}(\mathcal{T}; \pi, \theta) \right] \geq \Delta \cdot \sum_{m=1}^{M} \frac{t_m - t_{m-1}}{10\sqrt{d}} \exp\left(-\frac{16 t_{m-1} \Delta^2}{d^2} \right) \tag{46}$$

Step 1. Instance Construction. We will now assume that θ is sampled from a distribution, that is, $\theta \sim Q_0 = \text{Unif}\left(\Delta \mathbb{S}^{d-1}\right)$ uniformly distributes on the d-dimensional sphere centered at the origin with some small radius $\Delta > 0$. Let the contexts $x_{t,a} \sim \mathcal{N}(0, I_d/d)$ be independent over all $a \in [K]$, $t \in [T]$. We have:

$$\sup_{\theta: \|\theta^\star\|_2 \leq 1} \mathbb{E}_x \left[\mathcal{R}(\mathcal{T}; \pi, \theta) \right] \geq \sum_{t=1}^{T} \mathbb{E}_x \mathbb{E}_\theta \left(\mathbb{E} \left[\max_{i \in \{1,2\}} (x_{t,i} - x_{t,a_t})^\top \theta \right] \right).$$

Step 2. Separation Verification. Let $\langle x_{t,1} - x_{t,2}, \theta \rangle_+ \triangleq \max\{(x_{t,1} - x_{t,2})^\top \theta, 0\}$ and $\langle x_{t,1} - x_{t,2}, \theta \rangle_- \triangleq \max\{-(x_{t,1} - x_{t,2})^\top \theta, 0\}$. For each $t \in [T]$, we have

$\max_{i\in\{1,2\}} (x_{t,i} - x_{t,a_t})^\top \theta = \langle x_{t,1} - x_{t,2}, \theta\rangle_+ \cdot \mathbf{1}(a_t = 2) + \langle x_{t,1} - x_{t,2}, \theta\rangle_- \cdot \mathbf{1}(a_t = 1)$. Since we have infinitely many instances in this construction, the standard separation condition does not apply. Instead, we consider the single-period expected loss over random vector x, θ and the joint rewards and action distribution under θ and x:

$$\begin{aligned}
&\mathbb{E}_x \mathbb{E}_\theta \mathbb{E}_{\mathbb{P}^t_{\theta,x}} \left[\max_{i\in\{1,2\}} (x_{t,i} - x_{t,a_t})^\top \theta \right] \\
&= \mathbb{E}_x \mathbb{E}_\theta \left[\langle x_{t,1} - x_{t,2}, \theta\rangle_+ \cdot \mathbb{P}^t_{\theta,x}(a_t = 2) + \langle x_{t,1} - x_{t,2}, \theta\rangle_- \cdot \mathbb{P}^t_{\theta,x}(a_t = 1) \right] \\
&\overset{(i)}{=} \mathbb{E}_x Z_0 \cdot \left(\underbrace{\mathbb{E}_{Q_1} \mathbb{P}^t_{\theta,x}(a_t = 2) + \mathbb{E}_{Q_2} \mathbb{P}^t_{\theta,x}(a_t = 1)}_{\text{(a)}} \right),
\end{aligned}$$

where in (i), we define two new probability distributions of θ via

$$\frac{\mathrm{d}Q_1}{\mathrm{d}Q_0}(\theta) = \frac{\langle x_{t,1} - x_{t,2}, \theta\rangle_+}{Z_0}, \quad \frac{\mathrm{d}Q_2}{\mathrm{d}Q_0}(\theta) = \frac{\langle x_{t,1} - x_{t,2}, \theta\rangle_-}{Z_0},$$

with Z_0 denoting the common normalization factor, and $\mathbb{E}_{Q_i} \mathbb{P}^t_{\theta,x}$ denotes the mixture distribution of $z \sim \mathbb{P}^t_{\theta,x}$ where $\theta \sim Q_i$ for $i \in \{1, 2\}$. Let $x_{t,1} - x_{t,2} = r_t u_t$ with $r_t \geq 0$, and $\|u_t\|_2 = 1$ denote the direction vector of $x_{t,1} - x_{t,2}$. We note that Q_1, Q_2 and Z_0 enjoys some good properties as presented in Appendix D, Lemma D.1. Later, we will directly use these properties, with details left behind.

Note that here Z_0 can be interpreted as the separation parameter under the two instances $\mathbb{E}_{Q_1} \mathbb{P}^t_{\theta,x}$ and $\mathbb{E}_{Q_2} \mathbb{P}^t_{\theta,x}$. Z_0 is a function of context realized at period t, and according to Lemma D.1, we have $Z_0 \geq \frac{r_t \Delta}{5\sqrt{d}}$.

Step 3. Applying Information-Theoretical Tool. We have:

$$\begin{aligned}
\text{(a)} &\geq 1 - \|\mathbb{E}_{Q_1} \mathbb{P}^t_{\theta,x} - \mathbb{E}_{Q_2} \mathbb{P}^t_{\theta,x}\|_{\mathrm{TV}} \\
&\overset{(i)}{\geq} \frac{1}{2} \exp\left(-\mathrm{D}_{\mathrm{KL}} \left(\mathbb{E}_{Q_1} \mathbb{P}^t_{\theta,x} \| \mathbb{E}_{Q_2} \mathbb{P}^t_{\theta,x} \right) \right) \\
&\overset{(ii)}{=} \frac{1}{2} \exp\left(-\mathrm{D}_{\mathrm{KL}} \left(\mathbb{E}_{Q_1} \mathbb{P}^t_{\theta,x} \| \mathbb{E}_{Q_1} \mathbb{P}^t_{\theta - 2\left(u_t^\top \theta\right) u_t, x} \right) \right) \\
&\overset{(iii)}{\geq} \frac{1}{2} \exp\left(-\mathbb{E}_{Q_1} \mathrm{D}_{\mathrm{KL}} \left(\mathbb{P}^t_{\theta,x} \| \mathbb{P}^t_{\theta - 2\left(u_t^\top \theta\right) u_t, x} \right) \right),
\end{aligned}$$

where (i) follows from the Bretagnolle–Huber inequality; (ii) is according to the property of Q_1 and Q_2 such that $\theta \sim Q_1$ if and only if $\theta - 2\left(u_t^\top \theta\right) u_t \sim Q_2$ in Lemma D.1; (iii) is due to the joint convexity of the KL divergence.

Step 4. Calculating KL between Environments. For $t \in (t_{m-1}, t_m]$, the learner can only observe rewards up to time t_{m-1}, therefore:

$$\begin{aligned}
\mathbb{E}_{Q_1}\left[\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}^t_{\theta,x} \| \mathbb{P}^t_{\theta-2\left(u_t^\top \theta\right)u_t,x}\right)\right] &= \frac{1}{2}\mathbb{E}_{Q_1}\left[\sum_{i=1}^{2}\sum_{\tau=1}^{t_{m-1}}\left[x_{\tau,i}^\top\left[\theta-\left(\theta-2\left(u_t^\top\theta\right)u_t\right)\right]\right]^2\right] \\
&= 2\mathbb{E}_{Q_1}\left[\sum_{i=1}^{2}\sum_{\tau=1}^{t_{m-1}}\left(u_t^\top\theta\right)^2\left(u_t^\top x_{\tau,i}\right)^2\right] \\
&\overset{(i)}{=} \frac{4\Delta^2}{d+1}\sum_{i=1}^{2}\sum_{\tau=1}^{t_{m-1}}\left(u_t^\top x_{\tau,i}\right)^2,
\end{aligned}$$

where (i) is by $\mathbb{E}_{Q_1}[(u_t^\top\theta)^2] = \frac{2\Delta^2}{d+1}$ (see Lemma D.1). Combining the results together, and using the independence of $\{x_{\tau,i}\}_{\tau<t}$ and (r_t, u_t), the convexity of $x \mapsto \exp(-x)$ and $\mathbb{E}\left[x_{\tau,i}x_{\tau,i}^\top\right] = I_d/d$, we arrive at:

$$\begin{aligned}
\mathbb{E}_x\mathbb{E}_\theta\mathbb{E}_{\mathbb{P}^t_{\theta,x}}\left[\max_{i\in\{1,2\}}\left(x_{t,i}-x_{t,a_t}\right)^\top\theta\right] &\geq \frac{\mathbb{E}\left[r_t\right]\Delta}{10\sqrt{d}}\exp\left(-\frac{8\Delta^2 t_{m-1}}{d(d+1)}\right) \\
&\overset{(i)}{\geq} \frac{\Delta}{10\sqrt{d}}\exp\left(-\frac{16\Delta^2 t_{m-1}}{d^2}\right),
\end{aligned}$$

where (i) comes from

$$\mathbb{E}\left[r_t\right] = \mathbb{E}\left\|x_{t,1}-x_{t,2}\right\|_2 \geq \frac{\mathbb{E}\left\|x_{t,1}-x_{t,2}\right\|_1}{\sqrt{d}} = \frac{2}{\sqrt{\pi}} > 1.$$

Summing the single-period cost from $t = 1$ to T and by its batched structure, we derive Lemma 46.

Finally, for any grid $\mathcal{T} = \{t_1, \cdots, t_M\}$, define $s = \min\{m \in [M] : t_m \geq d^2\}$, which always exists due to our assumption that $T \geq d^2$. Choosing some candidates of $\Delta \in \left\{1, \frac{d}{\sqrt{t_s}}, \frac{d}{\sqrt{t_{s+1}}}, \cdots, \frac{d}{\sqrt{T}}\right\} \subset [0,1]$ in Lemma 46 gives

$$\sup_{\theta:\|\theta\|_2\leq 1}\mathbb{E}\left[\mathcal{R}(\mathcal{T};\pi,\theta)\right] \geq c\cdot\max\left\{\frac{t_s}{\sqrt{d}}, t_{s+1}\sqrt{\frac{d}{t_s}}, t_{s+2}\sqrt{\frac{d}{t_{s+1}}}, \cdots, T\sqrt{\frac{d}{t_{M-1}}}\right\}$$

Then, by setting $t_s = \frac{t_m}{\sqrt{t_{m-1}}} = c, \forall\ m > s$ and applying the same analysis as in Theorem 5.2 (see Appendix D), we conclude:

$$\sup_{\theta:\|\theta\|_2\leq 1}\mathbb{E}\left[\mathcal{R}(\mathcal{T};\pi,\theta^\star)\right] \geq c\sqrt{dT}\cdot\left(\frac{T}{d^2}\right)^{\frac{1}{2(2^{M-s+1}-1)}} \geq c\sqrt{dT}\cdot\left(\frac{T}{d^2}\right)^{\frac{1}{2(2^M-1)}}.$$

□

Application 19: Dynamic Batched Linear Contextual Bandits

Extending our previous discussion on dynamic batched learning, we now consider its application in the linear contextual bandit setting. Similar to the two-armed case, dynamic batching allows the learner to adaptively determine the lengths of future batches based on information gathered in earlier rounds. This flexibility can potentially lead to better performance compared to static batching, as it enables more efficient allocation of exploration effort.

In the context of linear contextual bandits, however, the challenge is amplified by the high-dimensional nature of the decision space. Each action is associated with a context vector in $\mathbb{R}^d$, and the learner must infer the unknown parameter vector θ from noisy linear rewards. The goal is to balance exploration across different directions in the feature space while making informed decisions in each batch.

The lower bound result we present below shows that: even with the added flexibility of dynamic batching, the regret remains fundamentally constrained by the statistical difficulty of learning θ under limited feedback. The proof follows the same high-level structure as in the dynamic batched two-armed case but requires careful decomposition of the parameter space and instance construction along each coordinate, as we have previously discussed in the static batched linear bandit setting. The following Theorem restates theorem 1 of Ren and Zhou (2024).

Theorem 5.8. *Fix $K = 2$, for any $T \geq d^2$ and $M \in [T]$, the regret of any dynamic batch learning policy $(\mathcal{T}, \pi)$ under stochastic context can be lower bounded by:*

$$\sup_{\theta:\|\theta\|_2\leq 1} \mathbb{E}_x \left[\mathcal{R}(\mathcal{T}; \pi, \theta)\right] = \Omega\left(\max\left(M^{-4}2^{-M}\sqrt{dT}\left(\frac{T}{d^2}\right)^{\frac{1}{2(2^M-1)}}, \sqrt{dT}\right)\right).$$

Proof of Theorem 5.8. Now we utilize both the idea in the proof of Theorem 5.3 and 5.7 by introducing the "ideal" design of grid $\{T_1, \cdots, T_M\}$ and the "bad" event $B_m = \{t_{m-1} \leq T_{m-1} < T_m \leq t_m\}$ for each $m \in [M]$ as well, with a more involved analysis borrowing from the static batched linear bandits:

Step 1. Instance Construction. We still assume $\theta \sim Q_0 = \text{Unif}\left(\Delta_m \mathbb{S}^{d-1}\right)$ be uniformly distributed on the d-dimensional sphere centered at the origin with some radius $\Delta > 0$ during each batch $T_{m-1} \leq t < T_m$ for any $m \in [M]$. Specifically, we construct the benchmark grid for each $m \in [M]$ through:

$$T_m = \left\lfloor d^2 \cdot \left(\frac{T}{d^2}\right)^{\frac{1-2^{-m}}{1-2^{-M}}} \right\rfloor.$$

Let the contexts $x_{t,a} \sim \mathcal{N}(0, I_d/d)$ be independent over all $a \in [K]$, $t \in [T]$. We suppose $M = O(\log\log(T/d^2))$ (if not, we simply adopt $\Omega(\sqrt{dT})$ for linear contextual bandits).

Step 2. Separation Verification. The single-period expected loss is the same as that of the static batched linear contextual bandits, that is,

$$\mathbb{E}_x\mathbb{E}_\theta\mathbb{E}_{\mathbb{P}^t_{\theta,x}}\left[\max_{i\in\{1,2\}}\left(x_{t,i}-x_{t,a_t}\right)^\top\theta\right]=\mathbb{E}_xZ_0\left[\mathbb{E}_{Q_1}\mathbb{P}^t_{\theta,x}\left(a_t=2\right)+\mathbb{E}_{Q_2}\mathbb{P}^t_{\theta,x}\left(a_t=1\right)\right],$$

with Z_0, Q_1 and Q_2 as defined in the proof of Theorem 5.7. Next, we introduce the following lemma by the "bad" events B_m we've constructed before, with its proof left in Appendix D. Note that the idea is similar to that in the proof for the dynamic batched two-armed bandits, with a more involved analysis.

Lemma 5.3. *If there exists $m\in[M]$, such that*

$$\sum_{t=T_{m-1}+1}^{T_m}\mathbb{E}_x\left[Z_0\cdot\mathbb{E}_{Q_1}\mathbb{P}^t_{\theta,x}(B_m)\right]\geq\frac{\Delta\cdot(T_m-T_{m-1})}{4\cdot2^{\frac{M}{2}}M^2\sqrt{d}},$$

then for $\Delta=\frac{1}{96\cdot M^2\cdot2^{M/2}}\cdot\left(\frac{T}{d^2}\right)^{-\frac{1-2^{1-m}}{2(1-2^{-M})}}$, we have

$$\sup_{\theta:\|\theta\|_2\leq1}\mathbb{E}_x\left[\mathcal{R}(\mathcal{T};\pi,\theta)\right]=\Omega\left(M^{-4}2^{-M}\sqrt{dT}\left(\frac{T}{d^2}\right)^{\frac{1}{2(2^M-1)}}\right).$$

Step 3. Applying Information-Theoretical Tool. We have:

$$\begin{aligned}\sup_{\theta:\|\theta\|_2\leq1}\mathbb{E}_x\left[\mathcal{R}(\mathcal{T};\pi,\theta)\right]&\geq\sum_{t=T_{m-1}+1}^{T_m}\mathbb{E}_xZ_0\left(\mathbb{E}_{Q_1}\mathbb{P}^t_{\theta,x}\left(a_t=2\right)+\mathbb{E}_{Q_2}\mathbb{P}^t_{\theta,x}\left(a_t=1\right)\right)\\&\geq\sum_{t=T_{m-1}+1}^{T_m}\mathbb{E}_xZ_0\cdot(1-\|\mathbb{E}_{Q_1}\mathbb{P}^{T_m}_{\theta,x}-\mathbb{E}_{Q_2}\mathbb{P}^{T_m}_{\theta,x}\|_{\mathrm{TV}}).\end{aligned}$$

For its total variation distance, following the analysis in (41), we have

$$\begin{aligned}&1-\|\mathbb{E}_{Q_1}\mathbb{P}^{T_m}_{\theta,x}-\mathbb{E}_{Q_2}\mathbb{P}^{T_m}_{\theta,x}\|_{\mathrm{TV}}=\int\min\left(\mathrm{d}\mathbb{E}_{Q_1}\mathbb{P}^{T_m}_{\theta,x},\mathrm{d}\mathbb{E}_{Q_2}\mathbb{P}^{T_m}_{\theta,x}\right)\\&\geq\int_{B_m}\min\left(\mathrm{d}\mathbb{E}_{Q_1}\mathbb{P}^{T_m}_{\theta,x},\mathrm{d}\mathbb{E}_{Q_2}\mathbb{P}^{T_m}_{\theta,x}\right)\\&=\frac{1}{2}\int_{B_m}\left(\mathrm{d}\mathbb{E}_{Q_1}\mathbb{P}^{T_m}_{\theta,x}+\mathrm{d}\mathbb{E}_{Q_2}\mathbb{P}^{T_m}_{\theta,x}-|\mathrm{d}\mathbb{E}_{Q_1}\mathbb{P}^{T_m}_{\theta,x}-\mathrm{d}\mathbb{E}_{Q_2}\mathbb{P}^{T_m}_{\theta,x}|\right)\\&=\frac{1}{2}\int_{B_m}\left(\mathrm{d}\mathbb{E}_{Q_1}\mathbb{P}^{T_{m-1}}_{\theta,x}+\mathrm{d}\mathbb{E}_{Q_2}\mathbb{P}^{T_{m-1}}_{\theta,x}-|\mathrm{d}\mathbb{E}_{Q_1}\mathbb{P}^{T_{m-1}}_{\theta,x}-\mathrm{d}\mathbb{E}_{Q_2}\mathbb{P}^{T_{m-1}}_{\theta,x}|\right)\\&\geq\mathbb{E}_{Q_1}\mathbb{P}^{T_{m-1}}_{\theta,x}(B_m)-\frac{3}{2}\|\mathbb{E}_{Q_1}\mathbb{P}^{T_{m-1}}_{\theta,x}-\mathbb{E}_{Q_2}\mathbb{P}^{T_{m-1}}_{\theta,x}\|_{\mathrm{TV}}\\&\geq\mathbb{E}_{Q_1}\mathbb{P}^{T_{m-1}}_{\theta,x}(B_m)-\frac{3}{2}\sqrt{\frac{1}{2}\mathrm{D}_{\mathrm{KL}}(\mathbb{E}_{Q_1}\mathbb{P}^{T_{m-1}}_{\theta,x}\|\mathbb{E}_{Q_2}\mathbb{P}^{T_{m-1}}_{\theta,x})}.\end{aligned}$$

Step 4. Calculating KL between Environments. The calculation of KL-divergence directly follows that in the static batched linear contextual bandits. We omit the details for simplicity. Finally:

$$\begin{aligned}
\sup_{\theta:\|\theta\|_2\le 1} \mathbb{E}_\theta\left[\mathcal{R}(\mathcal{T};\pi,\theta)\right] &\ge \sum_{t=T_{m-1}+1}^{T_m} \mathbb{E}_x Z_0 \cdot \left(\mathbb{E}_{Q_1}\mathbb{P}_{\theta,x}^{T_{m-1}}(B_m) - \frac{3}{2}\sqrt{\frac{1}{2}\mathrm{D}_{\mathrm{KL}}\left(\mathbb{E}_{Q_1}\mathbb{P}_{\theta,x}^{T_{m-1}} \| \mathbb{E}_{Q_2}\mathbb{P}_{\theta,x}^{T_{m-1}}\right)}\right) \\
&\ge \sum_{t=T_{m-1}+1}^{T_m} \mathbb{E}_x Z_0 \cdot \left(\mathbb{E}_{Q_1}\mathbb{P}_{\theta,x}^{T_{m-1}}(B_m) - \frac{3}{2}\sqrt{\frac{4\Delta^2 T_{m-1}}{d^2}}\right) \\
&\overset{(i)}{\ge} \sum_{t=T_{m-1}+1}^{T_m} \mathbb{E}_x Z_0 \cdot \left(\mathbb{E}_{Q_1}\mathbb{P}_{\theta,x}^{T_{m-1}}(B_m) - \frac{1}{32\cdot 2^{\frac{M}{2}} M^2}\right) \\
&\overset{(ii)}{\ge} \sum_{t=T_{m-1}+1}^{T_m} \left[\mathbb{E}_x\left(Z_0 \cdot \mathbb{E}_{Q_1}\mathbb{P}_{\theta,x}^{T_{m-1}}(B_m)\right) - \frac{\Delta}{8\cdot 2^{\frac{M}{2}} M^2\sqrt{d}}\right],
\end{aligned}$$

where (i) follows from the choice of T_m and Δ (in Lemma 5.3); (ii) follows from the fact that $\mathbb{E}_x Z_0 \le 4\Delta/\sqrt{d}$. Then, by choosing the m to be that satisfying Lemma 5.3, we can derive the regret lower bound. □

5.4 Summary and Bibliographic Remarks

This section surveys regret lower bounds in a range of online bandit settings, from classical unstructured models to contextual and high-dimensional extensions. Table 7 summarizes the main results across these problem classes.

Table 7: Regret Lower Bounds in Online Bandits Decision Making

	Problem Setting	**Lower Bound**
App.12	Two-Armed	$\Omega(\sqrt{T})$
App.13	Static Batched; Two-Armed	$\Omega\left(T^{1/(2-2^{1-M})}\right)$
App.14	Dynamic Batched; Two-Armed	$\Omega\left(T^{1/(2-2^{1-M})}\right)$
App.15	K-Armed	$\Omega\left(\sqrt{KT}\right)$
App.16	Linear	$\Omega(d\sqrt{T})$
App.17	Linear Contextual (with finite arms)	$\Omega(\sqrt{dT})$
App.18	Static Batched; Linear Contextual	$\Omega\left(\sqrt{dT}\left(\frac{T}{d^2}\right)^{\frac{1}{2(2^M-1)}}\right)$
App.19	Dynamic Batched; Linear Contextual	$\Omega\left(\sqrt{dT}\left(\frac{T}{d^2}\right)^{\frac{1}{2(2^M-1)}}\right)$

Online decision making with bandit feedback has been extensively studied in the literature, see Dani et al. (2008), Rigollet and Zeevi (2010), Abbasi-Yadkori et al. (2011) and Kim et al. (2021) for some related research and Lattimore and Szepesvári (2020) for a comprehensive tutorial. Here we note several natural extensions that are not included in our survey but are worth mentioning.

- For the multi-armed bandits: although our discussion in this section covers batched settings for two-armed and linear contextual bandits, we do not include regret lower bounds for batched K-armed bandits. The main reason is methodological: the best-known results for this problem, such as the work by Gao et al. (2019), rely on Fano's inequality rather than Le Cam's method. Their approach involves constructing a packing of reward vectors and bounding the mutual information between the learner's observations and the unknown instance. This style of argument does not naturally translate to a two-point reduction as in Le Cam's method, and whether similar results can be derived under the Le Cam framework remains an open question. Besides, Esfandiari et al. (2021) also study the batched adversarial multi-armed bandit problem.

- For contextual bandits with general function classes: Our survey primarily focuses on linear contextual bandits, but a growing body of work has extended regret lower bounds to more general reward models. For example, Ren et al. (2020) consider the static batched learning in generalized linear contextual bandits with general decision space (i.e., infinite arms), which extends Application 16 to a batched and general linear feedback structure. Combining the lower bound proofing idea introduced before, they provide a regret lower bound of $\Omega\left(d\sqrt{T}\left(\frac{T}{d^2}\right)^{\frac{1}{2(2^M-1)}}\right)$.

Finally, we note that the tools and insights (pairwise Le Cam) developed here can also be translated to another widely applied information-theoretical tool named the Assouad's lemma, which we will later discuss in the subsequent section to see that it can also inform regret lower bounds in offline bandits learning problem.

6 Offline Policy Learning with Bandits Feedback

In contrast to the online bandit setting discussed in the previous section where decisions are made sequentially and feedback is revealed after each (batched) action, offline policy learning deals with the problem of selecting a decision rule from a fixed dataset, collected beforehand under some specific policies. This setting arises naturally in high-stakes domains such as healthcare, education, or public policy, where online experimentation may be costly or ethically infeasible, and decisions must instead be based on historical data.

The central question in offline policy learning is: *How accurately can we learn a good decision policy given only a finite set of historical data*? A key difficulty is that the data may be biased or selectively observed, especially when it was generated by an adaptive or stochastic logging policy. As a result, even evaluating a candidate policy can be statistically challenging, let alone optimizing over a policy class.

In this section, we study lower bounds for offline policy learning under several representative scenarios. Our goal is to characterize the fundamental limits of policy learning under various assumptions on the data collection process-ranging from i.i.d. data with two actions, to general multi-action contextual bandits with adaptively collected data. In each case, we apply Le Cam's two-point method to identify hard instances that limit the performance of any offline learner.

To make these lower bounds meaningful, we must also specify the complexity of the policy class from which the learner selects. For this, we will first introduce complexity measures such as the *VC dimension* (for binary action spaces), and the *Natarajan dimension* (for multi-action settings). These complexity measures capture the richness of the policy class and play a central role in determining the statistical hardness of the learning problem. Intuitively, the more complex the policy class, the harder it is to distinguish between competing policies using finite data, and hence the worse the minimax regret or estimation error.

Definition 6.1 (VC-dimension)**.** Given a class $\mathcal{F}$ of binary-valued functions, we say that the set $S(n) = (x_1, \ldots, x_n)$ is shattered by $\mathcal{F}$ if $\text{card}(\mathcal{F}(S(n))) = 2^n$. The VC dimension $v(\mathcal{F})$ is the largest integer n for which there is some collection $S(n) = (x_1, \ldots, x_n)$ of n points that is shattered by $\mathcal{F}$.

Definition 6.2 (Natarajan dimension)**.** Given a K-action policy class Π, we say a set of m points $\{x_1, x_2, \cdots, x_m\}$ is shattered by Π if there exist two functions $f_{-1}, f_1 : \{x_1, x_2, \cdots, x_m\} \mapsto [K]$ such that,

(a) for any $j \in [m]$, $f_{-1}(x_j) \neq f_1(x_j)$;

(b) for any $\sigma = (\sigma_1, \sigma_2, \cdots, \sigma_m) \in \{\pm 1\}^m$, there exists a policy $\pi \in \Pi$ such that for any $j \in [m]$,

$$\pi(x_j) = \begin{cases} f_{-1}(x_j) & \text{if } \sigma_j = -1 \\ f_1(x_j) & \text{if } \sigma_j = 1 \end{cases}$$

The Natarajan dimension of Π is defined to be the size of the largest set of points that can be shattered by Π.

At last, we summarize the notations tailored for Section 6, with its detailed definitions introduced later in the corresponding sections.

Table 8: Notation Used throughout Section 6

Symbol	Description
$x \in \mathcal{X}$	Feature vector
$y(a)$	Potential outcome under action a
$e(x; a)$	Propensity score: probability of assigning action a given x
κ	lower bound of propensity score
$Q(\pi)$	Expected reward under policy π
$\mathcal{U}_{\mathbb{P}_0}(\delta)$	Distributional uncertainty set around $\mathbb{P}_0$ with radius δ
v	VC-dimension of two-action policy class Π
d	Natarajan dimension of multi-action policy class Π

6.1 Treatment Choice for Welfare Maximization

Offline policy learning under binary treatment assignment serves as a foundational setting for understanding statistical limits in welfare-maximizing decision-making (Kitagawa and Tetenov, 2018). In this subsection, we examine how the minimax regret lower bound for welfare maximization using observational data can be constructed. Note that the problem falls into the category of the two-action policy learning under i.i.d. data.

Application 20: Binary Treatment Policy Learning

We consider a population indexed by a covariate space $\mathcal{X} \subset \mathbb{R}^{d_x}$. Each individual i is associated with $x_i \in \mathcal{X}$: the observed pre-treatment covariates; $a_i \in \{0, 1\}$: the treatment assignment indicator; and $y_i \in \mathbb{R}$: post-treatment observed outcome.

The decision-maker has access to a dataset of N i.i.d. samples $z_i = (y_i, a_i, x_i)$, drawn from distribution characterized by $\mathbb{P}$, a joint distribution of $(y_{0,i}, y_{1,i}, a_i, x_i)$, where $y_{0,i}$ and $y_{1,i}$ are potential outcomes that would have been observed if i's treatment status were $a_i = 0$ and $a_i = 1$, respectively. The decision-maker's goal is to determine a *treatment rule* from the historical data that maximizes expected social welfare, which is indexed by a policy $\pi : \mathcal{X} \to \{0, 1\}$ that determines whether the individuals should be treated ($a = 1$) or not ($a = 0$). We let Π denote a policy class consisting of such policy π.

The goal of our analysis is to empirically select a treatment assignment policy that gives the highest welfare to the target population. Define the social welfare $W(\pi)$ and the

conditional social welfare $W(\pi; x)$ given covariate x as follows:

$$W(\pi) \triangleq \mathbb{E}\left[y(\pi(x))\right], \tag{47}$$

$$W(\pi; x) \triangleq \mathbb{E}\left[y(\pi(x)) \mid x\right], \tag{48}$$

where $y(\pi(x))$ stands for the outcome under action $\pi(x)$ (i.e., y_0 for $\pi(x) = 0$ and y_1 for $\pi(x) = 1$), and the expectation in (47) and (48) is taken w.r.t. the joint distribution of (y_0, y_1, x) and $y(\pi(x))$ given x, respectively. The optimal achievable welfare within the policy class Π is given by:

$$W_\Pi^* \triangleq \sup_{\pi \in \Pi} W(\pi).$$

The average welfare loss (regret) under treatment choice rule $\hat{\pi}$ as a function of $(z_1, \cdots, z_N)$ is thus defined as:

$$\mathcal{R}(N; \hat{\pi}, \mathbb{P}) \triangleq W_\Pi^* - \mathbb{E}_{\mathbb{P}^N}[W(\hat{\pi})] = \mathbb{E}_{\mathbb{P}^N}\left[W_\Pi^* - W(\hat{\pi})\right] \geq 0,$$

where the expectation $\mathbb{E}_{\mathbb{P}^N}$ is taken over different realizations of the random sample $(z_1, \cdots, z_N)$. In order to construct the lower bound, we first make the following assumptions on the problem instance: (i) unconfoundedness: conditional on feature x, treatment assignment is independent of the potential outcomes; (ii) bounded outcome: $y \in [0, M]$ for some $M > 0$; (iii) bounded assignment probability: $e(x; 1) \triangleq \mathbb{P}(a = 1|x)$ satisfies $e(x) \in [\kappa, 1 - \kappa]$ for all $x \in \mathcal{X}$ for some $\kappa \in (0, 1/2)$; (iv) VC-class: the decision sets Π has finite VC-dimension $v < \infty$. We denote the distribution class satisfying these assumptions as $\mathcal{P}(M, \kappa)$. The following Theorem restates the theorem 2.2 of Kitagawa and Tetenov (2018).

Theorem 6.1. *For any treatment policy $\hat{\pi}$, the regret can be lower bounded by:*

$$\sup_{\mathbb{P} \in \mathcal{P}(M,\kappa)} \mathcal{R}(N; \hat{\pi}, \mathbb{P}) = \Omega\left(M\sqrt{\frac{v}{N\kappa}}\right) \quad \textit{for all } N \geq 3v/\kappa.$$

Proof of Theorem 6.1.
Step 1. Instance Construction. Since Π has a finite VC-dimension v, there exist v points $\{x_1, \cdots, x_v\} \subset \mathcal{X}$ shattered by Π. We first construct the marginal distribution $\mathbb{P}_x$ to place equal mass $1/v$ on these v points. Treatment assignments are independent of all other variables, with $a \sim \text{Bern}(\kappa)$ independent of x.

Then, we construct 2^v different instances indexed by bit vectors $\mathbf{b} \in \{0, 1\}^v$. For each $j = 1, \cdots, v$, the conditional distribution of y_1 and y_0 given $x = x_j$ is:

$$y_0 \sim M \cdot \text{Bern}\left(\frac{1}{2}\right), \; y_1 \sim M \cdot \text{Bern}\left(\frac{1 + (2b_j - 1) \cdot \Delta}{2}\right), \tag{49}$$

for some small $\Delta > 0$ that will be determined later, and b_j is the j-th element of $\mathbf{b}$. Note that although we have constructed 2^v instances, we will later compare these instances **pair-wisely**, following Le Cam's routine.

Step 2. Separation Verification. According to our construction, for any given x_j, we further use $b(x_j) \in \{0, 1\}$ to denote the optimal treatment given x_j under distribution $\mathbb{P}_{\mathbf{b}}$ and $\hat{b}(x_j)$ to denote the treatment rule under any $\hat{\pi}$. In addition, we denote $M_j(\mathbf{b})$ as the vector that defers from $\mathbf{b}$ only in element j.

Then, for any two instances $\mathbf{b}$ and $M_j(\mathbf{b})$ given x_j, we have the optimal treatment decisions are different: for $b_j = 1$, it is optimal to treat in $\mathbf{b}$ and do not treat in $M_j(\mathbf{b})$, and vice versa. Thus, the separation condition holds with separation parameter $M\Delta$.

Step 3. Applying Le Cam. We can apply Le Cam pair-wisely for $\mathbf{b}$ and $M_j(\mathbf{b})$ given that $x = x_j$. Let $\pi^\star_{\mathbf{b}}$ denote the optimal policy under $\mathbb{P}_{\mathbf{b}}$ (clearly, given $x = x_j$, the optimal policy is to treat for $b_j = 1$ and do not treat for $b_j = 0$). To make it more clear on how the Le Cam is used, we present the full details as follows:

$$
\begin{aligned}
\sup_{\mathbb{P}\in\mathcal{P}(M,\kappa)} \mathcal{R}(N;\hat{\pi},\mathbb{P}) &\geq \sup_{\mathbb{P}_{\mathbf{b}}\in\mathcal{P}^*} \mathbb{E}_{\mathbb{P}^N_{\mathbf{b}}}\left[W\left(G^*_{\mathbf{b}}\right) - W(\hat{\pi})\right] \geq \frac{1}{2^v}\sum_{\boldsymbol{b}\in\{0,1\}^v} \mathbb{E}_{\mathbb{P}^N_{\mathbf{b}}}\left[W\left(\pi^\star_{\mathbf{b}}\right) - W(\hat{\pi})\right] \\
&= \frac{1}{2^v}\sum_{\mathbf{b}\in\{0,1\}^v} \frac{1}{v}\sum_{j=1}^{v} \mathbb{E}_{\mathbb{P}^N_{\mathbf{b}}}\left[W\left(\pi^\star_{\mathbf{b}}; x_j\right) - W(\hat{\pi}; x_j)\right] \\
&= \frac{M\Delta}{v2^{v+1}}\sum_{\mathbf{b}\in\{0,1\}^v}\sum_{j=1}^{v} \mathbb{P}^N_{\mathbf{b}}\left(\hat{b}(x_j) \neq b(x_j)\right) \\
&= \frac{M\Delta}{v2^{v+1}}\sum_{j=1}^{v}\sum_{\mathbf{b}:b_j=1}\left(\mathbb{P}^N_{\mathbf{b}}\left(\hat{b}(x_j) \neq 1\right) + \mathbb{P}^N_{M_j(\mathbf{b})}\left(\hat{b}(x_j) \neq 0\right)\right) \qquad (50) \\
&= \frac{M\Delta}{v2^{v+1}}\sum_{j=1}^{v}\sum_{\mathbf{b}:b_j=1}\left(\mathbb{P}^N_{\mathbf{b}}\left(\hat{b}(x_j) = 1\right) + 1 - \mathbb{P}^N_{M_j(\mathbf{b})}\left(\hat{b}(x_j) = 0\right)\right) \qquad (51) \\
&\geq \frac{M\Delta}{v2^{v+1}}\sum_{j=1}^{v}\sum_{\mathbf{b}:b_j=1}\left(1 - \|\mathbb{P}^N_{\mathbf{b}} - \mathbb{P}^N_{M_j(\mathbf{b})}\|_{\mathrm{TV}}\right) \\
&\geq \frac{M\Delta}{v2^{v+2}}\sum_{j=1}^{v}\sum_{\mathbf{b}:b_j=1}\exp\left(-\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}^N_{\mathbf{b}}\|\mathbb{P}^N_{M_j(\mathbf{b})}\right)\right). \qquad (52)
\end{aligned}
$$

Step 4. Calculating KL between Environments. The KL-divergence in (52) given

$b_j = 1$ can be further calculated through:

$$\begin{aligned}
\mathrm{D_{KL}}\left(\mathbb{P}_{\mathbf{b}}^N \| \mathbb{P}_{M_j(\mathbf{b})}^N\right) &= \frac{N}{v}\sum_{i=1}^{v}\mathbb{E}_{\mathbf{b}}\left[\log\frac{\mathrm{d}\mathbb{P}_{\mathbf{b}}(y,a,x)}{\mathrm{d}\mathbb{P}_{M_j(\mathbf{b})}(y,a,x)}\middle| x = x_i\right] \\
&= \frac{N}{v}\mathbb{E}_{\mathbf{b}}\left[\log\frac{\mathrm{d}\mathbb{P}_{\mathbf{b}}(y,a,x)}{\mathrm{d}\mathbb{P}_{M_j(\mathbf{b})}(y,a,x)}\middle| x = x_j\right] \\
&\overset{(i)}{=} \frac{N}{v}\mathbb{E}_{\mathbf{b}}\left[\mathbf{1}\{a=1\}\cdot\Delta\log\left(\frac{1+\Delta}{1-\Delta}\right)\middle| x = x_j\right] \\
&\leq \frac{3N\Delta^2}{v}\mathbb{P}(a=1|x=x_j) = \frac{3\kappa N\Delta^2}{v}.
\end{aligned} \tag{53}$$

Here, as the data $(z_1, \cdots, z_n)$ is i.i.d. generated, we omit the subscript i in $z_i = (y_i, a_i, x_i)$ and abbreviate it as $z = (y, a, x)$ to denote the observation. (i) is due to our construction that for the observations with $a = 0$, the two instances are identical (i.e., y_0 has the same distribution). By choosing $\Delta = \sqrt{\frac{v}{\kappa N}}$, we derive the lower bound in Theorem 6.1. □

6.2 Multi-action Policy Learning

In the previous subsection, we discuss the offline policy learning in the binary treatment setting, where the decision-maker chooses between two actions: treatment or control, for each individual based on their covariates. The minimax regret was shown to scale as $\Omega(\sqrt{v/N\kappa})$, with v the VC-dimension of the class of binary treatment rules. The proof used Le Cam's idea to construct a family of indistinguishable environments with distinct optimal actions, each corresponding to a binary labeling over a shattered covariate set.

We now generalize this framework to the *multi-action setting*, where the decision-maker must choose among $K > 2$ possible actions for each individual. This generalization captures a broader range of applications, including offline dosing, treatment assignment with multiple discrete options, etc. In this setting, the complexity of the policy class is captured not by the VC-dimension, but by its Natarajan dimension, which extends the notion of shattering to multi-class prediction.

Despite the added complexity, the fundamental idea for deriving lower bounds remains consistent: constructing pairs of distributions that are difficult to distinguish yet have different optimal policies. In the following sections, we formalize the multi-action offline policy learning problem, present the corresponding minimax lower bound, and discuss how Le Cam's two-point method can be adapted to accommodate the multi-action setting. We start with the i.i.d. data setting.

Application 21: Multi-action Policy Learning under i.i.d. Data

We apply the same notation as that in Section 6.1, with the difference in that now the action $a_i \in \{1, 2, \cdots, K\}$ takes K distinct values sampled from a multinomial distribution with probability $(e(x;1), \cdots, e(x;K))$; next the decision-maker receives a response y_i for each individual i from the chosen action generated via the model:

$$y_i = \mu(x_i; a_i) + \eta_i,$$

where $\{\eta_i\}_{i=1}^N$ is a sequence of i.i.d. random noise with zero-mean and σ^2-variance.

A policy π is a mapping from the covariate space to the action space, i.e., $\pi : \mathcal{X} \mapsto \mathcal{A}$. Given a policy π, we are interested in the expected reward incurred by taking actions according to π:

$$Q(\pi) \triangleq \mathbb{E}_x[\mu(x; \pi(x))].$$

For a class of policies Π, the optimal policy $\pi^\star \in \Pi$ is defined through

$$\pi^\star = \text{argmax}_{\pi \in \Pi} Q(\pi),$$

and we aim to learn a data-driven policy $\hat{\pi} \in \Pi$ from the given N i.i.d. samples $z_i = (y_i, a_i, x_i)$, drawn from distribution characterized by $\mathbb{P}$ as well. The regret of a policy $\hat{\pi}$ is defined through

$$\mathcal{R}(N; \hat{\pi}, \mathbb{P}) \triangleq Q(\pi^\star) - \mathbb{E}_{\mathbb{P}^N}[Q(\hat{\pi})],$$

which is a straightforward extension to that of the welfare maximization problem: when we choose a suboptimal action, we will incur an additional loss.

To characterize the fundamental limit of the offline policy learning problem, we establish a lower bound for the worst-case expected regret. The lower bound is stated in terms of the Natarajan dimension of a multi-action policy class, and again we introduce assumptions on the problem instance parallel to the two-action treatment setting: (i) bounded outcome with $|y_i| \leq M$ and $|\mu(x; a)| \leq M$ for all $(x, a) \in \mathcal{X} \times \mathcal{A}$; (ii) bounded assignment probability with $e(x; a) \geq \kappa$ for all $(x, a) \in \mathcal{X} \times \mathcal{A}$; and (iii) now the policy class Π has a finite Natarajan dimension $d < \infty$. We still denote the distribution class satisfying these assumptions as $\mathcal{P}(M, \kappa)$.

Theorem 6.2. *For any multi-action learning policy $\hat{\pi} \in \Pi$ under i.i.d. data, the regret can be lower bounded by:*

$$\sup_{\mathbb{P} \in \mathcal{P}(M,\kappa)} \mathcal{R}(N; \hat{\pi}, \mathbb{P}) = \Omega\left(M\sqrt{\frac{d}{N\kappa}}\right) \quad \textit{for all } N \geq 3d/\kappa.$$

Proof of Theorem 6.2.

Step 1. Instance Construction. We apply a similar construction as that in proving Theorem 6.1. Again, we can find d points $\{x_1, \cdots, x_d\} \subset \mathcal{X}$ shattered by Π, i.e., there

exist two functions f_0, $f_1 : \{x_1, \cdots, x_d\} \mapsto [K]$, such that $f_0(x_j) \neq f_1(x_j)$ for any $j \in [d]$, and for any $\mathbf{b} \in \{0,1\}^d$, there exists a policy $\pi \in \Pi$ such that $\pi(x_j) = f_{b_j}(x_j)$. Restrict the marginal distribution $\mathbb{P}_x$ to place equal mass $1/d$ on these d points. Conditional on $x = x_j$, a is chosen to be $f_1(x_j)$ w.p. κ and other decisions w.p. $(1-\kappa)/(K-1)$, the distribution of reward y under each decision is constructed as follows:

$$
\begin{aligned}
&y_{f_1(x_j)} \sim M \cdot \text{Bern}\left(\frac{1 + (2b_j - 1) \cdot \Delta}{2}\right), \\
&y_{f_0(x_j)} \sim M \cdot \text{Bern}\left(\frac{1}{2}\right), \\
&y_k = 0, \text{ for any } k \in [K] \backslash \{f_1(x_j), f_0(x_j)\},
\end{aligned}
$$

for some small $\Delta > 0$ that will be determined later.

Step 2. Separation Verification. For any given x_j, it is clear that by our construction, either $f_1(x_j)$ or $f_0(x_j)$ must be the optimal action, and the remaining action is the second best. Thus, the separation condition still holds follow the same analysis with separation parameter $M\Delta$ as well.

Step 3. Applying Le Cam. Note that all the analysis are the same as that in Theorem 6.1 when we are applying Le Cam by replacing the VC dimension v with Natarajan dimension d, except that now in (50) and (51), the actions should be $f_1(x_j)$ and $f_0(x_j)$ instead of 1 and 0, respectively. Also, it should be a "greater than or equal to" instead of equality when we move from (50) to (51) in the multi-action case. At last, we arrive at:

$$
\sup_{\mathbb{P} \in \mathcal{P}(M,\kappa)} \mathcal{R}(N; \hat{\pi}, \mathbb{P}) \geq \frac{M\Delta}{d2^{d+2}} \sum_{j=1}^{d} \sum_{\mathbf{b}: b_j = 1} \exp\left(-\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_{\mathbf{b}}^N \| \mathbb{P}_{M_j(\mathbf{b})}^N\right)\right).
$$

Step 4. Calculating KL between Environments. Now the KL-divergence can be upper bounded through:

$$
\begin{aligned}
\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_{\mathbf{b}}^N \| \mathbb{P}_{M_j(\mathbf{b})}^N\right) &= \frac{N}{d} \mathbb{E}_{\mathbf{b}}\left[\mathbf{1}\{a = f_1(x_j)\} \cdot \Delta \log\left(\frac{1+\Delta}{1-\Delta}\right) \middle| x = x_j\right] \\
&\leq \frac{3N\Delta^2}{d} \mathbb{P}(a = f_1(x_j) | x = x_j) = \frac{3\kappa N \Delta^2}{d}.
\end{aligned}
$$

By choosing $\Delta = \sqrt{\frac{d}{N\kappa}}$, we can derive the lower bound. □

Application 22: Multi-action Policy Learning under Adaptively Data

In the previous sections, we examined the offline policy learning problem under the assumption of i.i.d. data. This setting presumes that the data collection process remains

static throughout the experiment, with each action being assigned independently of past observations. However, in many real-world applications, data is collected adaptively, where the assignment of actions evolves based on accumulated information. This adaptive data collection introduces dependencies among samples and poses unique challenges for policy learning.

Almost all the settings and notations are the same as those in Section 6.2, except the following point: to emphasize the sequential and adaptive nature of data collection in such settings, we index the collected samples using $t = 1, 2, \cdots, T$, where T represents the total number of time steps or collected samples instead of using N as before. At each time step t, the experimenter observes a covariate x_t, selects an action a_t based on a policy that may depend on the history up to time $t-1$, and then observes an outcome y_t. The history up to time t is denoted by $\mathcal{H}_t = \{(x_s, a_s, y_s)\}_{s=1}^{t}$, and based on $\mathcal{H}_t$, the action a_{t+1} taken at period $t+1$ is sampled from the multinomial distribution with probability $\left(e_{t+1}\left(x_{t+1}; 1\right), \cdots, e_{t+1}\left(x_{t+1}; K\right)\right)$. All the assumptions in Application 21 remain unchanged, except that we make a slight modification on the assignment probability assumption to accommodate the adaptive data as follows: now $e_t(x; a)$ is lower bounded with probability one, over the sampling of $\{(x_s, a_s, y_s)\}_{s=1}^{t-1}$, by a positive and non-increasing deterministic sequence $\{\kappa_t\}$ for all $(x, a) \in \mathcal{X} \times \mathcal{A}$ and all $t \in [T]$. The following Theorem restates the theorem 1 of Zhan et al. (2024).

Theorem 6.3. *For any multi-action learning policy $\hat{\pi} \in \Pi$ under adaptive data, the regret can be lower bounded by:*

$$\sup_{\mathbb{P} \in \mathcal{P}(M,\kappa)} \mathcal{R}(T; \hat{\pi}, \mathbb{P}) = \Omega\left(M\sqrt{\frac{d}{\sum_{t=1}^{T} \kappa_t}}\right) \quad \textit{for all } T \geq 3d/\kappa.$$

Proof of Theorem 6.3.

Step 1. Instance Construction. Building upon the previous construction, we observe that the conditional probability of selecting a particular action only influences the analysis when we computing the KL-divergence between two instances. Therefore, we proceed to construct the distributions of x and y in the same manner as before, and at time period t, a_t is chosen to be $f_1(x_j)$ w.p. κ_t and other decisions w.p. $(1 - \kappa_t)/(K-1)$.

Step 2. Separation Verification. We still have the separation parameter $M\Delta$.

Step 3. Applying Le Cam. Everything remains the same to that under the i.i.d. data, and we have:

$$\sup_{\mathbb{P} \in \mathcal{P}(M,\kappa)} \mathcal{R}(N; \hat{G}, \mathbb{P}) \geq \frac{M\Delta}{d2^{d+2}} \sum_{j=1}^{d} \sum_{\mathbf{b}: b_j = 1} \exp\left(-\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_{\mathbf{b}}^{T} \| \mathbb{P}_{M_j(\mathbf{b})}^{T}\right)\right).$$

Step 4. Calculating KL between Environments. Reexamine the KL-divergence in (53): based on the chain rule of the KL-divergence, we can still derive the upper bound of the KL-divergence between the two worlds $\mathbb{P}_{\mathbf{b}}^T$ and $\mathbb{P}_{M_j(\mathbf{b})}^T$, the joint distribution over the sequence of T observations generated under $\mathbb{P}_{\mathbf{b}}$ and $\mathbb{P}_{M_j(\mathbf{b})}$, respectively, as follows.

$$\begin{aligned}
\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_{\mathbf{b}}^T \| \mathbb{P}_{M_j(\mathbf{b})}^T\right) &= \mathbb{E}_{\mathbf{b}}\left[\log \frac{\mathrm{d}\mathbb{P}_{\mathbf{b}}\left(x_1, a_1, y_1, \cdots, x_T, a_T, y_T\right)}{\mathrm{d}\mathbb{P}_{M_j(\mathbf{b})}\left(x_1, a_1, y_1, \cdots, x_T, a_T, y_T\right)}\right] \\
&= \mathbb{E}_{\mathbf{b}}\left[\sum_{t=1}^{T} \log \frac{\mathrm{d}\mathbb{P}_{\mathbf{b}}\left(y_t | a_t, x_t\right)}{\mathrm{d}\mathbb{P}_{M_j(\mathbf{b})}\left(y_t | a_t, x_t\right)}\right] \\
&= \sum_{t=1}^{T} \frac{1}{d} \sum_{i=1}^{v} \mathbb{E}_{\mathbf{b}}\left[\log \frac{\mathrm{d}\mathbb{P}_{\mathbf{b}}\left(y_t | a_t, x_t\right)}{\mathrm{d}\mathbb{P}_{M_j(\mathbf{b})}\left(y_t | a_t, x_t\right)} \middle| x_t = x_i\right] \\
&= \sum_{t=1}^{T} \frac{1}{d} \mathbb{E}_{\mathbf{b}}\left[\log \frac{\mathrm{d}\mathbb{P}_{\mathbf{b}}\left(y_t | a_t, x_t\right)}{\mathrm{d}\mathbb{P}_{M_j(\mathbf{b})}\left(y_t | a_t, x_t\right)} \middle| x_t = x_j\right] \\
&= \sum_{t=1}^{T} \frac{1}{d} \mathbb{E}_{\mathbf{b}}\left[\mathbf{1}\left\{a_t = f_1(x_j)\right\} \cdot \Delta \log\left(\frac{1+\Delta}{1-\Delta}\right) \middle| x_t = x_j\right] \\
&\leq \frac{3\Delta^2}{d} \sum_{t=1}^{T} \mathbb{P}(A_t = f_1(x_j) | x_t = x_j) = \frac{3\Delta^2}{d} \sum_{t=1}^{T} \kappa_t.
\end{aligned}$$

By choosing $\Delta = \sqrt{\frac{d}{\sum_{t=1}^{T} \kappa_t}}$, we can derive the lower bound. □

6.3 Distributionally Robust Policy Learning

In many high-stakes decision-making contexts, the historical data may not faithfully reflect the true underlying environment due to selection bias, unmeasured confounding, or misspecified policies. To address such distributional mismatch, a growing line of work adopts distributionally robust policy learning, where the goal is to learn policies that perform well under worst-case perturbations of the data-generating distribution. This perspective offers stronger generalization guarantees, particularly when models are misspecified or data is collected from heterogeneous sources. In this section, we study regret lower bounds for robust policy learning, focusing on the binary action case where the policy is chosen to guard against adversarial shifts within a KL divergence ball centered at the nominal distribution. Our goal is to understand how distributional robustness affects the fundamental difficulty of offline learning.

Application 23: Distributionally Robust Policy Learning

In contrast to the previous offline policy learning setting, this application considers a distributionally robust policy learning variant. The training data (x_i, a_i, y_i) is i.i.d. generated following a joint distribution $\mathbb{P}_0$, with the same assumptions as before (unconfoundedness, overlap, and bounded outcomes). For simplicity, we will focus on the two-action setting and we still assume the policy class has a finite VC-dimension $v < \infty$. Specifically, we denote $e_0(a; x_i)$ as the selection probability of action $a \in \{0, 1\}$ given feature x_i. Thus we express the distribution of observations as $\{(x_i, a_i, y_i)\}_{i=1}^N \sim (\mathbb{P}_0^{e_0})^N$, with $\mathbb{P}_0^{e_0} \in \mathcal{P}(M, \kappa)$ as well.

The key difference lies in the objective: instead of maximizing the expected welfare (reward) under a fixed distribution $\mathbb{P}_0$, we seek to ensure good performance under worst-case perturbations of $\mathbb{P}_0$ within a KL-divergence ball:

$$\mathcal{U}_{\mathbb{P}_0}(\delta) \triangleq \{\mathbb{P} \ll \mathbb{P}_0 \mid \mathrm{D}_{\mathrm{KL}}(\mathbb{P} \| \mathbb{P}_0) \leq \delta\}.$$

Note that the uncertainty set $\mathcal{U}_{\mathbb{P}_0}(\delta)$ is defined only over the outcome distribution $\mathbb{P}_0$, not the behavior policy e_0, which governs how actions were assigned in the data. This distinction is why we explicitly factor the data-generating distribution as $\mathbb{P}_0^{e_0}$: the robustness applies solely to $\mathbb{P}_0$, while e_0 is treated as known and fixed throughout (which only appears in the training data set).

For any policy π, we define its distributionally robust reward as:

$$Q_{\mathrm{DRO}}(\pi) \triangleq \inf_{\mathbb{P} \in \mathcal{U}_{\mathbb{P}_0}(\delta)} \mathbb{E}_{\mathbb{P}}[y(\pi(x))].$$

Optimizing this robust value allows the decision-maker to hedge against distributional misspecification: even when the true data-generating process differs from $\mathbb{P}_0$, the policy still achieves strong performance within the prescribed uncertainty set.

To facilitate analysis, we introduce the following lemma on the distribution robust expression of the expected reward.

Lemma 6.1 (Strong Duality)**.** *For any policy π, we have*

$$\inf_{\mathbb{P} \in \mathcal{U}_{\mathbb{P}_0}(\delta)} \mathbb{E}_{\mathbb{P}}[y(\pi(x))] = \sup_{\alpha \geq 0} \{-\alpha \log \mathbb{E}_{\mathbb{P}_0}[\exp(-y(\pi(x))/\alpha)] - \alpha\delta\}.$$

Given the historical data set, for any data-driven policy $\hat{\pi}$, its regret defined as:

$$\mathcal{R}_{\mathrm{DRO}}(N; \hat{\pi}, \mathbb{P}_0) \triangleq \mathbb{E}_{(\mathbb{P}_0^{e_0})^N}\left[\max_{\pi' \in \Pi} Q_{\mathrm{DRO}}(\pi') - Q_{\mathrm{DRO}}(\hat{\pi})\right].$$

Following Theorem restates that of theorem 3 in Si et al. (2023).

Theorem 6.4. *Let $\delta \leq 0.226$. For any data-driven distributionally robust policy $\hat{\pi} \in \Pi$, the regret can be lower bounded by*

$$\sup_{\mathbb{P}_0^{e_0} \in \mathcal{P}(M,\kappa)} \mathcal{R}_{\mathrm{DRO}}\left(N; \hat{\pi}, \mathbb{P}_0\right) = \Omega\left(M\sqrt{\frac{v}{N\kappa}}\right), \text{ for } N \geq 6.25v/\kappa.$$

Proof of Theorem 6.4.

Step 1. Instance Construction. Let $x_1, \cdots, x_v$ be v points that are shattered by policy Π. Let $\mathbf{b} = \{b_1, \cdots, b_v\} \in \{0,1\}^v$. By definition, for each $\mathbf{b} \in \{0,1\}^v$, there exist a $\pi \in \Pi$ such that $\pi(x_i) = b_i$ for $i = 1, \cdots, v$. We use $\pi = \{\pi_1, \cdots, \pi_v\} \in \{0,1\}^v$ to denote the policy π restricted in $\{x_1, \cdots, x_v\}$. Now, we still construct two distributions on the rewards as in (6.1) given $x = x_j$:

$$y_0 \sim M \cdot \mathrm{Bern}\left(\frac{1}{2}\right), \quad y_1 \sim M \cdot \mathrm{Bern}\left(\frac{1 + (2b_j - 1) \cdot \Delta}{2}\right),$$

for some small $\Delta > 0$ that will be determined later. The marginal distribution of x in $\mathbb{P}_{\mathbf{b}}$ supports uniformly on $\{x_1, \cdots, x_v\}$, with equal mass $1/v$. Further,

$$e_0(a = 1; x = x_i) = \kappa, \; e_0(a = 0; x = x_i) = 1 - \kappa,$$

That is, given x_i, if we can correctly specify b_i, then for $b_i = 1$, the optimal decision is $a = 1$; for $b_i = 0$, the optimal decision is $a = 0$. If we made a suboptimal decision, we will suffer an expected loss of $M\Delta$.

Step 2. Separation Verification. Now, as the objective functions are quite complex compared with those in all previous applications, we need to further simplify the distributionally robust reward and recognize the optimal policy under full information first. To emphasize the dependence on the underlying distribution $\mathbb{P}_{\mathbf{b}}$, we rewrite $Q_{\mathrm{DRO}}(\pi) = Q_{\mathrm{DRO}}(\pi, \mathbb{P}_{\mathbf{b}})$. Now, by the strong duality property in Lemma 6.1, we have

$$\begin{aligned}
\sup_{\pi \in \Pi} Q_{\mathrm{DRO}}(\pi, \mathbb{P}_{\mathbf{b}}) &= \sup_{\pi \in \Pi} \sup_{\alpha \geq 0} \left\{-\alpha \log\left(\mathbb{E}_{\mathbb{P}_{\mathbf{b}}}[\exp(-y(\pi(x))/\alpha)]\right) - \alpha\delta\right\} \\
&= \sup_{\alpha \geq 0} \sup_{\pi \in \Pi} \left\{-\alpha \log\left(\mathbb{E}_{\mathbb{P}_{\mathbf{b}}}[\exp(-y(\pi(x))/\alpha)]\right) - \alpha\delta\right\}.
\end{aligned}$$

Note that the optimal policy for the distribution $\mathbb{P}_{\mathbf{b}}$ is $\pi^*_{\mathbf{b}} = \mathbf{b}$ (i.e., correctly specifying $\mathbf{b}$). Further, the optimal expected reward depends on the number of $b_i = 1$ for $i = 1, \cdots, v$ in $\mathbf{b}$. For any given $\mathbf{b}$, we define $s \triangleq \dfrac{\sum_{i=1}^v b_i}{v} \in [0,1]$ as the proportion of 1 in the vector $\mathbf{b}$ and random variable $\tilde{Y}_1 = \mathrm{Bern}\left(\frac{1}{2} + \Delta s\right)$. Then the optimal distributionally robust reward

can be expressed as:

$$\begin{aligned}\sup_{\pi\in\Pi} Q_{\mathrm{DRO}}\left(\pi,\mathbb{P}_{\mathbf{b}}\right) &= Q_{\mathrm{DRO}}\left(\pi^*_{\mathbf{b}},\mathbb{P}_{\mathbf{b}}\right) = \sup_{\alpha\geq 0}\left\{-\alpha\log\left(\mathbb{E}\left[\exp\left(-\tilde{Y}_1/\alpha\right)\right]\right)-\alpha\delta\right\}\\ &= \inf_{\mathrm{D_{KL}}\left(\mathbb{P}\|\tilde{Y}_1\right)\leq\delta}\mathbb{E}_{Y\sim\mathbb{P}}[Y]\\ &= Mg(1/2+\Delta s),\end{aligned}$$

where the equality in second line again follows from Lemma 6.1, and the last equality follows from the construction of $\tilde{Y}_1$ and $g(q)\triangleq\inf_{\mathrm{D_{KL}}(p\|q)\leq\delta}p$, where $\mathrm{D_{KL}}(p\|q)$ denotes the KL-divergence of two Bernoulli distributions $\mathrm{Bern}(p)$ and $\mathrm{Bern}(q)$.

Besides, for any $\pi\in\Pi$ and any $\mathbb{P}_{\mathbf{b}}$, we define $d_H\left(\mathbf{b},\mathbf{b}'\right)=\sum_{i=1}^{v}|b_i-b'_i|$, i.e., the Hamming distance between $\mathbf{b}$ and $\mathbf{b}'$, and random variable $\tilde{Y}_0=\mathrm{Bern}\left(\frac{1}{2}-\Delta+\Delta s\right)$. Then we can express the distributionally robust reward under policy π as:

$$\begin{aligned}&Q_{\mathrm{DRO}}\left(\pi,\mathbb{P}_{\mathbf{b}}\right)=\sup_{\alpha\geq 0}-\alpha\log\left(\frac{1}{v}\sum_{i=1}^{v}\exp\left(-y(\pi_i)\right)/\alpha\right)-\alpha\delta\\ &=\sup_{\alpha\geq 0}-\alpha\log\left(\frac{1}{v}\left(\sum_{i=1,\pi_i=b_i}^{v}\mathbb{E}\left[\exp\left(-\frac{\tilde{Y}_1}{\alpha}\right)\right]+\sum_{i=1,\pi_i\neq b_i}^{v}\mathbb{E}\left[\exp\left(-\frac{\tilde{Y}_0}{\alpha}\right)\right]\right)\right)-\alpha\delta\\ &=\sup_{\alpha\geq 0}-\alpha\log\left(\frac{v-d_H(\mathbf{b},\pi)}{v}\mathbb{E}\left[\exp\left(-\frac{\tilde{Y}_1}{\alpha}\right)\right]+\frac{d_H(\mathbf{b},\pi)}{v}\mathbb{E}\left[\exp\left(-\frac{\tilde{Y}_0}{\alpha}\right)\right]\right)-\alpha\delta,\end{aligned}$$

where $\mathbb{E}\left[\exp\left(-\tilde{Y}_1/\alpha\right)\right]$ and $\mathbb{E}\left[\exp\left(-\tilde{Y}_0/\alpha\right)\right]$ can be regarded as the expected reward when we correctly specify or misspecify each component of $\mathbf{b}$, respectively. To simplify the notation, let $m=d_H(\mathbf{b},\pi)$. Then:

$$\begin{aligned}&Q_{\mathrm{DRO}}\left(\pi,\mathbb{P}_{\mathbf{b}}\right)\\ &=\sup_{\alpha\geq 0}\left\{-\alpha\log\left(\frac{v-m}{v}\mathbb{E}\left[\exp\left(-\tilde{Y}_1/\alpha\right)\right]+\frac{m}{v}\mathbb{E}\left[\exp\left(-\tilde{Y}_0/\alpha\right)\right]\right)-\alpha\delta\right\}\\ &=\sup_{\alpha\geq 0}\left\{-\alpha\log\left(\left(1/2+\frac{vs-m}{v}\Delta\right)\cdot\exp(-M/\alpha)+\left(1/2-\frac{vs-m}{v}\Delta\right)\right)-\alpha\delta\right\}.\end{aligned}$$

Accordingly, we construct a distribution of $\tilde{Y}_m$ as:

$$\tilde{Y}_m=\begin{cases}M & \text{w.p. } 1/2+\frac{vs-m}{v}\Delta\\ 0 & \text{w.p. } 1/2-\frac{vs-m}{v}\Delta\end{cases}$$

Then, $Q_{\mathrm{DRO}}\left(\pi,\mathbb{P}_{\mathbf{b}}\right)$ can be further simplified as

$$Q_{\mathrm{DRO}}\left(\pi,\mathbb{P}_{\mathbf{b}}\right)=\inf_{\mathrm{D_{KL}}(\mathbb{P}\|\tilde{Y}_m)\leq\delta}\mathbb{E}_{Y\sim\mathbb{P}}[Y]=Mg\left(1/2+\Delta s-\frac{m}{v}\Delta\right).$$

By the result that $g(q)$ is differentiable and $g'(q) \geq 1/2$ for $q \in [0.4, 0.6]$ and $\delta \leq 0.226$, see Lemma E.1 in the Appendix. We have for $0 < \Delta \leq 0.1$,

$$\begin{aligned}
&\sup_{\pi \in \Pi} Q_{\mathrm{DRO}}\left(\pi, \mathbb{P}_{\mathbf{b}}\right) - Q_{\mathrm{DRO}}\left(\pi, \mathbb{P}_{\mathbf{b}}\right) = Mg(1/2 + \Delta s) - Mg\left(1/2 + \Delta s - \frac{m}{v}\Delta\right) \\
&\geq \frac{m}{v} M\Delta \min_{x \in [1/2-\Delta, 1/2+\Delta]} g'(x) \geq \frac{d_H(\mathbf{b}, \pi) M\Delta}{2v}, \qquad (54)
\end{aligned}$$

which is independent of s. Now, we translate the problem into bounding the hamming distance $d_H(\mathbf{b}, \pi)$.

Step 3. Applying Le Cam. Now, we will apply Le Cam pair-wisely: denote $\mathbf{b}' = (b'_1, \cdots, b'_v)$ as the estimation under π, where $b'_j \in \{0, 1\}$ for any $j = 1, \cdots, v$; $\mathbf{b}_{-j} = (b_1, \cdots, b_{j-1}, b_{j+1}, \cdots, b_v)$ as $\mathbf{b}$ removing the j-th component. Following (54), we have

$$\begin{aligned}
\sup_{\mathbb{P}_0^{e_0} \in \mathcal{P}(M,\kappa)} \mathcal{R}_{\mathrm{DRO}}\left(N; \hat{\pi}, \mathbb{P}_0\right) &\geq \frac{M\Delta}{2v} \sup_{\mathbb{P}_0^{e_0} \in \mathcal{P}(M,\kappa)} \mathbb{E}_{\left(\mathbb{P}_0^{e_0}\right)^N}\left[d_H(\mathbf{b}, \pi)\right] \\
&\geq \frac{M\Delta}{2v} \cdot \frac{1}{2^v} \sum_{\mathbf{b} \in \{0,1\}^v} \mathbb{E}_{\left(\mathbb{P}_{\mathbf{b}}^{e_0}\right)^N}\left[d_H(\mathbf{b}, \mathbf{b}')\right] \\
&= \frac{M\Delta}{2v} \cdot \frac{1}{2^v} \sum_{\mathbf{b} \in \{0,1\}^v} \mathbb{E}_{\left(\mathbb{P}_{\mathbf{b}}^{e_0}\right)^N}[\sum_{j=1}^{v} |b_j - b'_j|] \\
&= \frac{M\Delta}{2v} \cdot \frac{1}{2^v} \sum_{j=1}^{v} \left(\sum_{\mathbf{b} \in \{0,1\}^v, b_j = 0} + \sum_{\mathbf{b} \in \{0,1\}^v, b_j = 1} \right) \mathbb{E}_{\left(\mathbb{P}_{\mathbf{b}}^{e_0}\right)^N}[|b_j - b'_j|] \\
&\geq \frac{M\Delta}{2v} \cdot \frac{1}{2^v} \sum_{j=1}^{v} \sum_{\mathbf{b}_{-j} \in \{0,1\}^{v-1}} \max_{b_j \in \{0,1\}} \mathbb{E}_{\left(\mathbb{P}_{\mathbf{b}}^{e_0}\right)^N}[|b_j - b'_j|] \\
&\overset{(i)}{\geq} \frac{M\Delta}{4v} \cdot \frac{1}{2^v} \sum_{j=1}^{v} \sum_{\mathbf{b}_{-j} \in \{0,1\}^{v-1}} \left(1 - \max_{d_H(\mathbf{b}, \mathbf{b}') = 1} \| \left(\mathbb{P}_{\mathbf{b}}^{e_0}\right)^N - \left(\mathbb{P}_{\mathbf{b}'}^{e_0}\right)^N \|_{\mathrm{TV}}\right) \\
&= \frac{M\Delta}{8} \cdot \left(1 - \max_{d_H(\mathbf{b}, \mathbf{b}') = 1} \| \left(\mathbb{P}_{\mathbf{b}}^{e_0}\right)^N - \left(\mathbb{P}_{\mathbf{b}'}^{e_0}\right)^N \|_{\mathrm{TV}}\right),
\end{aligned}$$

where (i) comes from the Le Cam's two-point method (note that $|1 - b'_j| + |b'_j| \geq 1$, and the separation parameter here is thus 1). By applying the Pinsker's inequality, we have

$$\begin{aligned}
\| \left(\mathbb{P}_{\mathbf{b}}^{e_0}\right)^N - \left(\mathbb{P}_{\mathbf{b}'}^{e_0}\right)^N \|_{\mathrm{TV}} &\leq \sqrt{\mathrm{D}_{\mathrm{KL}}\left(\left(\mathbb{P}_{\mathbf{b}}^{e_0}\right)^N \| \left(\mathbb{P}_{\mathbf{b}'}^{e_0}\right)^N\right)/2} \\
&= \sqrt{N \cdot \mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_{\mathbf{b}}^{e_0} \| \mathbb{P}_{\mathbf{b}'}^{e_0}\right)/2},
\end{aligned}$$

where the equality comes from the i.i.d. assumption on the historical data.

Step 4. Calculating KL between Environments. For $\mathbf{b}, \mathbf{b}'$ such that $d_H(\mathbf{b}, \mathbf{b}') = 1$, Let $b_l \neq b'_l$ and without loss of generality, we can assume $b_l = 1$. We abbreviate $\mathbb{P}_{\mathbf{b}}^{e_0}(x = x_i, a = j, y = Mk)$ as $\mathbb{P}_{\mathbf{b}}^{e_0}(x_i, j, Mk)$. Then, we have

$$\begin{aligned}
\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_{\mathbf{b}}^{e_0} \| \mathbb{P}_{\mathbf{b}'}^{e_0}\right) &= \sum_{i=1}^{v} \sum_{j=0}^{1} \sum_{k=0}^{1} \mathbb{P}_{\mathbf{b}}^{e_0}(x_i, j, Mk) \log\left(\frac{\mathbb{P}_{\mathbf{b}}^{e_0}(x_i, j, Mk)}{\mathbb{P}_{\mathbf{b}'}^{e_0}(x_i, j, Mk)}\right) \\
&= \sum_{j=0}^{1} \sum_{k=0}^{1} \mathbb{P}_{\mathbf{b}}^{e_0}(x_l, j, Mk) \log\left(\frac{\mathbb{P}_{\mathbf{b}}^{e_0}(x_l, j, Mk)}{\mathbb{P}_{\mathbf{b}'}^{e_0}(x_l, j, Mk)}\right) \\
&\overset{(i)}{=} \frac{\kappa}{v}(1/2 + \Delta) \log\left(\frac{1/2 + \Delta}{1/2 - \Delta}\right) + \frac{\kappa}{v}(1/2 - \Delta) \log\left(\frac{1/2 - \Delta}{1/2 + \Delta}\right) \\
&= \frac{2\kappa}{v} \Delta \log\left(\frac{1 + 2\Delta}{1 - 2\Delta}\right) \\
&\leq \frac{8\kappa}{v}(\Delta^2 + \Delta^3),
\end{aligned}$$

where (i) holds since for $b_l = 1$, only when $a = 1$ will contribute to the KL-divergence, otherwise the reward distributions are identical (following $y_0 \sim \mathrm{Bern}(1/2)$). Choosing $\Delta = \frac{1}{4}\sqrt{\frac{v}{N\kappa}} \leq 0.1$, we derive the lower bound. □

Note that in Step 3 of the proof above, we used Le Cam's method in a pairwise fashion to lower bound the Hamming distance $d_H(\mathbf{b}, \pi)$. A simpler and more direct alternative is to apply Assouad's Lemma (Tsybakov, 2009), which immediately yields the similar conclusion as our Step 3. Assouad's Lemma can in fact be viewed as a natural extension of Le Cam's method from two-point to the hypercube, and we will further discuss this technique later in Section 7.

6.4 Summary and Bibliographic Remarks

We conclude this section by summarizing the regret lower bounds derived in the applications of offline policy learning. The following table collects the main results:

While our discussion in this section focuses on structured offline policy learning problems with bandit feedback, it is worth noting that this topic has recently gained increasing attention in the learning theory and decision-making communities. Other notable works include Luedtke and Van Der Laan (2016), Nie et al. (2018), Athey and Wager (2021), Bibaut et al. (2021), Hadad et al. (2021), which analyze various forms of offline policy evaluation and learning under bandit feedback, often motivated by applications in healthcare, recommendation, social policy, and online education. These works have advanced algorithmic

Table 9: Regret Lower Bounds in Offline Policy Learning

	Problem Setting	**Lower Bound**
App.20	Two-action; i.i.d. Data	$\Omega\left(M\sqrt{\frac{v}{N\kappa}}\right)$
App.21	Multi-action; i.i.d. Data	$\Omega\left(M\sqrt{\frac{d}{N\kappa}}\right)$
App.22	Multi-action; Adaptive Data	$\Omega\left(M\sqrt{\frac{d}{\sum_{t=1}^{T}\kappa_t}}\right)$
App.23	Robust Two-action; i.i.d. Data	$\Omega\left(M\sqrt{\frac{v}{N\kappa}}\right)$

design and understanding of bias-variance tradeoffs in offline learning, particularly when using inverse propensity weighting or model-based estimation techniques.

However, the literature on regret lower bounds in offline policy learning with bandit feedback remains relatively limited. Most existing lower bounds rely on information-theoretic techniques, especially Le Cam's two-point method and its extension to Assouad's lemma, as we have emphasized throughout this section. These tools provide sharp characterization of statistical hardness by reducing policy learning to hypothesis testing over carefully constructed instance families.

7 Parting Words on Le Cam

This section extends our survey by highlighting the continued relevance and adaptability of Le Cam's two-point method beyond its classical applications in stylized inventory models, parametric pricing, and policy learning. We organize the discussion around three central themes: two advanced applications that illustrate how Le Cam's method can be extended to handle either complex dynamic system behavior or implicit instance constructions; deeper theoretical developments that reinterpret its foundational role in statistical decision theory; and a broader landscape of related information-theoretic tools that build upon or generalize the classical two-point argument.

First, we begin with a dynamic scheduling and queuing system with abandonment, where customers of different classes may leave the system before receiving service. This setting—analyzed in Zhong et al. (2024)—features stochastic arrivals, state-dependent transitions, and partially observed dynamics, all of which pose unique challenges for learning-based control. Despite these complexities, we can still construct a tight regret lower bound by leveraging a carefully designed two-point instance, showcasing that Le Cam's method remains effective even in highly structured, non-tabular environments. While our treatment is illustrative rather than exhaustive, this example underscores the method's broad utility in operational settings that lie outside classical static frameworks.

Besides, we also include a second application from assortment optimization under non-stationary customer preferences following Han et al. (2021). In this model, an adversarial determines the preference vector in each round, generalizing the classical multinomial logit setting to time-varying environments. What makes this example particularly noteworthy is its methodological novelty: unlike previous lower bound constructions that explicitly construct two or more hard-to-distinguish instances, here the analysis hinges on an existential argument. By appealing to duality theory and geometric convexity, we establish the existence of a worst-case instance that satisfies the separation condition required by Le Cam's method even though this instance cannot be explicitly expressed. This abstract construction highlights a new dimension of Le Cam's method: that its applicability extends not only across different decision settings, but also across different styles of proof—ranging from constructive to non-constructive arguments.

Second, we highlight a theoretically insightful direction: the reinterpretation of Le Cam's method as a dual tool: not only for deriving lower bounds, but also for certifying upper bounds under certain structural assumptions. Classical applications of Le Cam's method emphasize its simplicity: by reducing learning problems to binary hypothesis tests between two carefully constructed distributions, one can obtain minimax lower bounds in a modular and interpretable way. However, profound works such as Donoho and Liu (1991), Polyanskiy and Wu (2019), and Juditsky and Nemirovski (2020), reveal that this simplicity masks a deeper geometric principle. Specifically, they show that in many cases, optimizing the two-point construction itself provides a sharp upper bound on minimax risk, particularly when the loss is linear and the statistical model satisfies convexity or

symmetry conditions. This duality perspective helps explain why Le Cam's method is not only simple but also powerful: it provides tight risk characterizations in many structured problems. In addition, state-of-the-art developments such as the DEC framework for interactive learning further reflect how careful extensions of the two-point method can yield optimal characterizations in general dynamic environments.

Finally, we broaden the scope by surveying a set of complementary information-theoretic tools that help characterize statistical limits in learning problems. Although Le Cam's method provides a foundational lens, it is not the only one. Alternatives such as Fano's inequality, Assouad's lemma, and van Trees' inequality offer flexible means of deriving lower bounds, particularly when the hypothesis space is high-dimensional, structured, or continuous. Throughout this survey, we have encountered these tools either explicitly or implicitly. In this section, we revisit their roles more directly, clarifying their connections to Le Cam's principle and discussing when they offer sharper or more general insights. As problems in operations management grow increasingly complex and data-rich, this broader toolkit becomes indispensable for identifying the fundamental statistical difficulties in these problems.

In summary, this section aims not only to illustrate how Le Cam's two-point method continues to play a central role in operations research, but also to shed light on its evolving theoretical depth, connections to broader principles, and utility in emerging domains. Together, the examples and developments discussed here reinforce the enduring power of the two-point method as both a practical and conceptual cornerstone of learning-theoretic analysis for OM.

Before proceeding, we first summarize the notations tailored for Section 7, with its detailed definitions introduced later in the corresponding sections. The notations for the additional application of MNL assortment optimization follows from that in Section 4.3.

Table 10: Notation Used throughout Section 7

Symbol	**Description**
$c_t \in \{1, 2\}$	Class of the customer arriving at time t
$\theta_j > 0$	Abandonment rate of class j customers
$a_j > 0$	Cost of class j customer abandoning the system
$\mathcal{Q}_t$	Queue at beginning of period t
$Q_j(t)$	Number of class j customers in queue at time t
$U(t)$	Remaining service time of customer in service at time t
$R_j(T)$	Number of class j customers who abandon over $[T]$
$C_T(\pi; \theta)$	Cumulative abandonment cost under policy π and rate θ
π_a	Priority policy that serves class with higher a_j
$Y(t)$	System state at time t

7.1 Scheduling in Queueing Systems and Advanced Application in Assortment Optimization

The first part of this section explores how Le Cam's two-point method continues to provide meaningful insight even in complex modern operational environments. As real-world systems grow more complex due to featuring stochastic transitions, evolving system states, and interdependent decision rules, characterizing learning limits becomes increasingly subtle. Yet, the core idea behind Le Cam's method remains surprisingly robust. We will see this point through the following applications on a D/D/1+M scheduling queuing system, where the first "D" denotes deterministic interarrival times, the second "D" indicates deterministic service times, "1" refers to a single server, and "+M" captures the fact that each customer may abandon the system after an exponential patience time; and an adversarial MNL assortment optimization problem.

Application 24: D/D/1+M Scheduling Queuing System

We consider a single-server queuing system operating in discrete time over a horizon of T periods. There are $J = 2$ customer classes, indexed by $j \in \{1, 2\}$. The arrival and service time of each customer are deterministic, and the customer may abandon the queue if not served in time with probability. The system is assumed to be non-preemptive and can process at most one customer per period. The key modeling features are as follows:

Arrival and Service

- At each period $t \in [T]$, one customer arrives to the system. Let $c_t \in \{1, 2\}$ denote the type of the customer arriving at time t.
- Each customer requires a unit period of service. The server can serve at most one customer per period.

Abandonment and Cost

- Upon arrival, each customer enters a first-in-first-out (FIFO) queue and waits to be served.
- A customer of class j has an independent exponential patience time $\tau \sim \text{Exp}(\theta_j)$, where $\theta_j > 0$ is the (unknown) abandonment rate for class j. We use g_j to denote its pdf.
- If a customer in class j is not served before their patience elapses, they abandon the system and incur a cost $a_j > 0$. In addition, $\theta_1 > \theta_2$ implies $a_1 > a_2$, and vice versa.

Scheduling Policy

- At each time t, the system decides whether and whom to serve among the customers currently in queue.
- Let π_t denote the scheduling decision at time t, specifying which customer (if any) is selected for service. A scheduling policy $\pi = (\pi_1, \cdots, \pi_T) \in \Pi$ is non-anticipating, i.e., at time t, π_t depends only on: (i) the arrival sequence $\{c_s\}_{s \le t}$; (ii) the history of service and abandonment outcomes up to time $t-1$; (iii) the queue state at time t.

Sequence of Events

Assume the system is initialized empty at time $t = 0$, and define the queue $\mathcal{Q}_t$ as the ordered list of customers waiting at the beginning of period t. At each time period $t = 1, \cdots, T$, the following events occur in order:

- A new customer of class c_t arrives and joins the queue $\mathcal{Q}_t$.
- The scheduler observes the current queue state and selects one customer for service (if any).
- One customer is served for one time unit and then leaves the system.
- All customers whose patience expires in this period abandon the queue and incur cost.
- The system transitions to time $t+1$ with updated queue and state information.

Performance Metric

Let $R_j(T)$ denote the number of class j customers who abandon the system over the time horizon $[T]$. Then the cumulative cost under policy π and abandonment rates $\theta = (\theta_1, \theta_2)$ is given by:

$$C_T(\pi; \theta) = \mathbb{E}\left[\sum_{j=1}^{J} a_j R_j(T)\right],$$

with the expectation taken w.r.t. the randomness of π and customers abandon the system under parameter θ. For the given θ, we have the optimal policy $\pi^\star$ following the simple rule: ranking classes according to the descending order of their indices a_j, $j \in \{1, 2\}$, and when a server becomes available, serves the head-of-the-line (HL) customer from the class with waiting customers that has the highest a_j (and idles only if no customers are waiting). We denote policy satisfying this simple rule as π_a.[3] The regret of any scheduling policy π is thus defined as:

$$\mathcal{R}(T; \pi, \theta) = C_T(\pi; \theta) - C_T(\pi^\star; \theta).$$

We further make the consistency assumption on policy π: $T - \mathbb{E}\left[T\left(\pi_a\right)\right] = o\left(T^\alpha\right)$ for every $\alpha > 0$, where $T(\pi_a)$ denotes the cumulative time that the rule π_a is applied over $[0, T]$.

[3]Note that π_a may be different from $\pi^\star$ when the parameters a_j are misspecified.

The assumption restricts the time of policy π not following π_a is sublinear in T: in fact, if this consistency assumption does not hold, we would have the regret grows linearly with T for any admissible policies. The following Theorem restates the theorem 1 of Zhong et al. (2024).

Theorem 7.1. *For the two-class $D/D/1+M$ queue introduced before, for any policy π that satisfies the consistency property, the regret is lower bounded by*

$$\sup_{\theta} \mathcal{R}(T;\pi,\theta) = \Omega(\log T).$$

Proof of Theorem 7.1.
Step 1. Instance Construction. We construct the two instances as follows:

(i) Instance 1: $\theta_1^{(1)} = 3.5$, $\theta_2^{(1)} = 3$. Consequently, we let $a_1^{(1)} = 2$, $a_2^{(1)} = 1$, and $\pi_a^{(1)} = \pi_{1>2}$ is the optimal scheduling policy.

(ii) Instance 2: $\theta_1^{(2)} = 2.5$, $\theta_2^{(2)} = 3$. Consequently, we let $a_1^{(2)} = 1$, $a_2^{(2)} = 2$, and $\pi_a^{(2)} = \pi_{2>1}$ is the optimal scheduling policy.

Thus, we must learn the ordering of θ_1 and θ_2, which determines the optimal scheduling policy.

Step 2. Separation Verification. Let $\mathbb{P}_i$ (which induces $\mathbb{E}_i$) denote the probability measures induced by the interconnection of policy π and instance $i \in \{1,2\}$ over $[0,T]$. Note that here we do not have an explicit "separation condition", and instead, here we aim to directly show that the regret can be lower bounded by the cumulative time of not applying $\pi_{1>2}$ in instance 1 (or not applying $\pi_{2>1}$ in instance 2).

First, notice that the regret can be decomposed by:

$$\mathcal{R}(T;\pi,\theta) = \sum_{t=2}^{T} [C_t(\pi;\theta) - C_{t-1}(\pi;\theta)] - [C_t(\pi^\star;\theta) - C_{t-1}(\pi^\star;\theta)].$$

We restrict attention to non-idling policies π,[4] which always schedule some customer when the queue is non-empty. Note that for each $t \in \{2,\cdots,T\}$, through some detailed calculations, we have

$$C_t(\pi^\star;\theta) - C_{t-1}(\pi^\star;\theta) = a_2 \left(1 - \exp^{-\theta_2(t-1)}\right),$$

and

$$C_t(\pi;\theta) - C_{t-1}(\pi;\theta) \geq a_2 \left(\exp^{-\theta_2} - \exp^{-\theta_2(t-1)}\right) + a_1 \left(1 - \exp^{-\theta_1}\right).$$

[4]It can be shown that for any idling policy, we can always find its non-idling counterpart which has a smaller regret, and we omit the detailed analysis.

Summing the results together, we conclude

$$\mathcal{R}(T;\pi,\theta) \geq c_1 \cdot \left(T - 2 - \mathbb{E}_1\left[T\left(\pi_{1>2}\right)\right]\right), \tag{55}$$

for $c_1 \triangleq a_1^{(1)}\left(1-\exp^{-\theta_1^{(1)}}\right) - a_2^{(1)}\left(1-\exp^{-\theta_2^{(1)}}\right)$.

Next, we will show how the expected number of class 1 customers $\mathbb{E}_1\left[R_1(T)\right]$ has abandoned from the system under instance 1. By queuing dynamics, we have:

$$R_1(T;\pi) = E_1(T) - X_1(T) - D_1(T),$$

where $E_1(T)$ is the cumulative number of class 1 customers arrivals, $X_1(T)$ is the number of class 1 customers in queue at time T, and $D_1(T)$ is the number of class 1 customers who have started service by time T. Taking expectation gives:

$$\mathbb{E}_\eta[R_1(T;\pi)] = T - \mathbb{E}_\eta[X_1(T)] - \mathbb{E}_\eta[D_1(T)].$$

Observe that class 1 customers only receive service when policy $\pi_{1>2}$ is applied. Therefore, the total number of class 1 customers that either start or remain in service (i.e., $X_1(T) + D_1(T)$) is upper bounded by the cumulative number of rounds that $\pi_{1>2}$ is applied:

$$\mathbb{E}_1[X_1(T) + D_1(T)] \leq \mathbb{E}_1[T(\pi_{1>2})].$$

Combining the above, we conclude that

$$\mathbb{E}_1\left[R_1(T)\right] \leq T - \mathbb{E}_1\left[T\left(\pi_{1>2}\right)\right]. \tag{56}$$

Step 3. Applying Information-theoretical Tool. Define $\mathcal{A}_T \triangleq \{T\left(\pi_{1>2}\right) < \frac{T}{2}\}$. Notice that by our consistency assumption on policy π, there exist some finite positive constants T_0 and $c_0 > 0$ such that the following inequalities hold for all $T > T_0$:

$$T - \mathbb{E}_1\left[T\left(\pi_{1>2}\right)\right] \leq c_0 T^\alpha,\ T - \mathbb{E}_2\left[T\left(\pi_{2>1}\right)\right] \leq c_0 T^\alpha.$$

Then by Markov's inequality, it follows that for all $T > T_0$,

$$\begin{aligned}
\mathbb{P}_1\left(\mathcal{A}_T\right) &= \mathbb{P}_1\left(T - T\left(\pi_{1>2}\right) \geq \frac{T}{2}\right) \leq \frac{2\mathbb{E}_1\left[T - T\left(\pi_{1>2}\right)\right]}{T} \leq \frac{2c_0}{T^{1-\alpha}}, \\
\mathbb{P}_2\left(\mathcal{A}_T^c\right) &= \mathbb{P}_2\left(T\left(\pi_{1>2}\right) \geq \frac{T}{2}\right) \leq \frac{2\mathbb{E}_2\left[T\left(\pi_{1>2}\right)\right]}{T} \leq \frac{2c_0}{T^{1-\alpha}}.
\end{aligned} \tag{57}$$

Using the Bretagnolle-Huber inequality,

$$\begin{aligned}
\mathbb{P}_1\left(\mathcal{A}_T\right) + \mathbb{P}_2\left(\mathcal{A}_T^c\right) &\geq \frac{1}{2}\exp\left(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1^T \| \mathbb{P}_2^T)\right) \\
&\triangleq \frac{1}{2}\exp\left(-\mathbb{E}_1\left[\log\left(\frac{\mathcal{L}_1(\{Y(t): t\in[0,T]\})}{\mathcal{L}_2(\{Y(t): t\in[0,T]\})}\right)\right]\right)
\end{aligned} \tag{58}$$

where $\mathcal{L}_i(\{Y(t) : t \in [0,T]\})$ represents the likelihood function in instance i under $\{Y(t) : t \in [0,T]\}$. Here $Y(t) = (Q_1(t), Q_2(t), U(t)) \in \mathbb{Y}$ is the system state (i.e., observation) at any time t, with $\mathbb{Y} = \mathbb{Z}_+^2 \times [0,1]$. $Q_j(t) \in \mathbb{Z}_+$ denotes the number of class j customers waiting in queue at time t; and $U(t) \in [0,1]$ denotes the remaining service time of the customer in service at time t. Note that this state representation is sufficient for computing the log-likelihood ratio: since under both instances, the system evolves as a controlled Markov process, and all randomness (to be specific: arrival sequence, service decisions, abandonment events) is conditionally independent given the history of past observations and decisions. Moreover, the abandonment mechanism is memoryless according to our D/D/1+M setting, which implies that the likelihood of abandonment depends only on the current waiting time and not on how long a customer has already waited. Therefore, by observing $\{Y(t) : t \in [0,T]\}$, we capture all necessary information to evaluate the likelihood under each instance. Combining (57) and (58) together, we conclude that the following inequalities holds for all $T \geq T_0$:

$$\mathbb{E}_1\left[\log\left(\frac{\mathcal{L}_1(\{Y(t) : t \in [0,T]\})}{\mathcal{L}_2(\{Y(t) : t \in [0,T]\})}\right)\right] \geq \log\left(\frac{1}{2\left(\mathbb{P}_1\left(\mathcal{A}_T\right) + \mathbb{P}_2\left(\mathcal{A}_T^c\right)\right)}\right)$$
$$\geq \log\left(\frac{1}{2\left(\frac{2c_0}{T^{1-\alpha}} + \frac{2c_0}{T^{1-\alpha}}\right)}\right) = (1-\alpha)\log T - \log 8c_0.$$

Step 4. Calculating KL between Environments. Note that according to our construction, the likelihoods in Instance 1 and Instance 2 only differ in the distribution of the patience time realizations for class 1 customers. Let $w_{1,i}$ denote the time spent in queue by the i-th class 1 customer, for $i \in \{1, \ldots, E_1(T)\}$, where $E_1(T)$ denotes the total number of class 1 customers who have arrived by time T.

We partition these customers into two groups based on their observed outcomes: let $\mathcal{M}_{T,1}$ represent the set of customers who have abandoned the system by time T, and let $\mathcal{M}_{T,2}$ denote the complementary set, i.e., those who either completed service and departed, or are still in the system at time T.

Since patience times are only partially observed-specifically, they are right-censored for customers in $\mathcal{M}_{T,2}$-this leads to a mixture of fully observed and censored data. Accordingly, when computing the KL divergence between the two instances, only the likelihood contributions from $\mathcal{M}_{T,1}$ differ, as these correspond to fully observed abandonments. Therefore, the KL divergence simplifies to an expression that solely depends on the abandonment

behavior of class 1 customers, as we formalize next.

$$
\begin{aligned}
&\mathbb{E}_1\left[\log\left(\frac{\mathcal{L}_1(\{Y(t):t\in[0,T]\})}{\mathcal{L}_2(\{Y(t):t\in[0,T]\})}\right)\right]\\
=&\mathbb{E}_1\left[\log\left(\prod_{i\in\mathcal{M}_{T,1}}\frac{g_1^{(1)}(w_{1,i})}{g_1^{(2)}(w_{1,i})}\right)\right]+\mathbb{E}_1\left[\log\left(\prod_{i\in\mathcal{M}_{T,2}}\frac{\int_{w_{1,i}}^{\infty}g_1^{(1)}(\mathrm{d}w)}{\int_{w_{1,i}}^{\infty}g_1^{(2)}(\mathrm{d}w)}\right)\right]\\
=&\mathbb{E}_1\left[\sum_{i\in\mathcal{M}_{T,1}}\log\left(\frac{g_1^{(1)}(w_{1,i})}{g_1^{(2)}(w_{1,i})}\right)\right]+\mathbb{E}_1\left[\log\left(\prod_{i\in\mathcal{M}_{T,2}}e^{-\left(\theta_1^{(1)}-\theta_1^{(2)}\right)w_{1,i}}\right)\right]\\
=&\mathbb{E}_1\left[R_1(T)\right]\cdot\mathrm{D}_{\mathrm{KL}}\left(g_1^{(1)}\|g_1^{(2)}\right)+\mathbb{E}_1\left[\log\left(e^{-\left(\theta_1^{(1)}-\theta_1^{(2)}\right)\sum_{i\in\mathcal{M}_{T,2}}w_{1,i}}\right)\right]\\
=&\mathbb{E}_1\left[R_1(T)\right]\cdot\mathrm{D}_{\mathrm{KL}}\left(g_1^{(1)}\|g_1^{(2)}\right)+\mathbb{E}_1\left[-\left(\theta_1^{(1)}-\theta_1^{(2)}\right)\sum_{i\in\mathcal{M}_{T,2}}w_{1,i}\right]\\
\overset{(i)}{\leq}&\mathbb{E}_1\left[R_1(T)\right]\cdot\mathrm{D}_{\mathrm{KL}}\left(g_1^{(1)}\|g_1^{(2)}\right)\\
=&\mathbb{E}_1\left[R_1(T)\right]\cdot\left(\log\theta_1^{(1)}-\log\theta_1^{(2)}+\frac{\theta_1^{(1)}}{\theta_1^{(2)}}-1\right),
\end{aligned}
$$

where $g_1^{(1)}$ and $g_1^{(2)}$ are the pdfs of exponential distributions $\mathrm{Exp}(\theta_1^{(1)})$ and $\mathrm{Exp}(\theta_1^{(2)})$, and we use $\mathrm{D}_{\mathrm{KL}}\left(g_1^{(1)}\|g_1^{(2)}\right)$ to represent the KL-divergence between $\mathrm{Exp}(\theta_1^{(1)})$ and $\mathrm{Exp}(\theta_1^{(2)})$. (i) follows from the fact that $\theta_1^{(1)}>\theta_1^{(2)}$, and the last equality follows from the KL-divergence between two exponential distributions. Combining the two inequalities on the expected log-likelihood ratio (KL-divergence), we have for all $T>T_0$,

$$
c_3\cdot\mathbb{E}_1\left[R_1(T)\right]\geq(1-\alpha)\log T-\log 8c_0, \tag{59}
$$

with $c_3\triangleq\log\theta_1^{(1)}-\log\theta_1^{(2)}+\frac{\theta_1^{(1)}}{\theta_1^{(2)}}-1$.

Combining (55), (56), and (59), we obtain the regret lower bound $\sup_\theta\mathcal{R}(T;\pi,\theta)=\Omega(\log T)$.

□

Application 25: Adversarial Capacitated Assortment

We now consider a variant of capacitated assortment optimization under adversarially generated customer preferences. This setting captures more challenging environments where customer behavior may shift arbitrarily across time, possibly in response to the retailer's past actions. Such a model is particularly relevant in recommendation systems and online marketplaces subject to strategic or non-stationary environments.

Specifically, we still let M be the total number of items and $K \leq M$ the cardinality constraint. In each round $t = 1, \cdots, T$, the retailer now selects an assortment $S_t \subseteq [M]$ of **exactly** size K. The customer either purchases one of the displayed items $i_t \in S_t$, or does not make a purchase ($i_t = 0$).

The main difference of the adversarial assortment optimization compared with the previous two assortment applications under the stochastic setting lies in that: now at the beginning of each round, an adaptive adversary chooses a preference vector $v_t = (v_{t1}, \cdots, v_{tM}) \in [0,1]^M$ based on past history, and we use $\mathcal{A}$ to denote the adaptive adversary. Given the offered assortment S_t and preference v_t, the customer selects item $j \in S_t \cup \{0\}$ with probability specified by the MNL model:

$$\mathbb{P}\left(i_t = j \mid S_t, v_t\right) = \begin{cases} \dfrac{v_{tj}}{1 + \sum_{k \in S_t} v_{tk}}, & j \in S_t, \\ \dfrac{1}{1 + \sum_{k \in S_t} v_{tk}}, & j = 0, \end{cases}$$

where now we use $\mathbb{P}\left(i_t = j \mid S_t, v_t\right)$ to emphasis its dependence on the time-varying preference v_t, in contrast to that (27) for the stationary preference. We still use $r_j \in [0,1]$ to denote the revenue parameter for any $j \in [M]$, which is time-invariant and known to the retailer as before. For each period t, the expected revenue for assortment S_t under preference parameter is defined as:

$$r\left(S_t, v_t\right) = \sum_{j \in S_t} \frac{r_j v_{tj}}{1 + \sum_{k \in S_t} v_{tk}},$$

and we use $S^\star$, the full-information optimal assortment decision, as the benchmark. The retailer's objective is to learn from observed customer choices and select assortment policy $\pi = (S_1, \cdots, S_T)$ to minimize the cumulative regret with an adaptive adversary $\mathcal{A} = \{v_{ti}\}_{t,i=1}^{T,N}$ defined as:

$$\mathcal{R}(T; \pi, v) = \mathbb{E} \sum_{t=1}^{T} \left[r\left(S^\star, v_t\right) - r\left(S_t, v_t\right)\right],$$

with the expectation taken w.r.t. both the assortment policy π and the adaptive adversary $\mathcal{A}$. The following Theorem restates the lemma 3 of Han et al. (2021).

Theorem 7.2. *For the adversarial capacitated MNL assortment optimization introduced before, for any assortment policy π, there exists a constant $c_K > 0$ depends only on K such that for any assortment policy π,*

$$\sup_{\mathcal{A}} \mathcal{R}(T; \pi, v) = \Omega\left(c_K \cdot \min\{\sqrt{M^K T}, T\}\right).$$

Before we proceed to the detailed proof, we remark that the original result in Han et al. (2021) is established under a more general framework of adversarial combinatorial bandits with non-linear reward feedback, where the expected reward is modeled via a generic link function $g : \mathbb{R}^+ \to [0, 1]$, i.e., a function that maps utility $\sum_{i \in S_t} v_{ti}$ to the expected reward. Our current setting corresponds to a special case with the MNL structure, where the link function is $g(x) = \frac{x}{1+x}$. Importantly, the lower bound still holds for any link function g that is either (i) a polynomial of degree at least K, or (ii) a non-polynomial function (like that in MNL). The proof technique applies more broadly, and the adversarial construction used in this section is a specialization of their more general result. For the link function g to be a d-degree polynomial with $d < K$, we have a regret lower bound of $\Omega\left(c_{d,K} \cdot \min\{\sqrt{M^d T}, T\}\right)$ for some constant $c_{d,K} > 0$ that only depends on d and K, and the proof is similar to that of Theorem 7.2.

Proof of Theorem 7.2. We note that the key difference between this proof and all previous applications in this survey lies in how the hard instances are handled. In earlier uses of Le Cam's method, we explicitly constructed two (or more) hard-to-distinguish instances with carefully designed parameters. However, such explicit constructions fall short in the current adversarial setting, as they fail to yield the desired regret lower bound of order $\Omega(M^K T)$. Instead, the proof here relies on existential arguments: we argue that a pair of hard instances must exist, without explicitly constructing them. This shift from constructive to existential reasoning is central to the analysis and reflects the increased technical subtlety of the adversarial setting for assortment optimization.

Step 1. Existence of Hard Instances. We still let $r_1 = \cdots = r_M = 1$ be fixed. In the proof, we will consider a more general case with $r(S_t, v_t) = g(\sum_{i \in S_t} v_{ti})$, for some known $g : \mathbb{R}^+ \to [0, 1]$ which is not a polynomial of degree at most $K - 1$. For the MNL model, we have $g(x) = \frac{x}{1+x}$, which is clearly not a polynomial. Next, we will show the existence of two instance distributions through the following technical lemma.

Lemma 7.1. *Let $g \in C^K([0, K])$ be a real-valued and K times continuously differentiable function on $[0, K]$. Then the following two statements are equivalent:*

1. *g is not a polynomial of degree at most $K - 1$.*
2. *there exists a random vector $(V_1, \cdots, V_K)$ supported on $[0, 1]^K$, which follows an exchangeable joint distribution μ, and a scalar $v_0 \in [0, 1]$, such that:*

$$\begin{aligned} &\mathbb{E}_\mu \left[g \left(V_1 + \cdots + V_{\ell-1} + (K - \ell + 1) v_0 \right) \right] \\ &\quad = \mathbb{E}_\mu \left[g \left(V_1 + \cdots + V_\ell + (K - \ell) v_0 \right) \right], \end{aligned} \tag{60}$$

for all $\ell = 1, 2, \cdots, K - 1$, and

$$\mathbb{E}_\mu \left[g \left(V_1 + \cdots + V_{K-1} + v_0 \right) \right] < \mathbb{E}_\mu \left[g \left(V_1 + \cdots + V_K \right) \right]. \tag{61}$$

The proof of Lemma 7.1 is left in Appendix C. Based on Lemma 7.1, we construct i.i.d. copies $(V_{t,1}, \cdots, V_{t,K})$ for each $t \in [T]$. The random strategy of the adversary $\mathcal{A}$ when the optimal subset is $S^\star$ (note that the adversary can determine what the optimal $S^\star$ is) is constructed as follows: at each time $t \in [T]$, nature assigns (random vector) $(V_{t,1}, \cdots, V_{t,K})$ to the restriction of v_t to $S^\star$ with probability $\Delta \in (0, 1]$, and assigns $(v_0, \cdots, v_0)$ otherwise, with the parameter Δ to be specified later; nature also assigns $v_{ti} = v_0$ for all $i \notin S^\star$.

Step 2. Separation Verification. Here we will show the expected single-period revenue loss $\mathbb{E}_{v_t \sim \mu}\left[r\left(S^\star, v_t\right) - r\left(S_t, v_t\right)\right]$ under the previous constructions. Suppose an algorithm selects subset $S_t \subseteq [M]$, $|S_t| = K$ at time t, such that $|S_t \cap S^\star| = \ell$. Then we have:

$$\begin{aligned}\mathbb{E}_{v_t}[r(S_t, v_t)] &= \Delta \cdot \mathbb{E}_\mu\left[g\left(V_1 + \cdots + V_\ell + (K - \ell)v_0\right)\right] + (1 - \Delta) \cdot g\left(Kv_0\right) \\ &= g\left(Kv_0\right) + \Delta\gamma \cdot \mathbf{1}\{\ell = K\},\end{aligned}$$

where we define $\gamma \triangleq \mathbb{E}_\mu\left[g\left(V_1 + \cdots + V_K\right)\right] - g\left(Kv_0\right) > 0$, with the inequality holds from (61). Thus, for any $S_t \neq S^*$, it will incur an expected revenue loss of $\Delta\gamma$.

Step 3. Applying Information-Theoretical Tool. We use $\mathbb{P}_0$ here to denote the joint distribution of past T observations of the purchase decisions with $v_{ti} \equiv v_0$ and $\mathbb{E}_0$ is the expectation under $\mathbb{P}_0$ (so does $\mathbb{P}_{S^\star}$ and $\mathbb{E}_{S^\star}$, which is under the optimal v_t), and $T_S \triangleq \sum_{t=1}^T \mathbf{1}\{S_t = S\}$ denotes the number of times assortment decision S is selected. The regret $\sup_{\mathcal{A}} \mathcal{R}(T; \pi, v)$ can be decomposed and lower bounded through:

$$\begin{aligned}\sup_{\mathcal{A}} \mathcal{R}(T; \pi, v) &\geq \frac{1}{\binom{M}{K}} \cdot \sum_{|S^\star| = K} \mathbb{E}\left[\sum_{t=1}^T \left(r\left(S^\star, v_t\right) - r\left(S_t, v_t\right)\right)\right] \\ &\overset{(i)}{=} \frac{\Delta\gamma}{\binom{M}{K}} \cdot \sum_{|S^\star| = K} \left(T - \mathbb{E}_{S^\star}[T_{S^\star}]\right) \\ &\geq \frac{\Delta\gamma}{\binom{M}{K}} \cdot \sum_{|S^\star| = K} \left(T - \mathbb{E}_0\left[T_{S^\star}\right] - \left|\mathbb{E}_{S^\star}\left[T_{S^\star}\right] - \mathbb{E}_0\left[T_{S^\star}\right]\right|\right) \\ &\overset{(ii)}{\geq} \frac{\Delta\gamma}{\binom{M}{K}} \cdot \left[\binom{M}{K} T - T - T \sum_{|S^\star| = K} \left\|\mathbb{P}_0 - \mathbb{P}_{S^\star}\right\|_{\mathrm{TV}}\right] \\ &\overset{(iii)}{\geq} \frac{\Delta\gamma}{\binom{M}{K}} \cdot \left[\binom{M}{K} T - T - T \sum_{|S^\star| = K} \sqrt{\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_0 \| \mathbb{P}_{S^\star})/2}\right], \end{aligned} \tag{62}$$

where (i) is from the separation condition that deviating from the true $S^\star$ would lead to a revenue loss of $\Delta\gamma$, (ii) is by the fact that $\sum_{|S^\star| = K} \mathbb{E}_0\left[T_{S^\star}\right] = T$ and $\left|\mathbb{E}_0\left[T_{S^\star}\right] - \mathbb{E}_{S^\star}\left[T_{S^\star}\right]\right| \leq T \cdot \left\|\mathbb{P}_0 - \mathbb{P}_{S^\star}\right\|_{\mathrm{TV}}$; (iii) is by the Pinsker's inequality. By the decomposition, we now come

to a familiar problem of upper bounding the KL-divergence $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_0\|\mathbb{P}_{S^\star})$.

Step 4. Calculating KL between Environments. Here we examine the KL-divergence $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_0\|\mathbb{P}_{S^\star})$. We first note that:

$$\begin{aligned}\mathbb{E}_{v_t}[\mathbb{P}\left(i_t = 0 \mid S_t, v_t\right)] &= 1 - \mathbb{E}_\mu[g\left(V_1 + \cdots + V_\ell + (K-\ell)v_0\right)] \\ &= 1 - g(Kv_0) = \frac{1}{1+Kv_0}. \end{aligned} \tag{63}$$

For every item $j \in S_t \backslash S^\star$, $v_{tj} \equiv v_0$, and therefore

$$\begin{aligned}\mathbb{E}_{v_t}[\mathbb{P}\left(i_t = j \mid S_t, v_t\right)] &= v_0(1 - \mathbb{E}_\mu[g\left(V_1 + \cdots + V_\ell + (K-\ell)v_0\right)]) \\ &= v_0(1 - g(Kv_0)) = \frac{v_0}{1+Kv_0}. \end{aligned} \tag{64}$$

Next, for any $j \in S_t \cap S^\star$. Because $V_1, \cdots, V_\ell$ are exchangeable, the probabilities $\mathbb{E}_{v_t}\left[\mathbb{P}\left(i_t = j \mid S_t, v_t\right)\right]$ are the same for all $j \in S_t \cap S^\star$. Define $\beta \triangleq \mathbb{E}_{v_t}\left[\mathbb{P}\left(i_t = j \mid S_t, v_t\right)\right]$ for any $j \in S_t \cap S^\star$. By the law of total probability and (63), (64):

$$1 = \sum_{j \in S_t \cup \{0\}} \mathbb{E}_{v_t}\left[\mathbb{P}\left(i_t = j \mid S_t, v_t\right)\right] = \ell\beta + \frac{1 + (K-\ell)v_0}{1+Kv_0},$$

thus we have

$$\beta = \frac{1}{\ell}\left[1 - \frac{1 + (K-\ell)v_0}{1+Kv_0}\right] = \frac{v_0}{1+Kv_0}. \tag{65}$$

Comparing the previous three expected probabilities, we conclude for any $|S_t| = K$ that $S_t \neq S^\star$, $\mathbb{E}_{v_t}\left[\mathbb{P}\left(i_t = 0 \mid S_t, v_t\right)\right] = 1/\left(1+Kv_0\right)$ and $\mathbb{E}_{v_t}\left[\mathbb{P}\left(i_t = j \mid S_t, v_t\right)\right] = v_0/\left(1+Kv_0\right)$ for all $j \in S_t$. This shows that all $|S_t| = K$, $S_t \neq S^\star$ are information theoretically indistinguishable. On the other hand, for $S_t = S^\star$: with probability $1-\Delta$ all elements of v_t are assigned v_0, for which $\mathbb{P}\left(\cdot \mid S_t = S^\star, v_t\right) = \mathbb{P}\left(\cdot \mid S_t \neq S^\star, v_t\right)$; with probability Δ, we have $\mathbb{P}_0$ and $\mathbb{P}_{S^\star}$ follows two categorical distributions with no-purchase probabilities to be $1 - g(Kv_0)$ and $1 - \mathbb{E}_\mu[g(\sum_{i=1}^K V_i)]$, purchase probabilities of item $j \in S^\star$ to be $\frac{g(Kv_0)}{K}$ and $\frac{\mathbb{E}_\mu[g(\sum_{i=1}^K V_i)]}{K}$, respectively, under the two instances. By applying Lemma C.3, we conclude:

$$\begin{aligned}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_0\|\mathbb{P}_{S^\star}) &\leq \Delta^2\gamma^2 \cdot \left(\frac{1}{\mathbb{E}_\mu[g(\sum_{i=1}^K V_i)]} + \frac{1}{1 - \mathbb{E}_\mu[g(\sum_{i=1}^K V_i)]}\right)\mathbb{E}_0[T_{S^\star}] \\ &\triangleq \Delta^2\gamma_K^2 \cdot \mathbb{E}_0[T_{S^\star}],\end{aligned}$$

where $\gamma_K = \gamma \cdot \sqrt{\frac{1}{\mathbb{E}_\mu[g(\sum_{i=1}^K V_i)]} + \frac{1}{1-\mathbb{E}_\mu[g(\sum_{i=1}^K V_i)]}}$, which is a constant that only depends on K and does not involve M or T. Taking the result back to (62), we further conclude:

$$\begin{aligned}
\sup_{\mathcal{A}} \mathcal{R}(T;\pi,v) &\geq \frac{\Delta\gamma}{\binom{M}{K}} \cdot \left[\binom{M}{K} T - T - T \sum_{|S^\star|=K} \sqrt{\mathrm{D_{KL}}(\mathbb{P}_0 \| \mathbb{P}_{S^\star})/2} \right] \\
&\geq \frac{\Delta\gamma}{\binom{M}{K}} \cdot \left[\binom{M}{K} T - T - \gamma_K T\Delta \sum_{|S^\star|=K} \sqrt{\mathbb{E}_0[T_{S^\star}]/2} \right] \\
&\overset{(i)}{\geq} \frac{\Delta\gamma}{\binom{M}{K}} \cdot \left[\binom{M}{K} T - T - \gamma_K T\Delta \sqrt{\binom{M}{K} T/2} \right],
\end{aligned}$$

where (i) is from the Cauchy-Schwarz inequality. By choosing $\Delta = \min\left\{1, \sqrt{\binom{M}{K}} / \left(\sqrt{2}\gamma_K\sqrt{T}\right)\right\}$, the regret is then lower bounded by the minimum between $\Omega(T)$ and

$$\frac{\Delta\gamma}{\binom{M}{K}} \left[\binom{M}{K} T - T - \binom{M}{K} \frac{T}{2} \right] = \Omega\left(\sqrt{\binom{N}{K} T} \right),$$

which leads to our result in the theorem. □

It is worth emphasizing that the lower bound established in this proof is with respect to a *randomized* adversary, and we note that it matches the known upper bounds up to constant factors, and thereby confirms the tightness of the lower bound.

7.2 From Lower to Upper Bounds: Duality in Le Cam's Method

Le Cam's two-point method is classically known as a tool for establishing lower bounds in statistical learning. By reducing a complex estimation problem to a binary hypothesis test between two distributions, one can derive minimax lower bounds in a structured and interpretable manner. However, advanced theoretical developments have revealed its duality: under suitable conditions, optimizing over these two-point constructions can also yield tight upper bounds on minimax risk. This reinterpretation—referred to as the dualizing Le Cam's method—offers a variational perspective that unifies lower and upper bounds within a common geometric framework. Here we will introduce the high-level idea following Polyanskiy and Wu (2019), along with its main results and insights.

A Dual Characterization of Minimax Risk via Le Cam's Method

To illustrate the idea, we consider a statistical setting that arises empirical Bayes, where the learner does not observe the parameters directly but instead sees samples drawn from a latent mixture model.

Let Θ be a measurable parameter space, $\mathcal{X}$ the observation space, and for each $\theta \in \Theta$, let $\mathbb{P}_\theta$ denote the distribution of the observable data conditioned on parameter θ. We are given a class of mixing distributions Π, and for any $\pi \in \Pi$, we define the marginal mixture distribution on $\mathcal{X}$ as:

$$\pi\mathbb{P} \triangleq \int_\Theta \mathbb{P}_\theta \pi(\mathrm{d}\theta),$$

i.e., the law of a random variable $X \sim \mathbb{P}_\theta$ when the latent parameter $\theta \sim \pi$. Intuitively, this corresponds to a two-stage data generation process: first draw a latent parameter $\theta \sim \pi$, then draw an observation $X \sim \mathbb{P}_\theta$.

Let $X_1, \cdots, X_N$ denote i.i.d. observations drawn from the marginal distribution $\pi\mathbb{P}$. The learner does not observe the latent θ_i, but only the data X_i, which are thus indirectly influenced by π. Based on the observed sample $X^N = (X_1, \cdots, X_N) \sim (\pi\mathbb{P})^{\otimes N}$, the goal is to estimate a **linear** functional $\varphi(\pi) \in \mathbb{R}$ of the underlying mixing distribution π.

Let $\hat{\varphi} : \mathcal{X}^N \to \mathbb{R}$ be any estimator of $\varphi(\pi)$, and define the squared error loss

$$\ell(\varphi(\pi), \hat{\varphi}) \triangleq \left(\hat{\varphi}\left(X^N\right) - \varphi(\pi)\right)^2$$

Then the minimax risk over the class Π is defined by

$$\mathcal{R}^\star \triangleq \inf_{\hat{\varphi}} \sup_{\pi \in \Pi} \mathbb{E}_{X^N \sim (\pi\mathbb{P})^{\otimes N}} \left[\ell(\varphi(\pi), \hat{\varphi})\right].$$

Directly analyzing $\mathcal{R}^*$ over an infinite-dimensional space of distributions Π is generally intractable. However, when φ is linear and Π is convex, the two-point construction from Le Cam's method can be optimized to yield not just a lower bound, but also a tight upper bound on $\mathcal{R}^*$, thus revealing a duality between minimax risk and hypothesis testing. Specifically, we have the result holds:

$$\frac{1}{(1+\sqrt{e})^2} \cdot \delta_{\chi^2}\left(\frac{1}{\sqrt{N}}\right)^2 \leq \mathcal{R}^\star \leq \delta_{\chi^2}\left(\frac{1}{\sqrt{N}}\right)^2, \tag{66}$$

where $\delta_{\chi^2}(t) \triangleq \sup\left\{\varphi\left(\pi'\right) - \varphi(\pi) : \chi^2\left(\pi'\mathbb{P} \| \pi\mathbb{P}\right) \leq t^2, \pi, \pi' \in \Pi\right\}$ and $\chi^2(\mathbb{P}\|\mathbb{Q}) \triangleq \int \left(\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}} - 1\right)^2 \mathrm{d}\mathbb{Q}$ denotes the χ^2-divergence for $\mathbb{P} \ll \mathbb{Q}$; and the upper bound can be achieved by an estimator of the form $\hat{\varphi} = \frac{1}{N}\sum_{i=1}^N g\left(X_i\right)$ for some $g \in \mathcal{F}$, where $\mathcal{F}$ is a suitably rich class of test function under some regularity conditions.

We note that in the LHS of (66), the lower bound is derived according to a specialized strengthen version of the Le Cam's method tailored for **the squared loss function** ℓ with a stronger χ-square divergence instead of TV distance by Brown and Low (1996). As can be seen, for the lower bound, the goal is to find the worst-case pair (π, π') that achieves the maximum gap in the functional under the χ-square divergence constraint, and we refer to this as the primal problem in our formulation.

For the RHS of (66): the dual problem is to find a (plug-in) estimator $\hat{\varphi}$ that minimizes the quadratic risk. Specifically, the objective is to maximize the following bias-variance tradeoff for $\hat{\varphi} = \frac{1}{N}\sum_{i=1}^{N} g(X_i)$:

$$\mathbb{E}_{X^N \sim (\pi\mathbb{P})^{\otimes N}}\left[\ell(\varphi(\pi), \hat{\varphi})\right] \leq \frac{1}{N}\operatorname{Var}_{\pi\mathbb{P}}[g] + |\varphi(\pi) - \pi\mathbb{P}g|^2.$$

Taking worst-case π and optimizing over g we get:

$$\sqrt{\mathcal{R}^\star} \leq \inf_{g\in\mathcal{F}} \sup_{\pi\in\Pi} \left\{ \frac{1}{\sqrt{N}}\sqrt{\operatorname{Var}_{\pi\mathbb{P}}[g]} + |\varphi(\pi) - \pi\mathbb{P}g| \right\} \triangleq \delta_{\mathrm{bv}}\left(\frac{1}{\sqrt{N}}\right).$$

And through some technical lemmas, we have that: for any $t > 0$, $\delta_{\mathrm{bv}}(t) \leq \delta_{\chi^2}(t)$ holds. This further leads to the upper bound of minimax risk in (66), with the supremum achieved by some $\pi_*, \pi'_* \in \Pi$ such that $\delta_{\chi^2}(t) = \varphi(\pi'_*) - \varphi(\pi_*)$ and $\chi^2(\pi'_*\mathbb{P}\|\pi_*\mathbb{P}) \leq t^2$.

This dual characterization offers a number of important insights. First, it demonstrates that Le Cam's method is not limited to producing lower bounds: under appropriate structural assumptions (e.g., a linear target functional and a convex family of distributions), it may also implicitly encodes an optimal upper bound strategy. The usual view of Le Cam as a worst-case two-point reduction can be reinterpreted through the lens of convex duality: the primal problem searches for a worst-case pair of distributions with maximal functional gap under a divergence constraint, while the dual problem minimizes the estimation risk over plug-in estimators based on linear statistics, and (66) shows that they are equivalent. This dual view reveals that the two-point construction is not merely a lower bound technique, but a variational certificate of optimality.

Despite these strengths, the framework is not without limitations. Most notably, it is currently restricted to smooth functional estimation problems under the quadratic risk, where the parameter of interest is a linear mapping over a convex set of distributions. In contrast, many real-world problems in OM would suffer from more complex risk measures and decision rules, which lie outside the domain of applicability of the current theory. Nevertheless, this duality perspective offers a compelling blueprint for future work. It suggests that minimax optimality in complex decision problems may be approached via a convex dual optimization over estimators, rather than worst-case instance constructions. This raises the possibility of deriving upper bounds in OM problems without requiring explicit estimator construction.

In summary, this development highlights once again the "simple but powerful" nature of Le Cam's method. What begins as a basic inequality based on two distributions can, under the right structure, support a full characterization of minimax risk. It provides not only clean lower bounds, but also guidance on estimator design and achievable performance, demonstrating that Le Cam's legacy reaches far beyond hypothesis testing and remains central in modern statistical theory.

7.3 Beyond Le Cam: Related Information-Theoretic Tools

This survey has centered on Le Cam's two-point method as a unifying framework for deriving minimax lower bounds across a wide spectrum of decision-making problems in operations management. Despite its simplicity—relying on the construction of just two problem instances—the method proves remarkably powerful. By identifying pairs of environments that are statistically indistinguishable yet induce different optimal decisions, Le Cam's approach enables us to establish sharp lower bounds in diverse domains, from inventory control to scheduling queuing.

While Le Cam's method serves as a foundational building block, it is by no means the only tool available in the information-theoretic arsenal for proving lower bounds. Several alternative techniques generalize this two-point construction to richer setups and broader estimation regimes:

- The Assouad's Lemma: it extends Le Cam's idea by constructing a hypercube of problem instances indexed by binary strings of length m. Instead of reducing the minimax problem to a single binary test, it decomposes it into m coordinate-wise binary testing problems. This modular decomposition is especially powerful in high-dimensional settings with a product structure. Assouad's construction trades global complexity for a sum of local ones.

- The Fano's Inequality: it provides another multi-instance reduction strategy, typically used when the parameter space admits a large packing set. Fano's Inequality bounds the minimax risk in terms of mutual information between the observations and the underlying parameter, and is particularly effective when the hypothesis space is finite but large. Unlike Assouad, which localizes via binary reductions, Fano focuses on a single information-theoretic tradeoff across all the m hypotheses.

- The van Trees Inequality: it is a Bayesian analogue of the Cramér-Rao bound, providing lower bounds that incorporate prior information over continuous parameter spaces. Van Trees inequality is especially useful in smooth parametric models where the Fisher information structure is tractable. This method captures problem-dependent difficulty and is commonly used to establish sharp bounds for parametric estimation problems.

Throughout this survey, we have seen traces of these alternative techniques either explicitly or implicitly. In Section 4, the bibliographic remarks highlight the use of the van Trees inequality in deriving problem-dependent lower bounds for parametric dynamic pricing problems. In Section 5, we observed that regret lower bounds for stochastic multi-armed bandits can often be recovered via Fano's inequality, offering more compact proofs in some cases; In Section 6, we pointed out that the Assouad's Lemma can be directly applied as a pair-wise extension of Le Cam's two-point method to prove a regret lower bound for the

distributionally robust policy learning, where the underlying uncertainty set has a natural product structure over covariate groups.

These examples demonstrate that while Le Cam's method is broadly applicable, other tools may offer advantages depending on the geometry of the parameter space, the nature of the decisions, and the structure of the feedback. As operations management problems grow more complex—especially in networked or partially observed environments—we expect these alternative techniques to play an increasingly prominent role for the sharp regret bounds.

A comprehensive synthesis of these methods and their respective domains of strength within OM remains an open direction for future research. We view this survey as a first step toward that broader objective.

7.4 Summary and Bibliographic Remarks

Again, we first summarize the regret lower bounds derived in this section. The following table collects the main results:

Table 11: Regret Lower Bounds in Advanced Applications

	Problem Setting	**Lower Bound**
App.24	D/D/1+M queue	$\Omega\left(\log T\right)$
App.25	Adversarial MNL assortment	$\Omega\left(\sqrt{M^K T}\right)$

In addition to the developments discussed above, we also highlight a recent line of research that extends Le Cam's two-point method through the Decision-Estimation Coefficient (DEC) framework. Introduced by Foster et al. (2021) and further developed in Foster et al. (2023), Glasgow and Rakhlin (2023) and Chen et al. (2024), the DEC framework generalizes the dualizing Le Cam approach to a broader class of risk measures beyond the quadratic risk. It is designed to capture the fundamental difficulty of interactive learning problems including bandits and Markov decision processes. At its core, DEC quantifies the minimum statistical information required to distinguish hypotheses under decision-relevant environments. This complexity quantity, rooted in Le Cam-style reductions, allows for both lower and upper bounds on regret, and offers a unified view of learning in interactive settings. However, how such theoretical tools can be used to analyze concrete OM problems remains an open and important direction. In particular, understanding how the DEC framework can benefit decision-making in OM: by helping to derive problem-specific lower or upper bounds; or by revealing structural insights about regret growth and learning difficulty, poses several conceptual and technical challenges. Bridging this gap between abstract statistical complexity and domain-specific models is a promising avenue for future research, and could potentially lead to a deeper understanding of the fundamental limits of learning in OM settings.

A Appendix of Section 2

Proof of Lemma 2.1.
(a) Consider the function

$$\psi(x) = x \log x - x + 1, \quad x \geq 0$$

where $0 \log 0 \triangleq 0$. Observe that $\psi(0) = 1, \psi(1) = 0, \psi'(1) = 0, \psi''(x) = 1/x \geq 0$, and $\psi(x) \geq 0, \forall x \geq 0$. In addition, we have

$$\left(\frac{4}{3} + \frac{2}{3}x\right) \psi(x) \geq (x-1)^2, \quad x \geq 0 \tag{67}$$

Obviously, the inequality holds for $x = 0$. For $x > 0$, the function

$$g(x) = (x-1)^2 - \left(\frac{4}{3} + \frac{2}{3}x\right) \psi(x)$$

satisfies

$$g(1) = 0, \quad g'(1) = 0, \quad g''(x) = -\frac{4\psi(x)}{3x} \leq 0.$$

Thus, for ξ satisfying $|\xi - 1| < |x - 1|$, we have

$$g(x) = g(1) + g'(1)(x-1) + \frac{g''(\xi)}{2}(x-1)^2 = -\frac{4\psi(\xi)}{6\xi}(x-1)^2 \leq 0,$$

which proves the inequality (67). From (67), we obtain that if $P \ll Q$,

$$\begin{aligned}
\|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}} = \frac{1}{2}\int |\mathrm{d}\mathbb{P} - \mathrm{d}\mathbb{Q}| &= \frac{1}{2}\int_{\mathrm{d}\mathbb{Q}>0} \left|\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}} - 1\right| \mathrm{d}\mathbb{Q} \\
&\leq \frac{1}{2}\int_{\mathrm{d}\mathbb{Q}>0} \mathrm{d}\mathbb{Q} \sqrt{\left(\frac{4}{3} + \frac{2\mathrm{d}\mathbb{P}}{3\mathrm{d}\mathbb{Q}}\right) \psi\left(\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}\right)} \\
&\leq \frac{1}{2}\sqrt{\int \left(\frac{4\mathrm{d}\mathbb{Q}}{3} + \frac{2\mathrm{d}\mathbb{P}}{3}\right)} \sqrt{\int_{\mathrm{d}\mathbb{Q}>0} \mathrm{d}\mathbb{Q}\ \psi\left(\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}\right)} \\
&= \sqrt{\frac{1}{2}\int_{\mathrm{d}\mathbb{P}\mathrm{d}\mathbb{Q}>0} \mathrm{d}\mathbb{P} \log \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}} = \sqrt{\frac{1}{2}\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q})}
\end{aligned}$$

where the second inequality follows from Cauchy–Schwarz inequality. If $P \not\ll Q$. the inequality is straightforward.

(b) WLOG assume that $\mathbb{P} \ll \mathbb{Q}$. Fix any $A \in \mathcal{F}$, we have

$$\mathbb{P}(A) + \mathbb{Q}(A^c) = \int_A \mathrm{d}\mathbb{P} + \int_{A^c} \mathrm{d}\mathbb{Q} \geq \int \min\{\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q}\}.$$

In addition, since $\int \min\{\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q}\} + \int \max\{\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q}\} = 2$, by Cauchy-Schwarz inequality, we have

$$\begin{aligned}\left(\int \sqrt{\mathrm{d}\mathbb{P}\mathrm{d}\mathbb{Q}}\right)^2 &= \left(\int \sqrt{\min\{\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q}\}\max\{\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q}\}}\right)^2 \\ &\leq \int \min\{\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q}\} \int \max\{\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q}\} \\ &\leq 2\int \min\{\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q}\}.\end{aligned}$$

In addition, by Jensen inequality, we have

$$\begin{aligned}\left(\int \sqrt{\mathrm{d}\mathbb{P}\mathrm{d}\mathbb{Q}}\right)^2 &= \exp\left(2\log \int_{\mathrm{d}\mathbb{P}\mathrm{d}\mathbb{Q}>0} \sqrt{\mathrm{d}\mathbb{P}\mathrm{d}\mathbb{Q}}\right) = \exp\left(2\log \int_{\mathrm{d}\mathbb{P}\mathrm{d}\mathbb{Q}>0} \sqrt{\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}}\mathrm{d}\mathbb{P}\right) \\ &\geq \exp\left(2\int_{\mathrm{d}\mathbb{P}\mathrm{d}\mathbb{Q}>0} \log \sqrt{\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}}\mathrm{d}\mathbb{P}\right) = \exp\left(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q})\right). \qquad (68)\end{aligned}$$

Therefore, by combining these results, we have

$$\mathbb{P}(A) + \mathbb{Q}(A^c) \geq \frac{1}{2}\left(\int \sqrt{\mathrm{d}\mathbb{P}\mathrm{d}\mathbb{Q}}\right)^2 \geq \frac{1}{2}\exp\left(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q})\right).$$

Applying Scheffé's Theorem and inequality (68), we have

$$\begin{aligned}1 - \|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}}^2 &= \left(1 - \|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}}\right)\left(1 + \|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}}\right) \\ &= \int \min(\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q}) \int \max(\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q}) \\ &\geq \left(\int \sqrt{\min(\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q})\max(\mathrm{d}\mathbb{P}, \mathrm{d}\mathbb{Q})}\right)^2 \\ &= \left(\sqrt{\mathrm{d}\mathbb{P}\mathrm{d}\mathbb{Q}}\right)^2 \geq \exp\left(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q})\right).\end{aligned}$$

That is,

$$\|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}} \leq \sqrt{1 - \exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q}))} \leq 1 - \frac{1}{2}\exp(-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{Q})),$$

where the last inequality follows from the fact that $\sqrt{1-c} \leq 1 - \frac{1}{2}c$ for all $c \in [0, 1]$. □

Proof of Lemma 2.2. Assume $\mathbb{P}_X \ll \mathbb{Q}_X$, and define the Radon–Nikodym derivative $r(x) \triangleq \frac{\mathrm{d}\mathbb{P}_X}{\mathrm{d}\mathbb{Q}_X}(x)$. Then the induced marginals satisfy:

$$\frac{\mathrm{d}\mathbb{P}_Y}{\mathrm{d}\mathbb{Q}_Y}(y) = \frac{\int \mathbb{P}_{Y|X}(y|x) r(x)\, \mathbb{Q}_X(\mathrm{d}x)}{\int \mathbb{P}_{Y|X}(y|x)\, \mathbb{Q}_X(\mathrm{d}x)} = \mathbb{E}_{\mathbb{Q}_{X|Y=y}}[r(X)],$$

where $\mathbb{Q}_{X|Y=y}$ is the posterior distribution of X given $Y = y$ under the joint distribution $\mathbb{Q}_{X,Y} = \mathbb{Q}_X \mathbb{P}_{Y|X}$. Since f is convex, Jensen's inequality implies that

$$f\left(\frac{\mathrm{d}\mathbb{P}_Y}{\mathrm{d}\mathbb{Q}_Y}(y)\right) = f\left(\mathbb{E}_{\mathbb{Q}_{X|Y=y}}[r(X)]\right) \leq \mathbb{E}_{\mathbb{Q}_{X|Y=y}}\left[f(r(X))\right].$$

Now integrate both sides with respect to $\mathbb{Q}_Y$, we have

$$\begin{aligned}
\mathrm{D}_f(\mathbb{P}_Y \| \mathbb{Q}_Y) &= \int f\left(\frac{\mathrm{d}\mathbb{P}_Y}{\mathrm{d}\mathbb{Q}_Y}(y)\right) \mathbb{Q}_Y(\mathrm{d}y) \\
&\leq \int \mathbb{E}_{\mathbb{Q}_{X|Y=y}}[f(r(X))]\, \mathbb{Q}_Y(\mathrm{d}y) \\
&= \int f(r(x))\, \mathbb{Q}_X(\mathrm{d}x) = D_f(\mathbb{P}_X \| \mathbb{Q}_X).
\end{aligned}$$

□

Proof of Lemma 2.3.

1. Non-negativity: Since the KL divergence is a special case of the f-divergence with $f(x) = x \log x$, which is convex on $(0, \infty)$, its non-negativity follows directly from Jensen's inequality. All f-divergences are non-negative, and equality holds if and only if $\mathbb{P} = \mathbb{Q}$ almost surely.

2. Additivity for Product Measures: Suppose $\mathbb{P} = \bigotimes_{i=1}^n \mathbb{P}_i$ and $\mathbb{Q} = \bigotimes_{i=1}^n \mathbb{Q}_i$. Then the Radon–Nikodym derivative factorizes as follows:

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}(x_1, \cdots, x_n) = \prod_{i=1}^n \frac{\mathrm{d}\mathbb{P}_i}{\mathrm{d}\mathbb{Q}_i}(x_i).$$

Taking the logarithm and integrating it, we have

$$\begin{aligned}
\mathrm{D}_{\mathrm{KL}}(\mathbb{P} \| \mathbb{Q}) &= \int \log\left(\prod_{i=1}^n \frac{\mathrm{d}\mathbb{P}_i}{\mathrm{d}\mathbb{Q}_i}(x_i)\right) \mathrm{d}\mathbb{P}(x_1, \cdots, x_n) \\
&= \sum_{i=1}^n \int \log\left(\frac{\mathrm{d}\mathbb{P}_i}{\mathrm{d}\mathbb{Q}_i}(x_i)\right) \mathrm{d}\mathbb{P}_i(x_i) \\
&= \sum_{i=1}^n \mathrm{D}_{\mathrm{KL}}(\mathbb{P}_i \| \mathbb{Q}_i).
\end{aligned}$$

3. Joint Convexity: Let $\mathbb{P}_1, \mathbb{P}_2$ and $\mathbb{Q}_1, \mathbb{Q}_2$ be probability distributions over a measurable space $(\mathcal{X}, \mathcal{F})$. For $0 < \lambda < 1$, define the mixtures:

$$\mathbb{P}_\lambda = \lambda \mathbb{P}_1 + (1 - \lambda)\mathbb{P}_2, \quad \mathbb{Q}_\lambda = \lambda \mathbb{Q}_1 + (1 - \lambda)\mathbb{Q}_2.$$

Then the KL divergence between $\mathbb{P}_\lambda$ and $\mathbb{Q}_\lambda$ is given by

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_\lambda \| \mathbb{Q}_\lambda) = \int \log\left(\frac{\mathbb{P}_\lambda}{\mathbb{Q}_\lambda}\right) \mathrm{d}\mathbb{P}_\lambda.$$

To establish convexity, we apply the log-sum inequality, which states that for all $a_1, a_2, b_1, b_2 \geq 0$ and $\lambda \in (0, 1)$,

$$(\lambda a_1 + (1-\lambda)a_2) \log\left(\frac{\lambda a_1 + (1-\lambda)a_2}{\lambda b_1 + (1-\lambda)b_2}\right) \leq \lambda a_1 \log\left(\frac{a_1}{b_1}\right) + (1-\lambda)a_2 \log\left(\frac{a_2}{b_2}\right).$$

Applying this inequality pointwise to the densities of $\mathbb{P}_i$ and $\mathbb{Q}_i$ (assuming all measures are absolutely continuous with respect to a common dominating measure), and integrating, we obtain:

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_\lambda \| \mathbb{Q}_\lambda) \leq \lambda \mathrm{D}_{\mathrm{KL}}(\mathbb{P}_1 \| \mathbb{Q}_1) + (1-\lambda)\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_2 \| \mathbb{Q}_2).$$

□

B Appendix of Section 3

Omitted Details in the Proof of Theorem 3.1. Here we provide the calculation details of the separation condition in (4) and (5). By the definitions of $\mathbb{P}_1, \mathbb{P}_2$, the corresponding optimal order quantities are given by $q^\star_{F_1} = 0,\ q^\star_{F_2} = M$, and the expected cost functions under each instance are:

$$\begin{aligned}
C\left(q;\mathbb{P}_1\right) &= hq(\rho+\Delta) + b(M-q)(1-\rho-\Delta), \quad \forall q > M/2,\\
C\left(q^\star_{F_1};\mathbb{P}_1\right) &= bM(1-\rho-\Delta),\\
C\left(q;\mathbb{P}_2\right) &= hq(\rho-\Delta) + b(M-q)(1-\rho+\Delta), \quad \forall q \le M/2,\\
C\left(q^\star_{F_2};\mathbb{P}_2\right) &= hM(\rho-\Delta).
\end{aligned}$$

Note that $\rho = \frac{b}{b+h}$. Using this, we compute the cost differences as follows:

$$\begin{aligned}
C(q;\mathbb{P}_1) - C(q^\star_{F_1};\mathbb{P}_1) &= hq(\rho+\Delta) - bq(1-\rho-\Delta)\\
&= (b+h)q\Delta \ge \frac{M\Delta(b+h)}{4},\ \forall q > M/2,\\
C(q;\mathbb{P}_2) - C(q^\star_{F_2};\mathbb{P}_2) &= h(q-M)(\rho-\Delta) + b(M-q)(1-\rho+\Delta)\\
&= (h+b)(M-q)\Delta \ge \frac{M\Delta(b+h)}{4},\ \forall q \le M/2,
\end{aligned}$$

□

Omitted Details in the Proof of Theorem 3.3. Here we provide the calculation details of the separation condition in (8) and (9). By the definitions of $\mathbb{P}_1, \mathbb{P}_2$, the corresponding optimal order quantities are given by $q^\star_{F_1} = \alpha,\ q^\star_{F_2} = 3\alpha + 1$, and the expected cost functions under each instance are:

$$\begin{aligned}
C\left(q;\mathbb{P}_1\right) &= h\int_0^q (q-x)\mathrm{d}F_1(x) + b\int_q^{4\alpha+1} (x-q)\mathrm{d}F_1(x), \quad \forall q\\
C\left(q;\mathbb{P}_2\right) &= h\int_0^q (q-x)\mathrm{d}F_2(x) + b\int_q^{4\alpha+1} (x-q)\mathrm{d}F_2(x), \quad \forall q.
\end{aligned}$$

We compute the cost differences as follows:

$$\begin{aligned}
C(q;\mathbb{P}_1) - C(q^\star_{F_1};\mathbb{P}_1) &= (b+h)\int_\alpha^q (q-x)\mathrm{d}F_1(x) \\
&\geq (b+h)\int_\alpha^{2\alpha+1/2} (2\alpha+1/2-x)\mathrm{d}F_1(x) \\
&= (b+h)\int_\alpha^{2\alpha} \varepsilon\cdot(x-2\alpha-1/2)\mathrm{d}x \\
&= \frac{\alpha^2+\alpha}{2}\varepsilon(b+h) \geq \frac{(b+h)\alpha\varepsilon}{2}, \quad \forall q > 2\alpha+1/2, \\
C(q;\mathbb{P}_2) - C(q^\star_{F_2};\mathbb{P}_2) &= (b+h)\int_q^{3\alpha+1} (x-q)\mathrm{d}F_2(x) \\
&\geq (b+h)\int_{2\alpha+1/2}^{3\alpha+1} (x-2\alpha-1/2)\mathrm{d}F_2(x) \\
&= (b+h)\int_{2\alpha+1}^{3\alpha+1} \varepsilon\cdot(x-2\alpha-1/2)\mathrm{d}x \\
&= \frac{\alpha^2+\alpha}{2}\varepsilon(b+h) \geq \frac{(b+h)\alpha\varepsilon}{2}, \quad \forall q \leq 2\alpha+1/2,
\end{aligned}$$

□

Proof of Claim 1. By our definitions of $\mathbb{P}_1, \mathbb{P}_2$, we know that $q_1^\star = \frac{M}{2}$, $q_2^\star = M$. To show the $(1+\frac{\epsilon}{20})$-optimal solution sets under $\mathbb{P}_1$ and $\mathbb{P}_2$ are disjoint, it suffices to show that

$$C\left(\frac{3M}{4};\mathbb{P}_1\right) \geq \left(1+\frac{\epsilon}{10}\right) C\left(\frac{M}{2};\mathbb{P}_1\right),\ C\left(\frac{3M}{4};\mathbb{P}_2\right) \geq \left(1+\frac{\epsilon}{10}\right) C(M;\mathbb{P}_2). \tag{69}$$

Since $C(q;\mathbb{P}_1),\ C(q;\mathbb{P}_2)$ are convex functions, their sets of minima are intervals. Over the domain $q \in [\frac{M}{2}, M]$, the cost functions $C(q;\mathbb{P}_1), C(q;\mathbb{P}_2)$ can be expressed as:

$$\begin{aligned}
C(q;\mathbb{P}_1) &= \frac{hb-h^2}{b+h}q + \frac{h^2(1+\epsilon)}{b+h}\left(q-\frac{M}{2}\right) + \frac{bh(1-\epsilon)}{b+h}(M-q) \\
C(q;\mathbb{P}_2) &= \frac{hb-h^2}{b+h}q + \frac{h^2(1-\epsilon)}{b+h}\left(q-\frac{M}{2}\right) + \frac{bh(1+\epsilon)}{b+h}(M-q)
\end{aligned}$$

Then, we have

$$
\begin{aligned}
C\left(\frac{M}{2};\mathbb{P}_1\right) &= \frac{hbM}{h+b}\left(1-\frac{h}{2b}-\frac{\epsilon}{2}\right)\\
C\left(\frac{3M}{4};\mathbb{P}_1\right) &= \frac{hbM}{h+b}\left(1-\frac{h}{2b}+\frac{h\epsilon}{4b}-\frac{\epsilon}{4}\right)\\
C\left(\frac{3M}{4};\mathbb{P}_2\right) &= \frac{hbM}{h+b}\left(1-\frac{h}{2b}-\frac{h\epsilon}{4b}+\frac{\epsilon}{4}\right)\\
C\left(M;\mathbb{P}_2\right) &= \frac{hbM}{h+b}\left(1-\frac{h}{2b}-\frac{h\epsilon}{2b}\right)
\end{aligned}
$$

Since $\epsilon, h, b, M > 0$, a straightforward calculation shows that

$$
\begin{aligned}
1-\frac{h}{2b}+\frac{h\epsilon}{4b}-\frac{\epsilon}{4} &\geq \left(1+\frac{\epsilon}{10}\right)\cdot\left(1-\frac{h}{2b}-\frac{\epsilon}{2}\right)\\
1-\frac{h}{2b}-\frac{h\epsilon}{4b}+\frac{\epsilon}{4} &\geq \left(1+\frac{\epsilon}{10}\right)\cdot\left(1-\frac{h}{2b}-\frac{h\epsilon}{2b}\right)
\end{aligned}
$$

thus establishing inequalities (69). □

Proof of Claim 2. By the definition of $\mathbb{P}_1, \mathbb{P}_2$, and the i.i.d. property of data-generating process, we have $\mathrm{D_{KL}}(\mathbb{P}_1^N \| \mathbb{P}_2^N) = N\mathrm{D_{KL}}(\mathbb{P}_1 \| \mathbb{P}_2)$ and the KL divergence $\mathrm{D_{KL}}(\mathbb{P}_1 \| \mathbb{P}_2)$ can be expressed as follows:

$$
\begin{aligned}
\mathrm{D_{KL}}(\mathbb{P}_1 \| \mathbb{P}_2) &= \frac{b-h}{b+h}\log\frac{b-h}{b-h} + \frac{h+h\epsilon}{b+h}\log\frac{h+h\epsilon}{h-h\epsilon} + \frac{h-h\epsilon}{b+h}\log\frac{h-h\epsilon}{h+h\epsilon}\\
&= \frac{2h\epsilon}{b+h}\log\left(1+\frac{2\epsilon}{1-\epsilon}\right) \leq \frac{4h\epsilon^2}{(b+h)(1-\epsilon)} \leq \frac{8h\epsilon^2}{b+h}
\end{aligned}
$$

where the first inequality follows from the fact that $\log(1+x) \leq x$ for all $x \geq -1$, and the last inequality holds by the assumption that $\epsilon < 1/2$. □

Proof of Claim 2′. By the definition of $\tilde{\mathbb{P}}_1, \tilde{\mathbb{P}}_2$, the KL divergence $\mathrm{D_{KL}}(\tilde{\mathbb{P}}_1 \| \tilde{\mathbb{P}}_2)$ can be expressed as follows:

$$
\begin{aligned}
\mathrm{D_{KL}}(\tilde{\mathbb{P}}_1 \| \tilde{\mathbb{P}}_2) &= \left(1-\kappa+\kappa\frac{b-h}{b+h}\right)\log\frac{1-\kappa+\kappa\frac{b-h}{b+h}}{1-\kappa+\kappa\frac{b-h}{b+h}} + \kappa\frac{h+h\epsilon}{b+h}\log\frac{h+h\epsilon}{h-h\epsilon} + \kappa\frac{h-h\epsilon}{b+h}\log\frac{h-h\epsilon}{h+h\epsilon}\\
&= \frac{2h\kappa\epsilon}{b+h}\log\left(1+\frac{2\epsilon}{1-\epsilon}\right)\\
&\leq \frac{4h\kappa\epsilon^2}{b+h(1-\epsilon)} \leq \frac{8h\kappa\epsilon^2}{b+h}
\end{aligned}
$$

where the first inequality follows from the fact that $\log(1+x) \leq x$ for all $x \geq -1$, and the last inequality holds by the assumption that $\epsilon < 1/2$. $\square$

C Appendix of Section 4

Omitted Details in the Proof of Theorem 4.3. In calculating the KL divergence of $\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}^s_{z_1} \| \mathbb{P}^s_{z_2} \mid \mathbf{y}_{s-1}\right)$, we rely on the following two lemmas, which are the common properties of KL-divergence:

Lemma C.1 (KL-divergence between two Bernoulli distributions)**.** *Suppose* $\mathrm{Bern}(q_1)$ *and* $\mathrm{Bern}(q_2)$ *are distributions of Bernoulli random variables with parameters* q_1 *and* q_2*, respectively, with* $q_1, q_2 \in (0,1)$*. Then*

$$\mathrm{D}_{\mathrm{KL}}\left(\mathrm{Bern}(q_1) \| \,\mathrm{Bern}(q_2)\right) \leq \frac{(q_1-q_2)^2}{q_2(1-q_2)}.$$

Proof of Lemma C.1. By applying the inequality $\log x \leq x-1, \forall x>0$, we have

$$\begin{aligned}
\mathrm{D}_{\mathrm{KL}}(\mathrm{Bern}(q_1) \| \,\mathrm{Bern}(q_2)) &= q_1 \log\left(\frac{q_1}{q_2}\right) + (1-q_1)\log\left(\frac{1-q_1}{1-q_2}\right) \\
&\leq q_1\left(\frac{q_1}{q_2}-1\right) + (1-q_1)\left(\frac{1-q_1}{1-q_2}-1\right) \\
&= \frac{(q_1-q_2)^2}{q_2(1-q_2)}
\end{aligned}$$

□

Lemma C.2 (Chain rule for KL-divergence)**.** *For joint probability mass function* $p(x,y)$ *and* $q(x,y)$*,*

$$\mathrm{D}_{\mathrm{KL}}\left(p(x,y) \| q(x,y)\right) = \mathrm{D}_{\mathrm{KL}}\left(p(x) \| q(x)\right) + \mathrm{D}_{\mathrm{KL}}\left(p(y|x) \| q(y|x)\right),$$

where

$$\begin{aligned}
\mathrm{D}_{\mathrm{KL}}\left(p(y|x) \| q(y|x)\right) &= \sum_x p(x) \sum_y p(y|x) \log \frac{p(y|x)}{q(y|x)} \\
&= \mathbb{E}_{p(x,y)} \log \frac{p(Y|X)}{q(Y|X)}.
\end{aligned}$$

Proof of Lemma C.2.

$$\begin{aligned}
\mathrm{D}_{\mathrm{KL}}(p(x,y) \| q(x,y)) &= \sum_x \sum_y p(x,y) \log \frac{p(x,y)}{q(x,y)} \\
&= \sum_x \sum_y p(x,y) \log \frac{p(x)p(y|x)}{q(x)q(y|x)} \\
&= \sum_x \sum_y p(x,y) \log \frac{p(x)}{q(x)} + \sum_x \sum_y p(x,y) \log \frac{p(y|x)}{q(y|x)} \\
&= \mathrm{D}_{\mathrm{KL}}(p(x) \| q(x)) + \mathrm{D}_{\mathrm{KL}}(p(y|x) \| q(y|x)).
\end{aligned}$$

□

Omitted Details in the Proof of Theorem 4.5. Note that in Lemma C.1, we have built the upper bound of KL-divergence for two Bernoulli distributions. As the structure of observations under the assortment problems are more complex than that for the dynamic pricing problems (in which the observations are binary), here we present the following lemma in bounding the KL-divergence between two assortment S and S' decisions.

Lemma C.3. *Suppose $\mathbb{P}$ and $\mathbb{Q}$ are two categorical distributions with parameters $(p_1, \cdots, p_n)$ and $(q_1, \cdots, q_n)$ with $\sum_{i=1}^n p_i = \sum_{i=1}^n q_i = 1$ and $0 \le p_i, q_i \le 1$ for any $i \in [n]$. Then:*

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P} \| \mathbb{Q}) \le \sum_{i=1}^{n} \frac{(p_i - q_i)^2}{q_i}.$$

Proof of Lemma C.3. By definition, we have:

$$\mathrm{D}_{\mathrm{KL}}(\mathbb{P} \| \mathbb{Q}) = \sum_{i=1}^{n} p_i \log \frac{p_i}{q_i} \overset{(i)}{\le} \sum_{i=1}^{n} p_i \cdot \frac{p_i - q_i}{q_i} = \sum_{i=1}^{n} \frac{(p_i - q_i)^2}{q_i},$$

where (i) uses the fact that $\log(1 + x) \le x$ for all $x > -1$. □

D Appendix of Section 5

Omitted Details in the Proof of Theorem 5.2. Here we present the details in calculating T_m and Δ_m: by letting $\frac{t_m}{\sqrt{t_{m-1}}} = c$, we have

$$t_M = c\sqrt{t_{M-1}} = c^{1+\frac{1}{2}} t_{M-2}^{(\frac{1}{2})^2} = \cdots = c^{1+\frac{1}{2}+\cdots+(\frac{1}{2})^{M-2}} t_1^{(\frac{1}{2})^{M-1}} = c^{2-2^{2-M}} t_1^{2^{1-M}}.$$

Since $t_M = T$ and $t_1 \ge 1$, we obtain

$$c = T^{1/(2-2^{2-M})} t_1^{1/(2-2^M)}.$$

Therefore, the regret is lower bounded by

$$\begin{aligned}
\sup_v \mathcal{R}(\mathcal{T}; \pi, v) &\ge \frac{1}{4\sqrt{e}} T^{1/(2-2^{2-M})} t_1^{1/(2-2^M)} \\
&\ge \frac{1}{4\sqrt{e}} T^{1/(2-2^{1-M})} t_1^{1/(2-2^M)} = \Omega\left(T^{1/(2-2^{1-M})}\right).
\end{aligned}$$

□

Omitted Details in the Proof of Theorem 5.4. Note that $\mathbb{P}_{v_j}^T$ denotes the marginal distribution over the trajectory $\{(a_t, r_t)\}_{t=1}^T$ generated by policy π under environment v_j. Because all arms except for arm i are the same in both environments, we have the following decomposition of KL-divergence:

$$\begin{aligned}
\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_{v_1}^T \| \mathbb{P}_{v_2}^T) &= \mathbb{E}_{v_1}\left[\sum_{t=1}^T \log \frac{\mathrm{d}\mathbb{P}_{v_1}(r_t|a_t)}{\mathrm{d}\mathbb{P}_{v_2}(r_t|a_t)}\right] \\
&= \mathbb{E}_{v_1}\left[\sum_{t=1}^T \mathbf{1}(a_t = i) \log \frac{\mathrm{d}\mathbb{P}_{v_1}(r_t|i)}{\mathrm{d}\mathbb{P}_{v_2}(r_t|i)}\right] \\
&= \mathbb{E}_{v_1}[T_i(T;\pi)] \cdot \mathrm{D}_{\mathrm{KL}}(\mathcal{N}(0,1)\|\mathcal{N}(2\Delta,1))
\end{aligned}$$

□

Proof of Lemma 5.1. Assume that $\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_k, \mathbb{P}'_k\right) < \infty$ for all $k \in [K]$. It follows that $\mathbb{P}_k \ll \mathbb{P}'_k$. Define $\lambda = \sum_{k=1}^K (\mathbb{P}_k + \mathbb{P}'_k)$, which is the measure defined by $\lambda(A) = \sum_{k=1}^K \left(\mathbb{P}_k(A) + \mathbb{P}'_k(A)\right)$ for any measurable set A. It is known that if $\frac{\mathrm{d}\mathbb{P}_v}{\mathrm{d}\mathbb{P}_{v'}} < +\infty$,

$$\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_v \| \mathbb{P}_{v'}\right) = \mathbb{E}_v\left[\log\left(\frac{\mathrm{d}\mathbb{P}_v}{\mathrm{d}\mathbb{P}_{v'}}\right)\right]$$

Let ρ be the counting measure over $[K]$, the Radon-Nikodym derivative of $\mathbb{P}_v$ with respect to the product measure $(\rho \times \lambda)^T$ with respect to the product measure is given by

$$p_{v\pi}\left(a_1, r_1, \ldots, a_T, r_T\right) = \prod_{t=1}^T \pi_t\left(a_t \mid a_1, r_1, \ldots, a_{t-1}, r_{t-1}\right) p_{a_t}\left(r_t\right)$$

The density of $\mathbb{P}_{v'}$ is identical except that p_{a_t} is replaced by p'_{a_t}. Then,

$$\log \frac{\mathrm{d}\mathbb{P}_v}{\mathrm{d}\mathbb{P}_{v'}}\left(a_1, r_1, \cdots, a_T, r_T\right) = \sum_{t=1}^T \log \frac{p_{a_t}\left(r_t\right)}{p'_{a_t}\left(r_t\right)}$$

where we used the chain rule for Radon-Nikodym derivatives and the fact that the terms involving the policy cancel. Taking expectations of both sides,

$$\mathbb{E}_v\left[\log \frac{\mathrm{d}\mathbb{P}_v}{\mathrm{d}\mathbb{P}_{v'}}\left(a_1, r_1, \ldots, a_T, r_T\right)\right] = \sum_{t=1}^T \mathbb{E}_v\left[\log \frac{p_{a_t}\left(r_t\right)}{p'_{a_t}\left(r_t\right)}\right]$$

and

$$\mathbb{E}_v\left[\log \frac{p_{a_t}\left(r_t\right)}{p'_{a_t}\left(r_t\right)}\right] = \mathbb{E}_v\left[\mathbb{E}_v\left[\log \frac{p_{a_t}\left(r_t\right)}{p'_{a_t}\left(r_t\right)}\middle|\ a_t\right]\right] = \mathbb{E}_v\left[\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_{a_t} \| \mathbb{P}'_{a_t}\right)\right]$$

where in the second equality we used that under $\mathbb{P}_v(\cdot \mid a_t)$, the distribution of r_t is $\mathrm{d}\mathbb{P}_{a_t} = p_{a_t}\mathrm{d}\lambda$. Plugging back into the previous result,

$$\begin{aligned}
\mathbb{E}_v\left[\log\frac{\mathrm{d}\mathbb{P}_v}{\mathrm{d}\mathbb{P}_{v'}}(a_1, r_1, \cdots, a_T, r_T)\right] &= \sum_{t=1}^{T}\mathbb{E}_v\left[\log\frac{p_{a_t}(r_t)}{p'_{a_t}(r_t)}\right] \\
&= \sum_{t=1}^{T}\mathbb{E}_v\left[\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_{a_t}\|\mathbb{P}'_{a_t}\right)\right] = \sum_{k=1}^{K}\mathbb{E}_v\left[\sum_{t=1}^{T}\mathbf{1}\left\{a_t = k\right\}\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_{a_t}\|\mathbb{P}'_{a_t}\right)\right] \\
&= \sum_{k=1}^{K}\mathbb{E}_v\left[T_k(T;\pi)\right]\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}_k\|\mathbb{P}'_k\right).
\end{aligned}$$

When the right-hand side of the statement is infinite, i.e., $\mathbb{P}_k \not\ll \mathbb{P}'_k$ for some $k \in [K]$, by our previous calculation, it is not hard to see that the left-hand side will also be infinite. □

Omitted Details in the Proof of Theorem 5.7.

Lemma D.1 (Properties of Q_1, Q_2 and Z_0)**.** *Let $x_{t,1} - x_{t,2} = r_t u_t$ with $r_t \geq 0, \|u_t\|_2 = 1$. Then $\theta \sim Q_1$ if and only if $\theta - 2\left(u_t^\top\theta\right)u_t \sim Q_2$ (for the current x_t). Furthermore, we have*

$$Z_0 = r_t\Delta \cdot \begin{cases} \frac{2^d}{\pi d}\binom{d}{d/2}^{-1}, & \text{if } d \text{ is even} \\ \frac{1}{2^d}\binom{d-1}{(d-1)/2}, & \text{if } d \text{ is odd} \end{cases} \geq \frac{r_t\Delta}{5\sqrt{d}}. \tag{70}$$

$$\mathbb{E}_{Q_1}(u_t^\top\theta)^2 = \mathbb{E}_{Q_2}(u_t^\top\theta)^2 = \frac{2\Delta^2}{d+1}. \tag{71}$$

Proof. Let $v_1, \cdots, v_d$ be an orthonormal basis of $\mathbb{R}^d$ with $v_1 = u_t$. By rotational invariance of the uniform distribution on spheres, we have $\left(v_1^\top\theta, v_2^\top\theta, \cdots, v_d^\top\theta\right) \sim \mathrm{Unif}\left(\Delta\mathbb{S}^{d-1}\right)$ under Q_0. Recall that

$$\frac{\mathrm{d}Q_1}{\mathrm{d}Q_0}(\theta) = \frac{r_t\langle v_1, \theta\rangle_+}{Z_0}, \quad \frac{\mathrm{d}Q_2}{\mathrm{d}Q_0}(\theta) = \frac{r_t\langle v_1, \theta\rangle_-}{Z_0}.$$

We conclude that if $\theta' = \theta - 2\left(v_1^\top\theta\right)v_1$, we have

$$\frac{\mathrm{d}Q_1}{\mathrm{d}Q_0}(\theta) = \frac{\mathrm{d}Q_2}{\mathrm{d}Q_0}\left(\theta'\right).$$

As a result, it is equivalent to have $\theta \sim Q_1$ or $\theta' = \theta - 2\left(v_1^\top\theta\right)v_1 \sim Q_2$. For the identity (70), recall that the density of $\theta = (\theta_1, \cdots, \theta_d) \sim \mathrm{Unif}\left(\Delta\mathbb{S}^{d-1}\right)$ is

$$\begin{aligned}
f(\theta) &= f(\theta_2, \cdots, \theta_d) \\
&= \left(\frac{d\pi^{d/2}\Delta^{d-1}}{\Gamma\left(\frac{d}{2}+1\right)}\right)^{-1}\frac{2\Delta}{\sqrt{\Delta^2 - \theta_2^2 - \cdots - \theta_d^2}} \cdot \mathbf{1}\left(\sum_{i=2}^{d}\theta_i^2 \leq \Delta^2\right),
\end{aligned}$$

where $\Gamma(t) = \int_0^\infty x^{t-1}e^{-x}\mathrm{d}x$ is the Gamma function. Hence, by rotational invariance, we have

$$\begin{aligned}
Z_0 = \frac{r_t}{2}\mathbb{E}_{Q_0}\left[|\theta_1|\right] &= r_t\Delta \cdot \int_{\sum_{i=2}^d \theta_i^2 \le \Delta^2} \left(\frac{d\pi^{d/2}\Delta^{d-1}}{\Gamma\left(\frac{d}{2}+1\right)}\right)^{-1} \mathrm{d}\theta_2 \cdots \mathrm{d}\theta_d \\
&= r_t\Delta \left(\frac{d\pi^{d/2}\Delta^{d-1}}{\Gamma\left(\frac{d}{2}+1\right)}\right)^{-1} \cdot \frac{\Delta^{d-1}\pi^{\frac{d-1}{2}}}{\Gamma\left(\frac{d-1}{2}+1\right)} \\
&= r_t\Delta \cdot \begin{cases} \frac{2^d}{\pi d}\binom{d}{d/2} - 1, & \text{if } d \text{ is even} \\ \frac{1}{2^d}\binom{d-1}{(d-1)/2}, & \text{if } d \text{ is odd} \end{cases}
\end{aligned}$$

Using Stirling's approximation $\sqrt{2\pi n}\left(\frac{n}{e}\right)^n \le n! \le e\sqrt{n}\left(\frac{n}{e}\right)^n$ for any $n \ge 1$, we have

$$\frac{2}{e^2}\sqrt{\frac{\pi}{n}} \le \frac{1}{2^{2n}}\binom{2n}{n} \le \frac{e}{\pi\sqrt{2n}}$$

for all $n \ge 1$, and the rest of (70) follows from the above inequalities.

As for the second moment in (71), we use the spherical coordinates

$$\begin{cases}
\theta_2 = r\cos\varphi_1 \\
\theta_3 = r\sin\varphi_1\cos\varphi_2 \\
\vdots \\
\theta_{d-1} = r\sin\varphi_1\sin\varphi_2\cdots\sin\varphi_{d-3}\cos\varphi_{d-2} \\
\theta_d = r\sin\varphi_1\sin\varphi_2\cdots\sin\varphi_{d-3}\sin\varphi_{d-2}
\end{cases}$$

to obtain

$$\begin{aligned}
&\mathbb{E}_{Q_1}\left[\left(v_1^\top\theta\right)^2\right] = \frac{r_t}{2Z_0}\cdot\mathbb{E}_{Q_0}\left[|\theta_1|^3\right] = \frac{r_t\Delta}{Z_0}\cdot\int_{\sum_{i=2}^d\theta_i^2\le\Delta^2}\left(\frac{d\pi^{d/2}\Delta^{d-1}}{\Gamma\left(\frac{d}{2}+1\right)}\right)^{-1}\left(\Delta^2-\theta_2^2-\cdots-\theta_d^2\right)\mathrm{d}\theta_2\cdots\mathrm{d}\theta_d \\
&= \frac{r_t\Delta}{Z_0}\cdot\int_0^\Delta\int_0^\pi\cdots\int_0^\pi\int_0^{2\pi}\left(\frac{d\pi^{d/2}\Delta^{d-1}}{\Gamma\left(\frac{d}{2}+1\right)}\right)^{-1}\left(\Delta^2-r^2\right)\cdot r^{d-2}\sin^{d-3}\left(\varphi_1\right)\cdots\sin\left(\varphi_{d-3}\right)\mathrm{d}r\mathrm{d}\varphi_1\cdots\mathrm{d}\varphi_{d-2} \\
&= \frac{r_t\Delta}{Z_0}\left(\frac{d\pi^{d/2}\Delta^{d-1}}{\Gamma\left(\frac{d}{2}+1\right)}\right)^{-1}\cdot\frac{2\Delta^{d+1}}{d^2-1}\cdot\frac{\Gamma\left(\frac{d-2}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{d-1}{2}\right)}\cdot\frac{\Gamma\left(\frac{d-3}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{d-2}{2}\right)}\cdots\frac{\Gamma(1)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{2}\right)}\cdot 2\pi \\
&= \frac{r_t\Delta}{Z_0}\left(\frac{d\pi^{d/2}\Delta^{d-1}}{\Gamma\left(\frac{d}{2}+1\right)}\right)^{-1}\cdot\frac{2\Delta^{d+1}}{d^2-1}\cdot\frac{2\pi^{\frac{d-1}{2}}}{\Gamma\left(\frac{d-1}{2}\right)} = \frac{2\Delta^2}{d+1}.
\end{aligned}$$

□

Omitted Details in the Proof of Theorem 5.8.

Proof of Lemma 5.3. Since the union of $\{B_m\}_{m\in[M]}$ is the whole space, by a union bound, we have $\sum_{m=1}^{M}\mathbb{P}\left(B_m\right)\geq\mathbb{P}\left(\cup_{m=1}^{M}B_m\right)=1$ for any probability measure $\mathbb{P}$. Hence $\mathbb{P}\left(B_m\right)\geq 1/M$ for at least one m. For any $m\in[M]$,

$$\begin{aligned}&\sum_{t=T_{m-1}+1}^{T_m}\mathbb{E}_x\left[Z_0\cdot\mathbb{E}_{Q_1}\mathbb{P}_{\theta,x}(B_m)\right]\\&=\Delta\cdot\sum_{t=T_{m-1}+1}^{T_m}\mathbb{E}_x\mathbb{E}_{\tilde{Q}_0}\left[\langle x_{t,1}-x_{t,2},\theta_m\rangle_+\cdot\mathbb{P}_{\theta,x}(B_m)\right],\end{aligned}$$

where θ_m are independently sampled from $\tilde{Q}_0=\mathrm{Unif}\left(\mathbb{S}^{d-1}\right)$. Conditional on $\{x_{t,a}\}_{t\leq T_{m-1},a\in[K]}$, we have $\mathbf{1}\{B_m\}$ is independent of $\{x_{t,a}\}_{t>T_{m-1},a\in[K]}$. Hence,

$$\mathbb{P}_{\theta,x}(B_m)=\mathbb{P}_\theta\left(t_{m-1}\leq T_{m-1}<T_m\leq t_m\mid\{x_{1,a}\}_{a\in[K]},\cdots,\left\{x_{T_{m-1},a}\right\}_{a\in[K]}\right).$$

Then, based on the independence between context across t, we have

$$\begin{aligned}\mathbb{E}_x\mathbb{E}_{\tilde{Q}_0}\left[\langle x_{t,1}-x_{t,2},\theta_m\rangle_+\cdot\mathbb{P}_{\theta,x}(B_m)\right]&=\mathbb{E}_{\tilde{Q}_0}\left[\mathbb{E}_x\left[\langle x_{t,1}-x_{t,2},\theta_m\rangle_+\right]\cdot\mathbb{E}_x\left[\mathbb{P}_{\theta,x}\left(B_m\right)\right]\right]\\&=\mathbb{E}_{\tilde{Q}_0}\left[\mathbb{E}_x\left[\langle x_{T,1}-x_{T,2},\theta_m\rangle_+\right]\cdot\mathbb{E}_x\left[\mathbb{P}_{\theta,x}\left(B_m\right)\right]\right]\\&=\mathbb{E}_x\mathbb{E}_{\tilde{Q}_0}\left[\langle x_{T,1}-x_{T,2},\theta_m\rangle_+\cdot\mathbb{P}_{\theta,x}(B_m)\right].\end{aligned}$$

Then, we can write

$$\begin{aligned}\sum_{t=T_{m-1}+1}^{T_m}\mathbb{E}_x\left[Z_0\cdot\mathbb{E}_{Q_1}\mathbb{P}_{\theta,x}(B_m)\right]&=\Delta\sum_{t=T_{m-1}+1}^{T_m}\mathbb{E}_x\mathbb{E}_{\tilde{Q}_0}\left[\langle x_{T,1}-x_{T,2},\theta_m\rangle_+\cdot\mathbb{P}_{\theta,x}(B_m)\right]\\&\geq\Delta\sum_{t=T_{m-1}+1}^{T_m}\mathbb{E}_x\mathbb{E}_{\tilde{Q}_0}\left[\min_m\{\langle x_{T,1}-x_{T,2},\theta_m\rangle_+\}\cdot\mathbb{P}_{\theta,x}(B_m)\right]\\&=\Delta\sum_{t=T_{m-1}+1}^{T_m}\tilde{Z}_0\cdot\mathbb{E}_{\tilde{Q}_1}\mathbb{P}_{\theta,x}(B_m)\\&=\Delta(T_m-T_{m-1})\tilde{Z}_0\cdot\mathbb{E}_{\tilde{Q}_1}\mathbb{P}_{\theta,x}(B_m)\end{aligned}$$

where $\tilde{Q}_1$ is the changed measure such that

$$\frac{\mathrm{d}\tilde{Q}_1}{\mathrm{d}\tilde{Q}_0\times\mathrm{d}P_x}(x,\theta)=\frac{\min_m\langle x_{T,1}-x_{T,2},\theta_m\rangle_+}{\tilde{Z}_0},$$

and $\tilde{Z}_0 = \mathbb{E}_x \mathbb{E}_{\tilde{Q}_0}[\min_m \langle x_{T,1} - x_{T,2}, \theta_m \rangle_+]$ is a normalizing constant. Further, we have

$$\tilde{Z}_0 = \mathbb{E}_x \left[\|x_{T,1} - x_{T,2}\|_2 \cdot \mathbb{E}_{\tilde{Q}_0} \left[\min_{m \in [M]} (\theta_{m,1})_+ \right] \right],$$

and by the property of Unif $\left(\mathbb{S}^{d-1}\right)$, we have

$$\mathbb{E}_{\tilde{Q}_0} \left[\min_{m \in [M]} (\theta_{m,1})_+ \right] \geq \frac{1}{2^{\frac{M+1}{2}}(M+1)\sqrt{d}}.$$

Thus $\tilde{Z}_0 \geq \frac{1}{2^{\frac{M+1}{2}} \cdot (M+1)\sqrt{d}}$. Because $\sum_{m=1}^{M} \mathbb{E}_{\tilde{Q}_1} \mathbb{P}_{\theta,x}(B_m) \geq 1$, there exists $m \in [M]$ such that $\mathbb{E}_{\tilde{Q}_1} \mathbb{P}_{\theta,x}(B_m) \geq 1/M$. Hence for this m we have $\tilde{Z}_0 \cdot \mathbb{E}_{\tilde{Q}_1} \mathbb{P}_{\theta,x}(B_m) \geq \frac{1}{2^{\frac{M}{2}+2} M^2}$. As a result,

$$\sum_{t=T_{m-1}+1}^{T_m} \mathbb{E}_x \left[Z_0 \cdot \mathbb{E}_{Q_1} \mathbb{P}_{\theta,x}(B_m) \right] \geq \frac{\Delta(T_m - T_{m-1})}{2^{\frac{M}{2}+2} M^2 \sqrt{d}}.$$

□

E Appendix of Section 6

Proof of Lemma 6.1. We introduce the Lagrangian multiplier $\alpha \geq 0$ for the constraint $\mathrm{D}_{\mathrm{KL}}(\mathbb{P}\|\mathbb{P}_0) \leq \delta$, and define the Lagrangian function

$$\mathcal{L}(\mathbb{P}, \alpha) = \mathbb{E}_{\mathbb{P}}[y(\pi(x))] + \alpha \left(\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}\|\mathbb{P}_0\right) - \delta\right).$$

Then, the primal problem can be reformulated as

$$\inf_{\mathbb{P}\in\mathcal{U}_{\mathbb{P}_0}(\delta)} \mathbb{E}_{\mathbb{P}}[y(\pi(x))] = \inf_{\mathbb{P}\ll\mathbb{P}_0} \sup_{\alpha\geq 0} \mathcal{L}(\mathbb{P}, \alpha) = \sup_{\alpha\geq 0} \inf_{\mathbb{P}\ll\mathbb{P}_0} \mathcal{L}(\mathbb{P}, \alpha),$$

where the last equality follows from strong duality.
Now, for a fixed $\alpha \geq 0$, we consider the inner minimization over over $\mathbb{P} \ll \mathbb{P}_0$. Write the KL divergence as $\mathrm{D}_{\mathrm{KL}}\left(\mathbb{P}\|\mathbb{P}_0\right) = \mathbb{E}_{\mathbb{P}}\left[\log \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{P}_0}\right]$ and define the Radon-Nikodym derivative $r(x) \triangleq \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{P}_0}(x)$ that satisfies $r \geq 0,\ \mathbb{E}_{\mathbb{P}_0}[r(x)] = 1$. This leads to

$$\begin{aligned} g(\alpha) &\triangleq \inf_{\mathbb{P}\ll\mathbb{P}_0} \mathcal{L}(\mathbb{P}, \alpha) = \inf_{\mathbb{P}\ll\mathbb{P}_0} \mathbb{E}_{\mathbb{P}}\left[y(\pi(x)) + \alpha \log\left(\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{P}_0}\right)\right] - \alpha\delta \\ &= \inf_{r\geq 0, \mathbb{E}_{\mathbb{P}_0}[r]=1} \mathbb{E}_{\mathbb{P}_0}[r(x)(y(\pi(x)) + \alpha \log r(x))] - \alpha\delta \end{aligned}$$

We now introduce another Lagrangian multiplier λ for the constraint $\mathbb{E}_{\mathbb{P}_0}[r(x)] = 1$, and define

$$\mathcal{L}(r, \lambda) = \mathbb{E}_{\mathbb{P}_0}[r(x)(y(\pi(x)) + \alpha \log r(x))] + \lambda\left(1 - \mathbb{E}_{\mathbb{P}_0}[r(x)]\right) - \alpha\delta.$$

By strong duality, we obtain

$$g(\alpha) = \inf_{r\geq 0} \sup_{\lambda} \mathcal{L}(r, \lambda) = \sup_{\lambda} \inf_{r\geq 0} \mathcal{L}(r, \lambda).$$

Fixing λ, we take the derivatives of $\mathcal{L}(r, \lambda)$ with respect to r:

$$\frac{\partial \mathcal{L}}{\partial r}(r, \lambda) = \mathbb{E}_{\mathbb{P}_0}[y(\pi(x)) + \alpha \log r(x) + \alpha - \lambda],\ \frac{\partial^2 \mathcal{L}}{\partial r^2}(r, \lambda) = \alpha \mathbb{E}_{\mathbb{P}_0}\left[\frac{1}{r(x)}\right] \geq 0.$$

The first-order condition implies that the minimizer is given by

$$r^\star(x) = \exp\left(\frac{\lambda - \alpha - y(\pi(x))}{\alpha}\right).$$

Substituting $r^\star$ into the Lagrangian, we obtain

$$g(\alpha) = \sup_{\lambda} \mathcal{L}(r^\star, \lambda) = \sup_{\lambda} \mathbb{E}_{\mathbb{P}_0}\left[-\alpha \exp\left(\frac{\lambda - \alpha - y(\pi(x))}{\alpha}\right)\right] + \lambda - \alpha\delta.$$

Taking the derivatives of $\mathcal{L}(r^\star, \lambda)$ with respect to λ, we obtain

$$\frac{\partial \mathcal{L}}{\partial \lambda}(r^\star, \lambda) = \mathbb{E}_{\mathbb{P}_0}\left[1 - \exp\left(\frac{\lambda - \alpha - y(\pi(x))}{\alpha}\right)\right],$$
$$\frac{\partial^2 \mathcal{L}}{\partial \lambda^2}(r^\star, \lambda) = \mathbb{E}_{\mathbb{P}_0}\left[-\frac{1}{\alpha}\exp\left(\frac{\lambda - \alpha - y(\pi(x))}{\alpha}\right)\right] \leq 0$$

Then, the maximum of $\mathcal{L}(r^\star, \lambda)$ is taken at

$$\lambda^\star = \alpha - \alpha \log \mathbb{E}_{\mathbb{P}_0}\left[\exp\left(-\frac{1}{\alpha}y(\pi(x))\right)\right],$$

and the function $g(\alpha)$ becomes

$$g(\alpha) = \mathcal{L}(r^\star, \lambda^\star) = -\alpha \log \mathbb{E}_{\mathbb{P}_0}\left[\exp\left(-\frac{1}{\alpha}y(\pi(x))\right)\right] - \alpha\delta,$$

which completes the proof. □

Lemma E.1. *Let $g(q) \triangleq \inf_{\mathrm{D_{KL}}(p\|q)\leq\delta} p$. For $\delta \leq 0.226, g(q)$ is differentiable and $g'(q) \geq 1/2$ for $q \in [0.4, 0.6]$.*

Proof. Since $\delta \leq 0.226$, $q \geq 0.4$ and $\mathrm{D_{KL}}(\mathrm{Bern}(0.1)\|\,\mathrm{Bern}(0.4)) \approx 0.2264 > \delta$, we have $q \geq g(q) \geq 0.1$. By Lemma B.12 in Yang et al. (2022), we have

$$g'(q) = \frac{\log(q/g(q)) - \log((1-q)/(1-g(q)))}{q/g(q) - (1-q)/(1-g(q))}$$

and $g'(q)$ is increasing. Therefore, $g'(q) \geq g'(0.4) \geq 0.5$. □

F Appendix of Section 7

Omitted Details in the Proof of Theorem 7.2. Here we present the proof for Lemma 7.1.

Proof of Lemma 7.1. We first show statement 2 ⇒ statement 1, i.e., if g is a polynomial of degree at most $m-1$, then (60) and (61) cannot hold simultaneously.

To show the point, define $s(x) \triangleq g(x + bv_0) - g(bv_0)$, and $Y_i \triangleq V_i - x_0$. Using (60), we have $\mathbb{E}[s(Y_1 + \cdots + Y_\ell)] = 0$ for all $\ell = 1, \cdots, K-1$. By exchangeability of Y_i, we also have $\mathbb{E}[s_\ell(Y_1, \cdots, Y_K)] = 0$ for $s_\ell(Y_1, \cdots, Y_K)$ defined through:

$$s_\ell(Y_1, \cdots, Y_K) \triangleq \frac{1}{\binom{K}{\ell}} \sum_{S\subseteq[K]:|S|=\ell} s\left(\sum_{i\in S} Y_i\right).$$

However, since s is a polynomial of degree at most $K-1$, and $s(0)=0$, we can conclude that for any reals $y_1, \cdots, y_K$,

$$\sum_{\ell=0}^{m}(-1)^{\ell}\binom{m}{\ell}s_{\ell}\left(y_1, \cdots, y_K\right)=0, \tag{72}$$

which contradicts (61). Here (72) comes from simple algebra: due to its linearity, it's enough to prove for monomials $s(x)=x^d$ for $d=0,1,\cdots,K-1$. We can see that the coefficient of $\prod_{i=1}^{K} y_i^{r_i}$ with $r_i \geq 0, \sum_{i=1}^{K} r_i = d$ in $s_{\ell}\left(y_1, \cdots, y_K\right)$ is:

$$\frac{d!}{\prod_{i=1}^{K}\left(r_i!\right)} \cdot \frac{\binom{K-I(r)}{\ell-I(r)}}{\binom{K}{\ell}},$$

with $I(r) \triangleq \sum_{i=1}^{K} \mathbf{1}\left\{r_i>0\right\}$. By summing these coefficient, we have

$$\frac{d!}{\prod_{i=1}^{K}\left(r_i!\right)}\sum_{\ell=0}^{K}(-1)^{\ell}\binom{K-I(r)}{\ell-I(r)}=\frac{d!}{\prod_{i=1}^{K}\left(r_i!\right)}\sum_{\ell=I(r)}^{K}(-1)^{\ell}\binom{K-I(r)}{\ell-I(r)}=0.$$

We next show statement 1 $\Rightarrow$ statement 2: after fixing v_0, the problem is to find an exchangeable distribution of $(V_1, \cdots, V_K)$ from the convex set of all such distributions. As both constraints (60) and (61) are linear in the joint distribution, and the Dirac point measure on $(v_0, \cdots, v_0)$ satisfies (60) as well as (61) with $<$ replaced by $\leq$. Therefore, there are two convex sets in total, one being the family of all exchangeable joint distributions and one from the constraints (60) and (61), and their closure has an intersection point, i.e. the Dirac point measure at $(v_0, \cdots, v_0)$. Now our target is to show that these sets have a non-empty intersection without taking the closure, and we need to introduce the following technical lemma which characterizes a sufficient condition via duality.

Lemma F.1. *For $g \in C([0,K])$ and a fixed $x_0 \in [0,1]$, there exists an exchangeable Borel distribution on $[0,1]^K$ such that both (60) and (61) hold, if the following condition holds: for every non-zero vector $\lambda=(\lambda_1, \cdots, \lambda_K)$ and the functions $g_{\ell}\left(v_1, \cdots, v_K\right)$ defined as:*

$$g_{\ell}\left(v_1, \cdots, v_K\right)=\frac{1}{\binom{K}{\ell}}\sum_{S \subseteq[K]:|S|=\ell} g\left(\sum_{i \in S} v_i+(K-\ell)v_0\right),$$

*for $\ell=1,2,\cdots,K$, the point $(v_0, \cdots, v_0)$ is **not** a global maxima of the function $\sum_{\ell=1}^{K}\lambda_{\ell} \cdot g_{\ell}(v)$ over $[0,1]^K$.*

Proof of Lemma F.1. We prove the contrapositive of this result. Let X be the topological vector space of all finite Borel signed measures on $[0,1]^K$, and $A \subseteq X$ be the collection

of all exchangeable Borel probability measures (i.e. invariant to permutations). Moreover, for $\varepsilon > 0$, we define $B_\varepsilon \subseteq X$ as the collection of all signed Borel measures μ such that:

$$\int_{[0,1]^K} \left(g_\ell(v) - g\left(Kv_0\right)\right) \mu(\mathrm{d}v) = 0,$$

for $\ell = 1, 2, \cdots, K-1$ and

$$\int_{[0,1]^K} \left(g_K(v) - g\left(Kv_0\right) - \varepsilon\right) \mu(\mathrm{d}v) = 0.$$

Therefore, the non-existence of such an exchangeable distribution implies that $A \cap B_\varepsilon = \emptyset$ for all $\varepsilon > 0$. Since $[0,1]^K$ is compact, the set A is a closed subset of the unit weak-* ball and is thus compact due to Banach–Alaoglu according to Rudin (1991) theorem 3.15. Besides, as $g_\ell \in C([0,1]^K)$, the set B_ε is also closed under the weak-* topology. Finally, as both sets are convex, according to Rudin (1991) theorem 3.10, the dual of X under the weak-* topology is $C([0,1]^K)$. Rudin (1991) theorem 3.4 further implies that there exists some $f_\varepsilon \in C([0,1]^K)$ and $\gamma_\varepsilon \in \mathbb{R}$ such that

$$\sup_{\mu \in A} \int f_\varepsilon \mathrm{d}\mu < \gamma_\varepsilon \leq \inf_{\nu \in B_\varepsilon} \int f_\varepsilon \mathrm{d}\nu. \tag{73}$$

As B_ε is a linear subspace of X, the RHS of (73) must be 0, otherwise it would be $-\infty$. By Rudin (1991) lemma 3.9 we must $f_\varepsilon(v) = \sum_{\ell=1}^{K-1} \lambda_{\varepsilon,\ell} \left(g_\ell(v) - g\left(Kv_0\right)\right) + \lambda_{\varepsilon,K} \left(g_K(v) - g\left(Kv_0\right) - \varepsilon\right)$ for some vector $\lambda_\varepsilon = (\lambda_{\varepsilon,1}, \cdots, \lambda_{\varepsilon,K})$. Plugging this back into the first inequality of (73), and defining $\mu_0 \in A$ as the Dirac point measure on $(v_0, \cdots, v_0)$, we arrive at

$$\sup_{\mu \in A} \int \left(\sum_{\ell=1}^{K} \lambda_{\varepsilon,\ell} g_\ell(x)\right) \left(\mu(\mathrm{d}x) - \mu_0(\mathrm{d}x)\right) < \lambda_{\varepsilon,K} \varepsilon. \tag{74}$$

Since $g_\varepsilon(x) = \sum_{\ell=1}^{m} \lambda_{\varepsilon,\ell} g_\ell(x)$ is a symmetric function, the LHS of (74) is $\max_{v \in [0,1]^K} g_\varepsilon(v) - g_\varepsilon\left(v_0, \cdots, v_0\right) \geq 0$. Then $\lambda_{\varepsilon,K} > 0$, and by multiplying a positive constant to all entries of λ_ε, we may assume that $\max_\ell |\lambda_{\varepsilon,\ell}| = 1$ and $\max_{v \in [0,1]^K} g_\varepsilon(v) - g_\varepsilon\left(v_0, \cdots, v_0\right) < \varepsilon$. Choosing $\varepsilon_n \to 0$, the compactness of $[-1,1]^K \ni \lambda_{\varepsilon_n}$ implies some subsequence $\lambda_{\varepsilon_{n_k}} \to \lambda$ as $k \to \infty$. Taking the limit along this subsequence, it is clear that λ is a non-zero vector with $\lambda_K \geq 0$, and $(v_0, \cdots, v_0)$ is a global maxima of the function $g(v) = \sum_{\ell=1}^{K} \lambda_\ell g_\ell(v)$. □

Now it remains to choose a suitable $v_0 \in [0,1]$ such that the condition in Lemma F.1 holds. We choose v_0 to be any point in $(0,1)$ such that $g^{(\ell)}\left(Kv_0\right) \neq 0$ for all $\ell \in [K]$, whose existence is ensured by the following lemma from the real analysis in the textbook of Donoghue (1969), and we thus omit the detailed proof.

Lemma F.2 (A Variant of Donoghue (1969), Page 35.)**.** *If $g \in C^K$ satisfies $g^{(n_x)}(x) = 0$ for some $n_x \in [K]$ and every $x \in (0,1)$, then g is a polynomial of degree at most $K-1$ on $(0,1)$.*

Assume by by contradiction that the function $f(v) \triangleq \sum_{\ell=1}^{K} \lambda_\ell \cdot g_\ell(v)$ attains its maximum at $(v_0, \cdots, v_0)$ for some non-zero vector λ, We will prove by induction on $k = 1, 2, \cdots, K$ the following claims:

1. $D^\alpha f(v_0, \cdots, v_0) = 0$ for all multi-indices $\alpha = (\alpha_1, \cdots, \alpha_k) \in \mathbb{N}^k$ with $|\alpha| \triangleq \sum_{\ell=1}^{k} \alpha_\ell = k$, where $D^\alpha f = \partial^{|\alpha|} f / \prod_{i=1}^{k} \partial v_i^{\alpha_i}$;
2. $\sum_{\ell=1}^{K} \lambda_\ell \cdot \binom{K-k}{\ell-k} / \binom{K}{\ell} = 0$, with $\binom{0}{0} \triangleq 1$ and $\binom{a}{b} \triangleq 0$ for $b < 0$.

For the base case $k = 1$, the first claim is exactly the first-order condition for the maxima, and the second claim follows from the first claim, the identity $\nabla g_\ell(v_0, \cdots, v_0) = g'(Kv_0)\mathbf{1} \cdot \binom{K-1}{\ell-1} / \binom{K}{\ell}$, and our choice of v_0 that $g'(Kv_0) \neq 0$. Now suppose that these claims hold for $1, 2, \cdots, k-1$. Then for $|\alpha| = k$ with $I(\alpha) \triangleq \sum_{\ell=1}^{m} \mathbf{1}(\alpha_\ell > 0) \leq k-1$, simple algebra gives:

$$D^\alpha f(v_0, \cdots, v_0) = g^{(k)}(Kv_0) \cdot \sum_{\ell=1}^{K} \lambda_\ell \frac{\binom{K-I(\alpha)}{\ell-I(\alpha)}}{\binom{K}{\ell}} = 0,$$

As $\mathbb{E}\left[\sum_{S \subseteq [M]: |S| = k} \prod_{i \in S} \varepsilon_i\right] = 0$ when $\varepsilon_1, \cdots, \varepsilon_K$ are Radamacher random variables, and the term inside the expectation is not identically zero, we conclude that this term could be either positive or negative after carefully choosing $(\varepsilon_1, \cdots, \varepsilon_m)$. As a result, in order for $(v_0, \cdots, v_0)$ to be a maxima of f, the above expansion implies that $\sum_{\ell=1}^{m} \lambda_\ell \cdot \binom{m-k}{\ell-k} / \binom{m}{\ell} = 0$ (recall that $g^{(k)}(Kv_0) \neq 0$), which is exactly the second claim for k. The remainder of the first claim also follows from the second claim, and the induction is complete.

Therefore, we derive a contradiction, which shows that such a non-zero vector λ cannot exist.

□

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